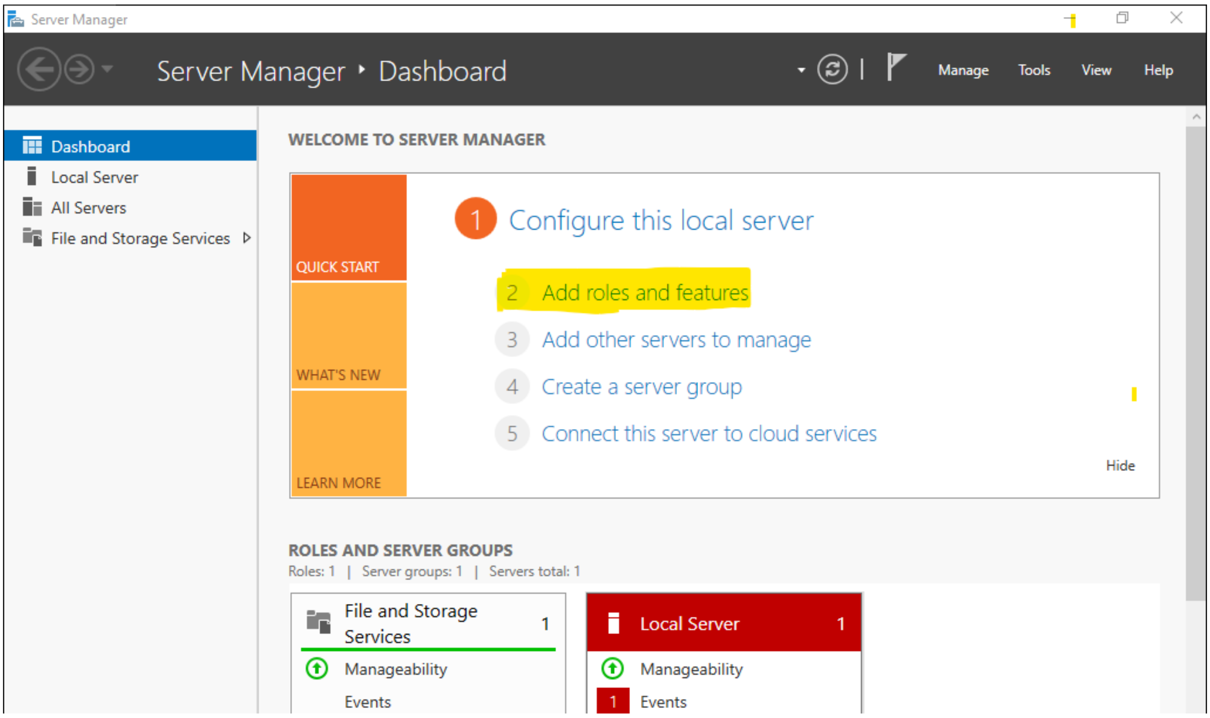
Practical 7

Aim: Protecting Workloads using Data Protection Manager

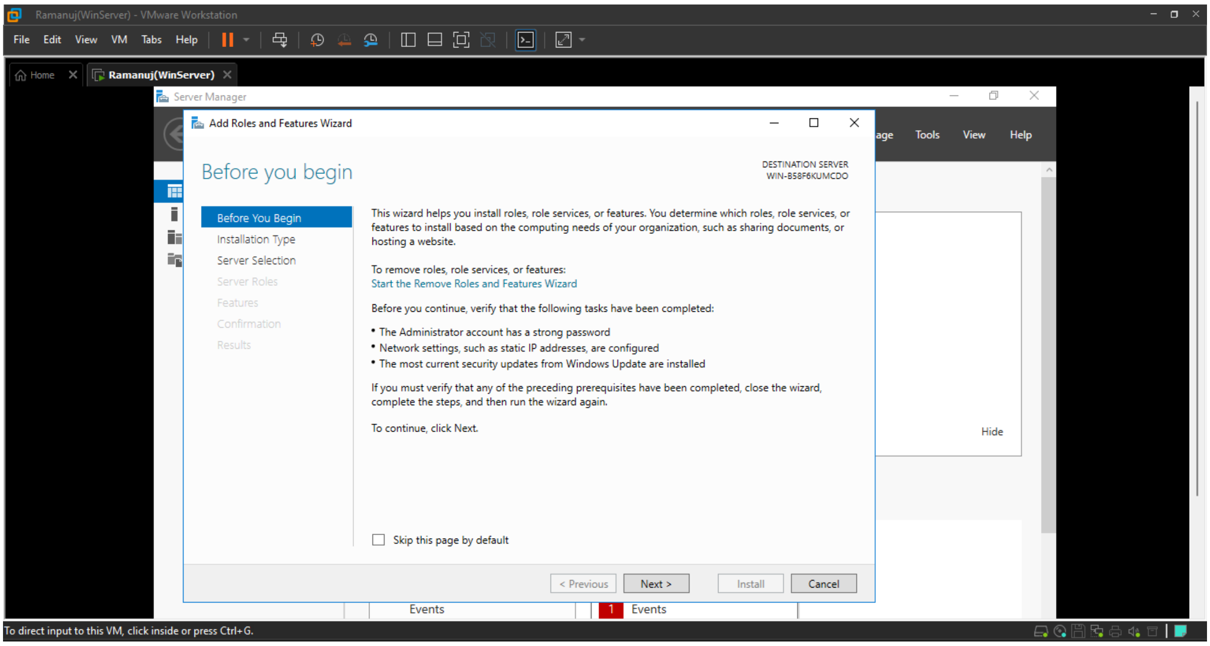
Writeup:

7-A: Back up Hyper-V Virtual Machine
Note: This is to be installed in your Domain Controller
Step 1: Installing Hyper-V on Server Manager

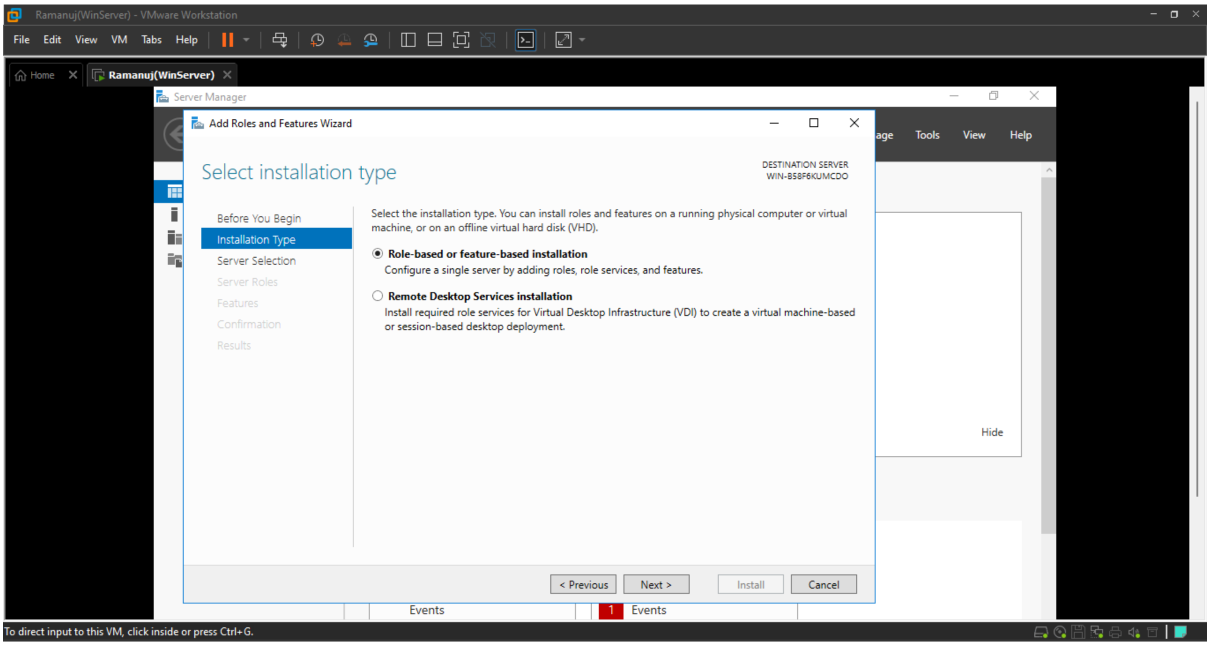
- Within Server Manager Click on **Add Roles and Features**



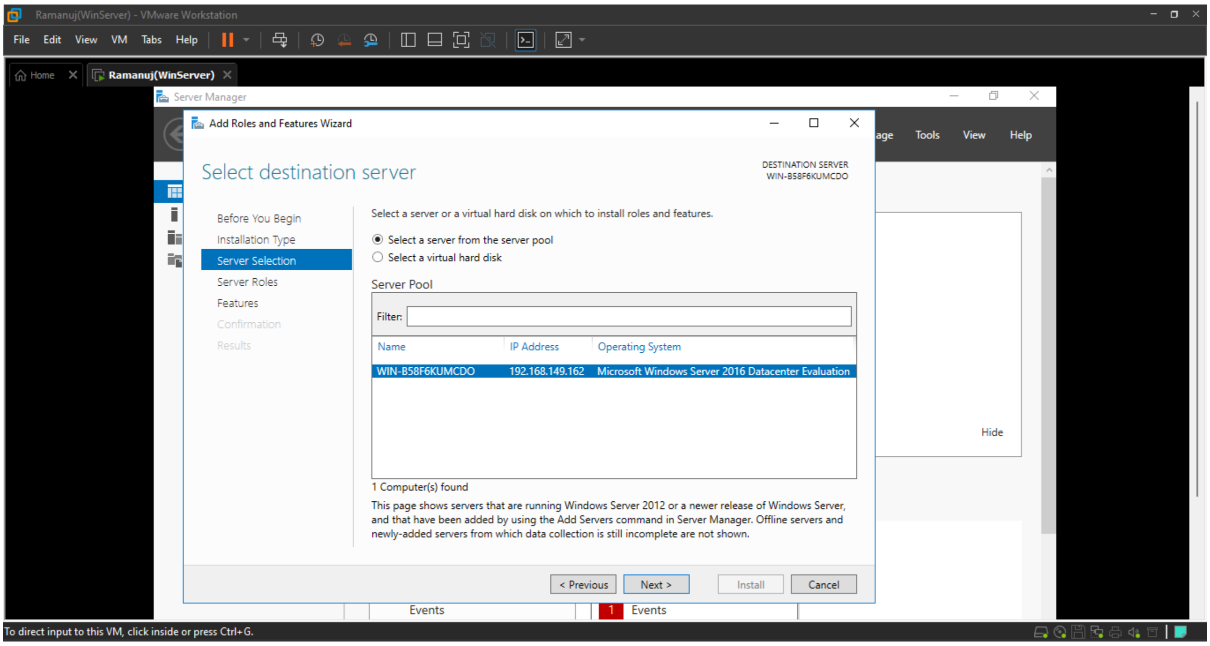
- Click **Next**



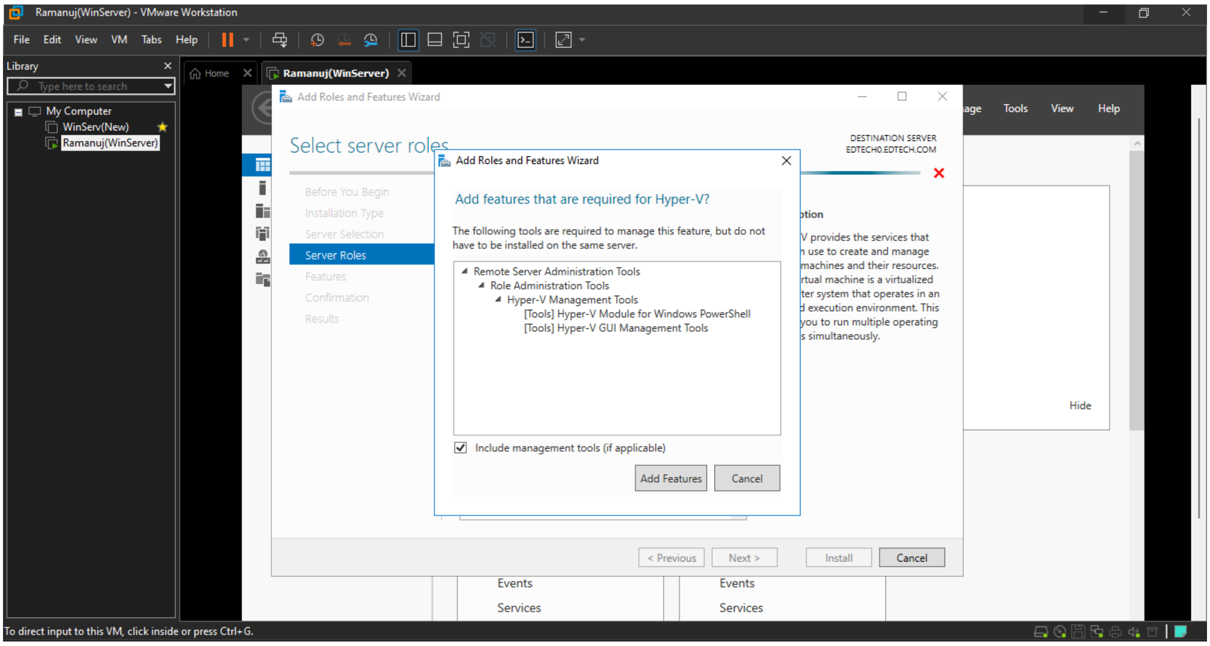
- Make sure **Role-based or Feature-based installation** is selected and Click **Next**



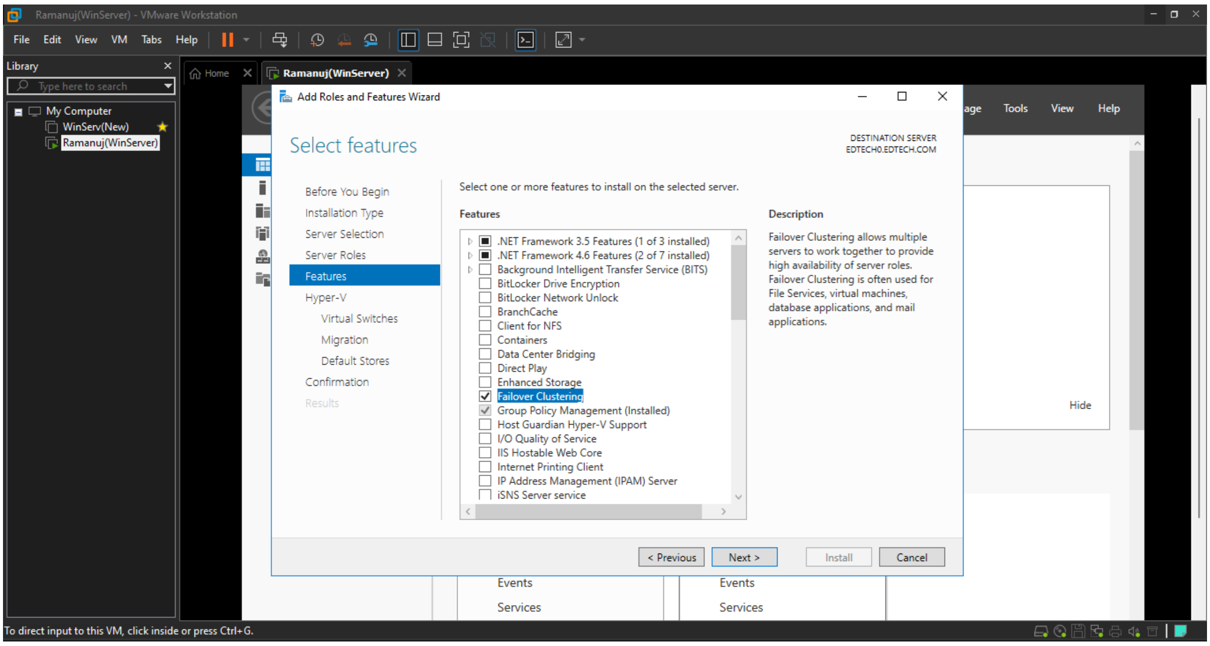
- Keep default settings and Click **Next**



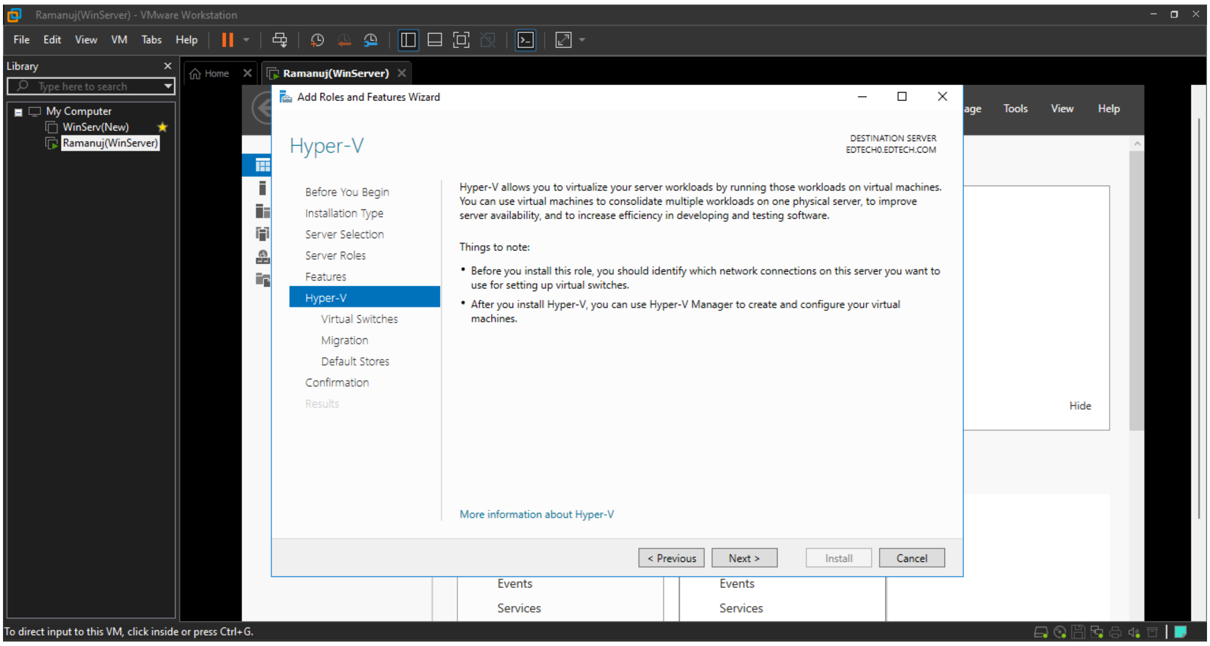
- Select **Hyper-V** and Click **Add Features** and Click **Next**



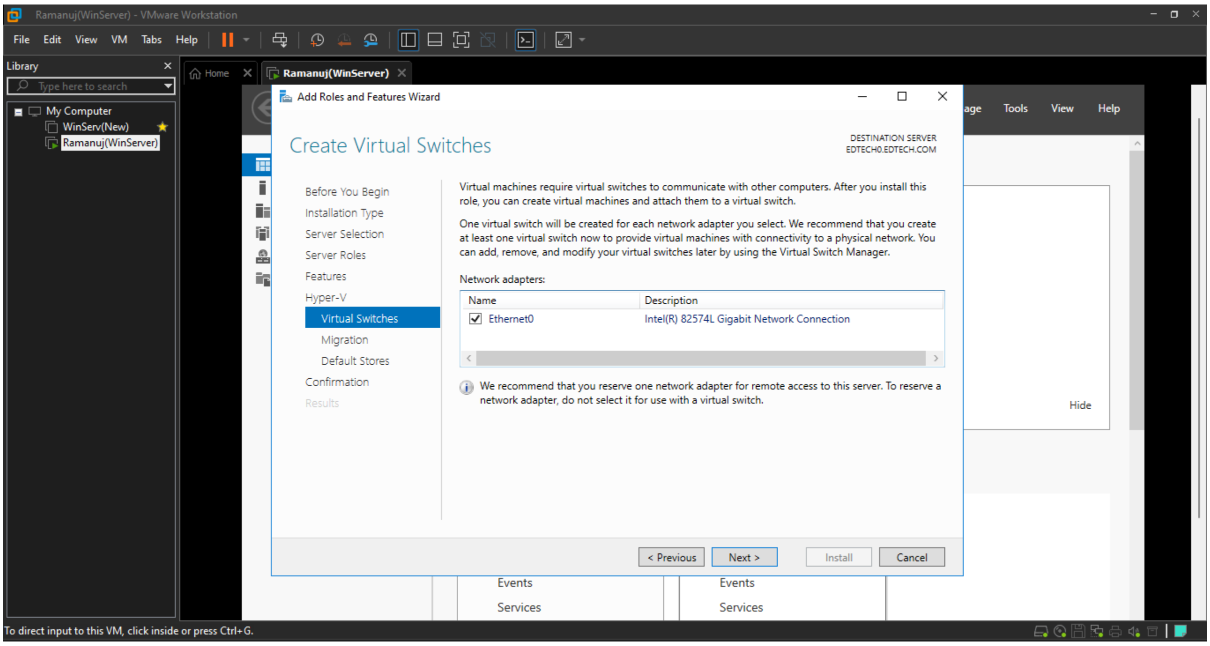
- Select **Failover Clustering** and Click **Add Features** and Click **Next**
- Note: Make sure **Group policy management** is also selected



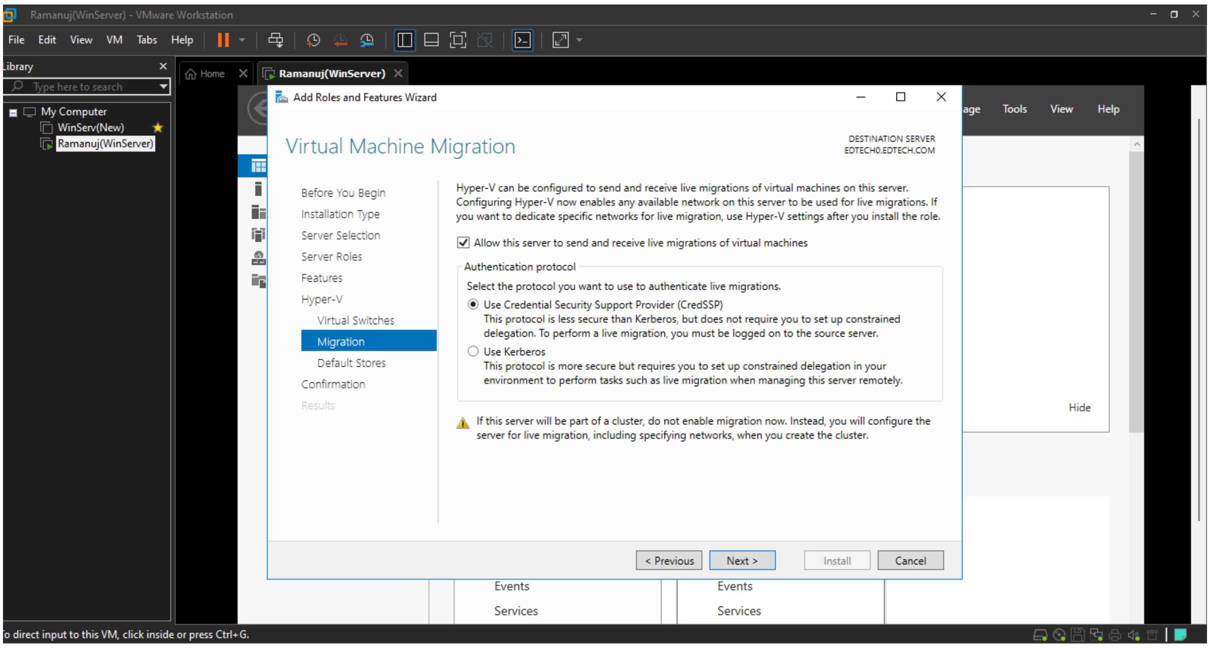
- Keep default settings and Click **Next**



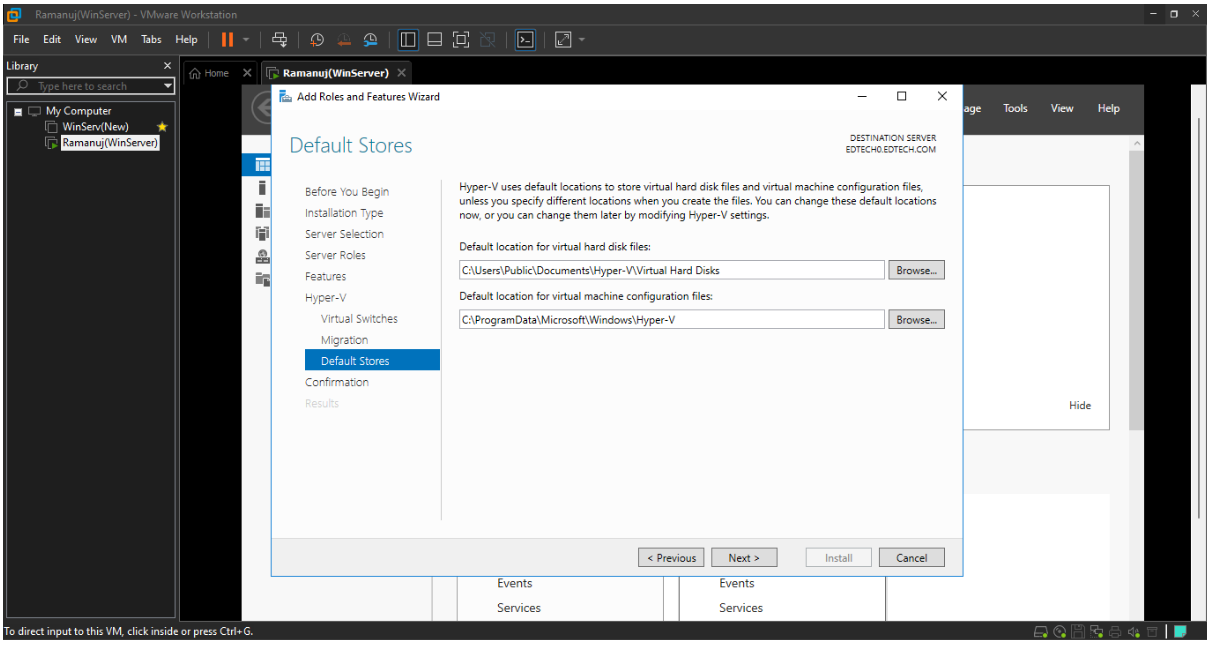
- Select **Ethernet0** and Click **Next**



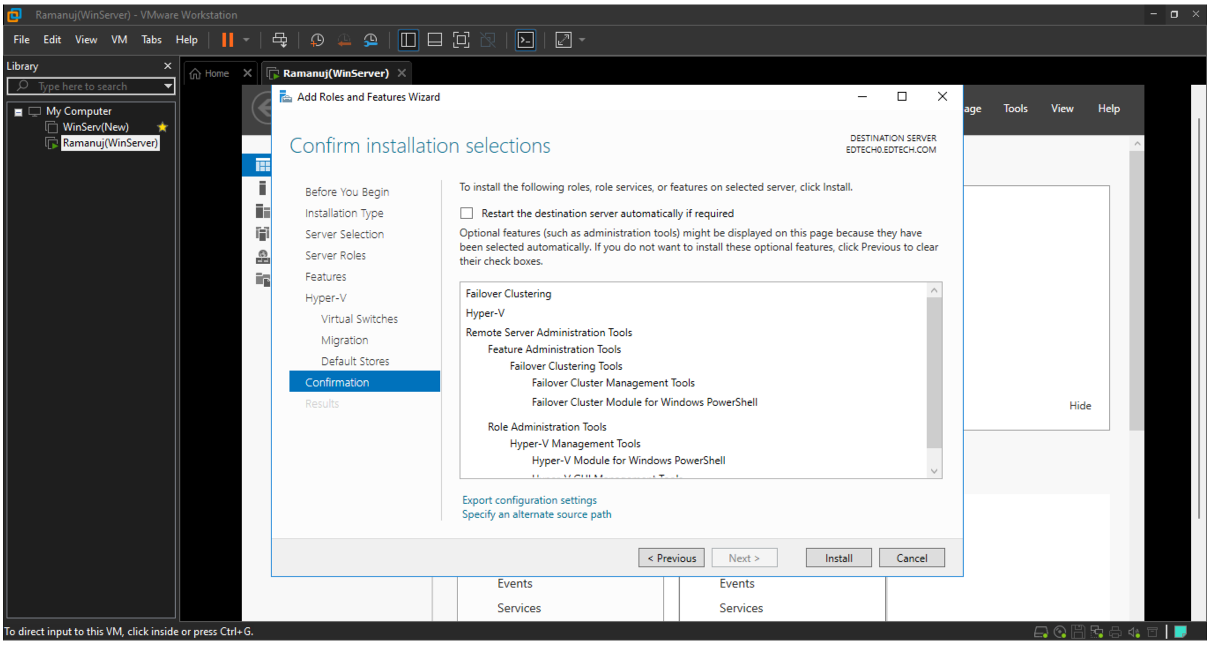
- Select **Allow to server to send and receive Live Migration of virtual machines**



- Keep default setting and Click **Next**

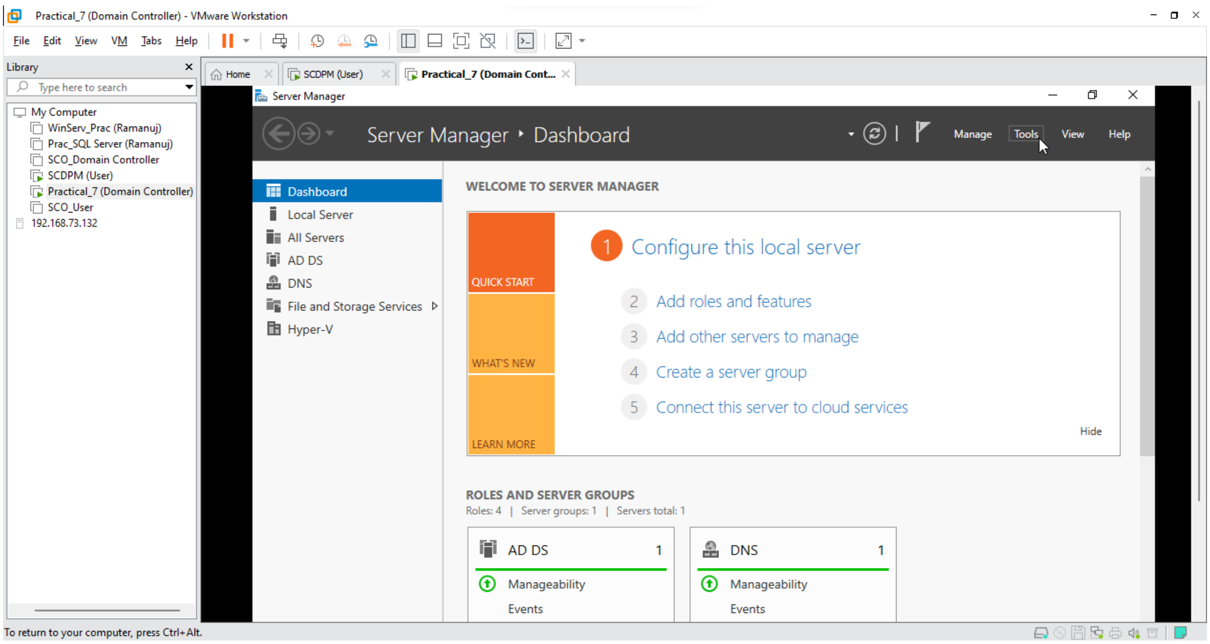


- Click on **Install,** After installation is complete restart the machine for the changes to be taken place.

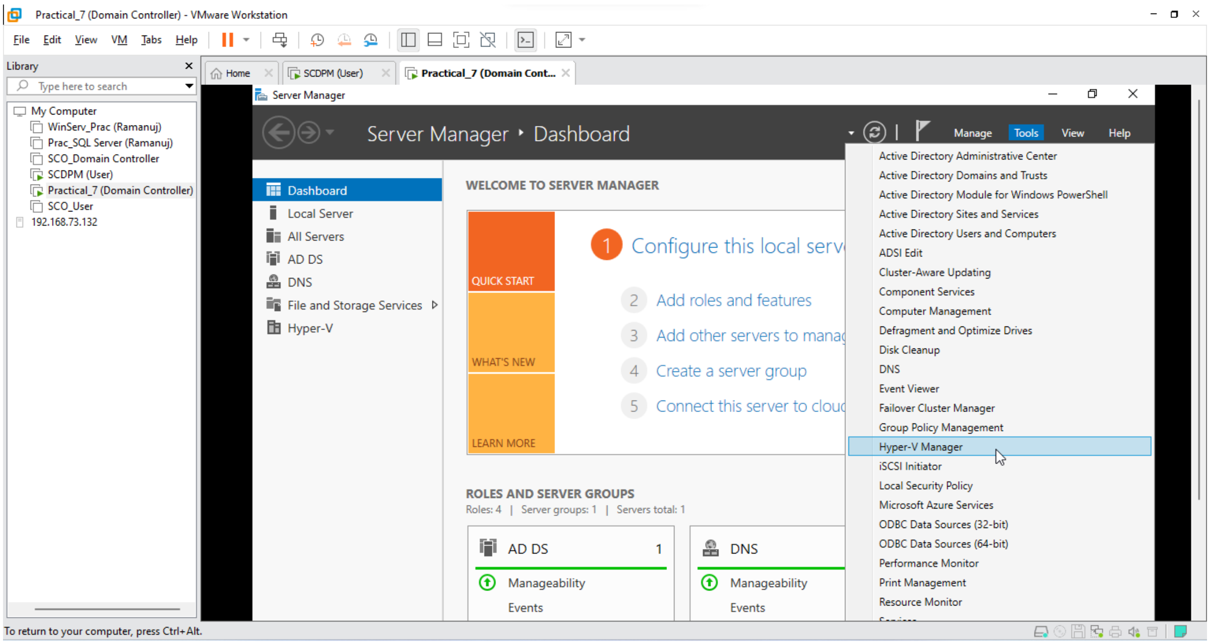


Step 2: Creating a Virtual Machine in Hyper-V

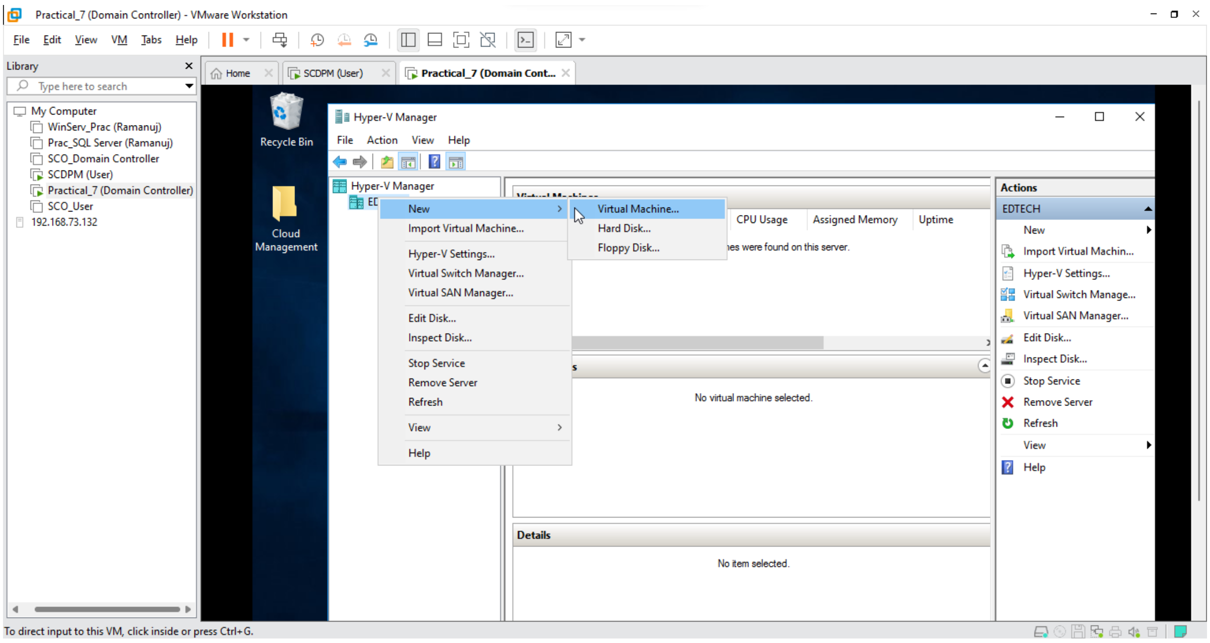
- Within Server Manager Click on **Tools**



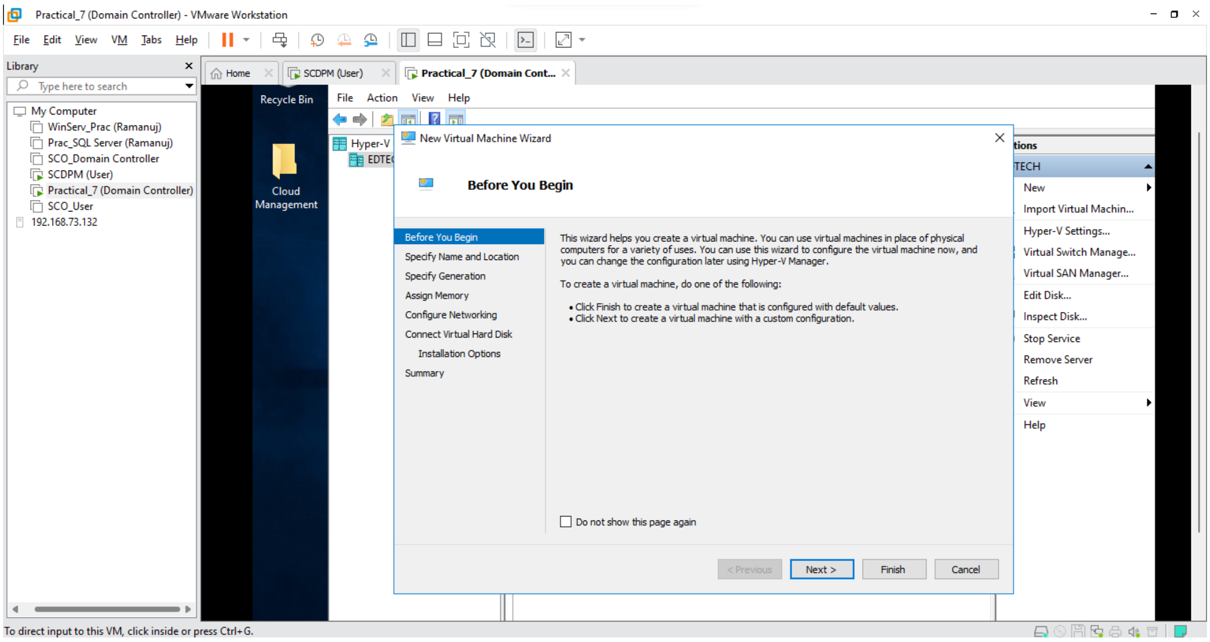
- Select **Hyper-V Manager**



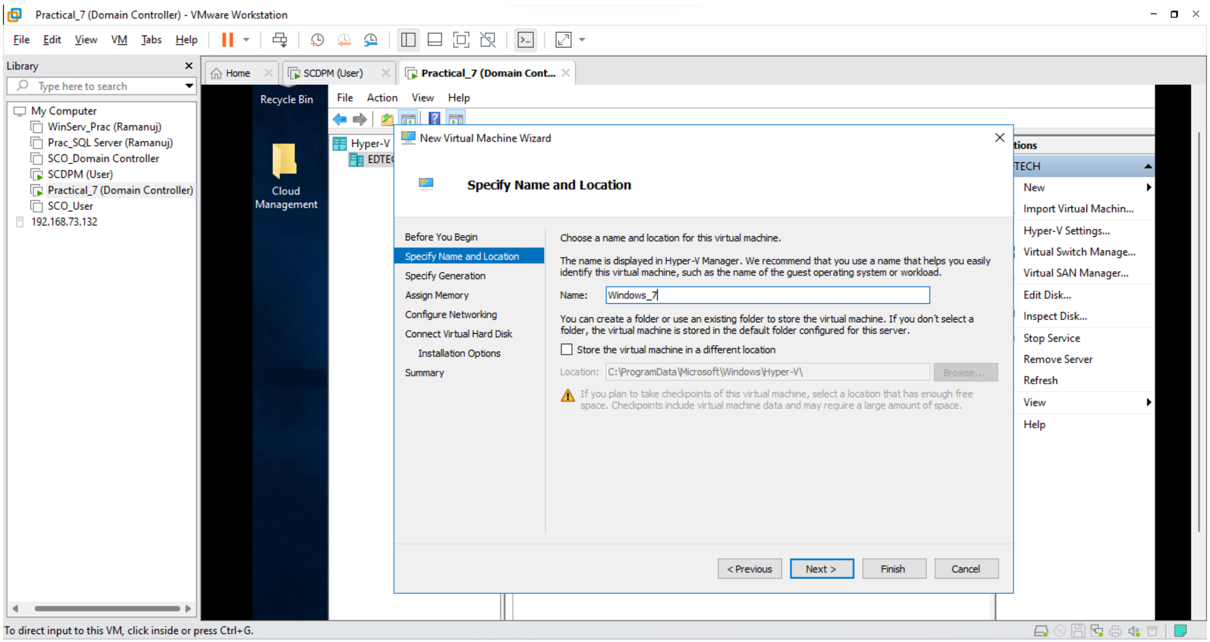
- Within Hyper-V Manager Right-Click on your **Domain Controller** and Select **New** and Click on **Virtual Machine**



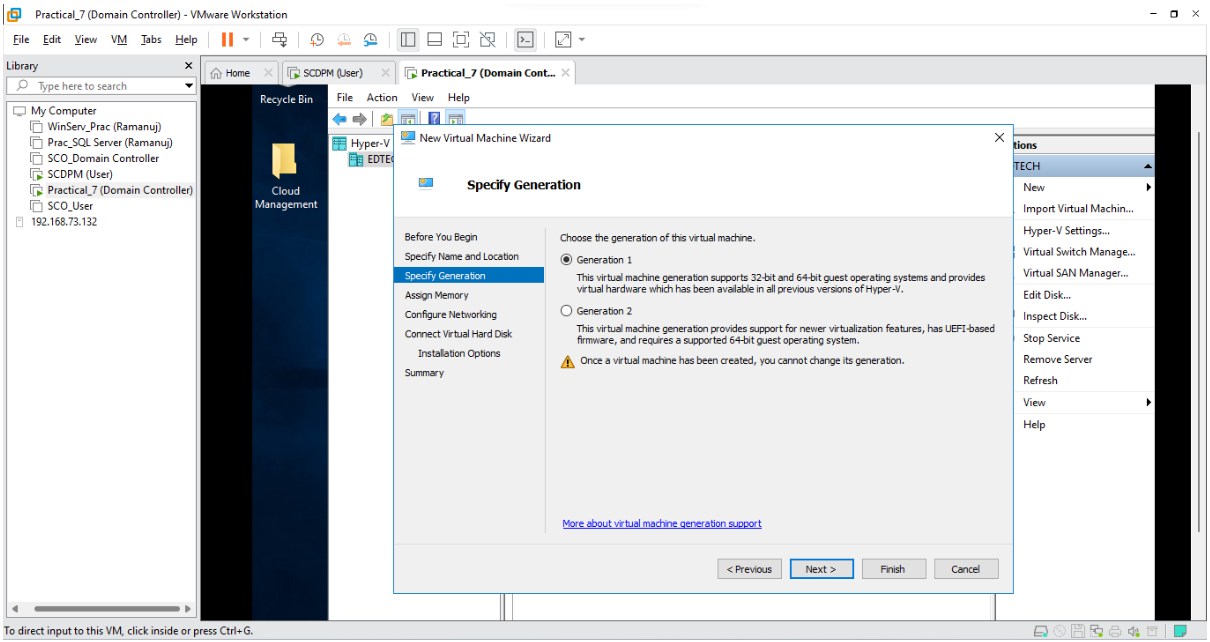
- Click **Next**



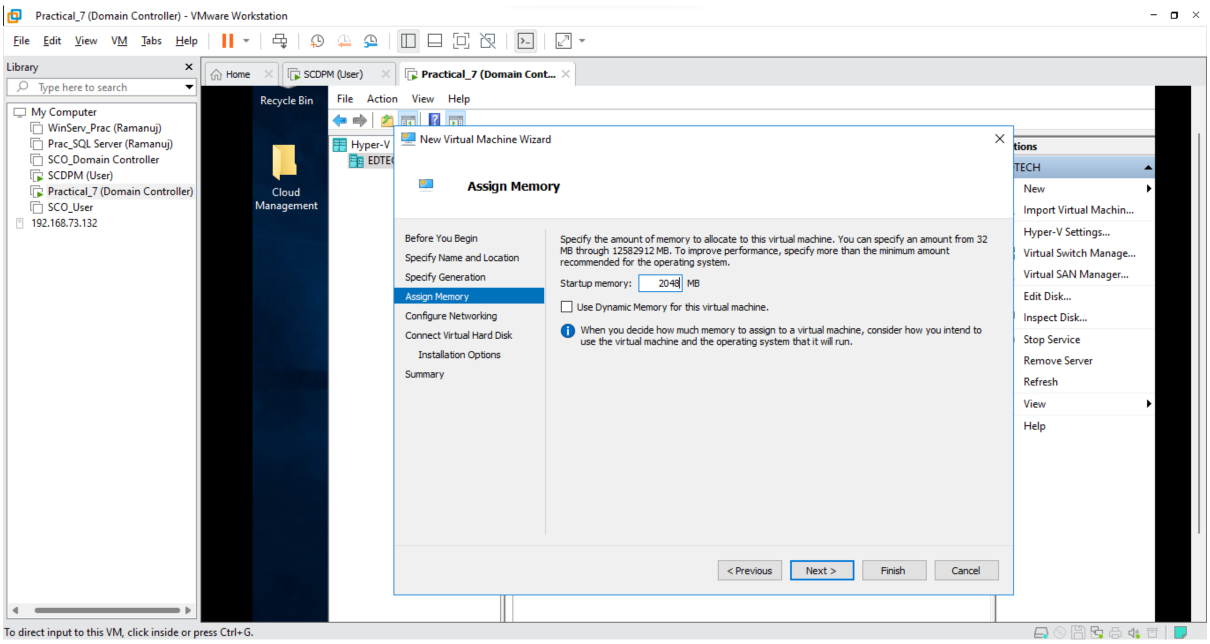
- Name your virtual machine and Click **Next**



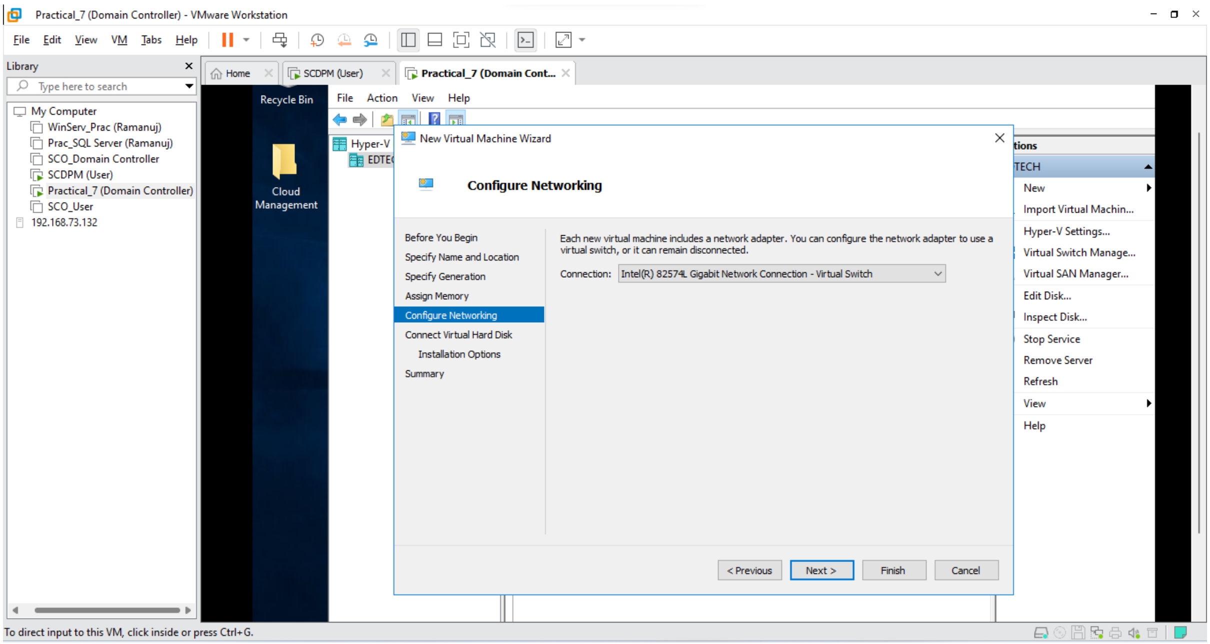
- Select **Generation 1**and Click **Next**



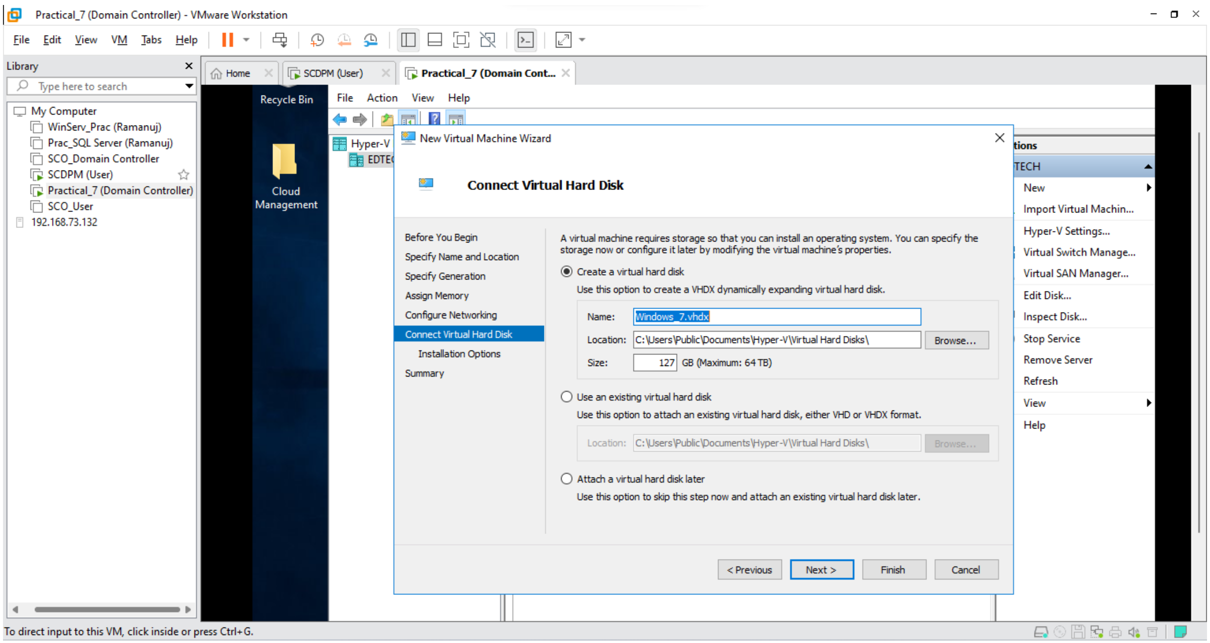
- Select Start-up memory as **2048 MB** and Click **Next**



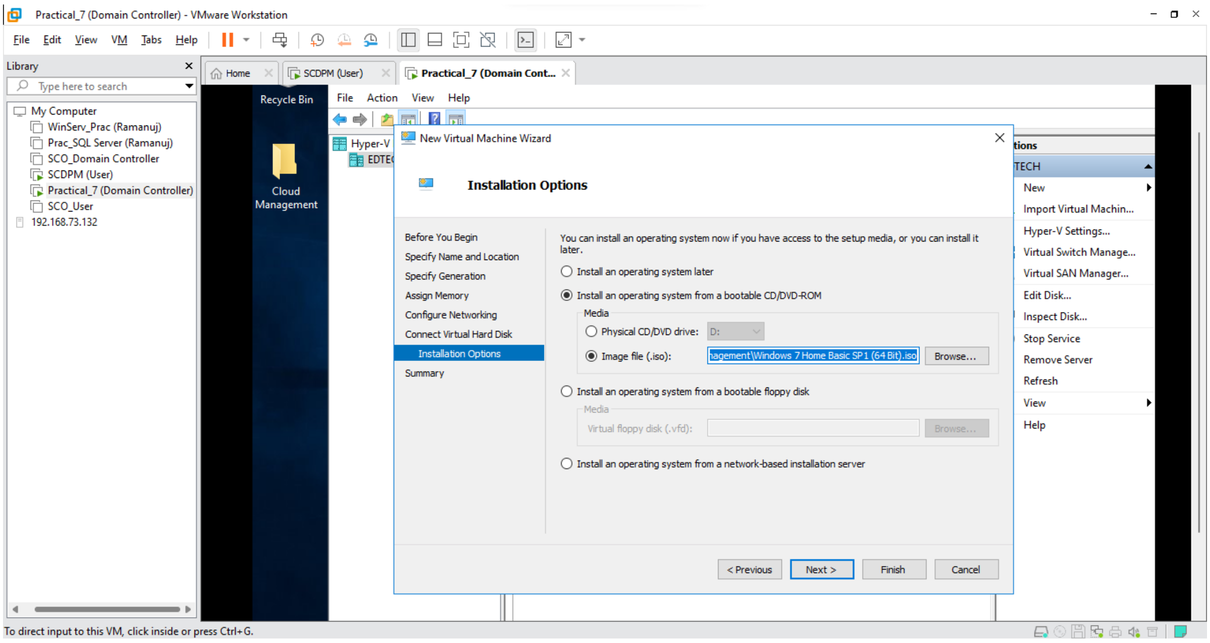
- In Connection Select your **network adapter** (Here it is Intel (R) Gigabit) and Click **Next**



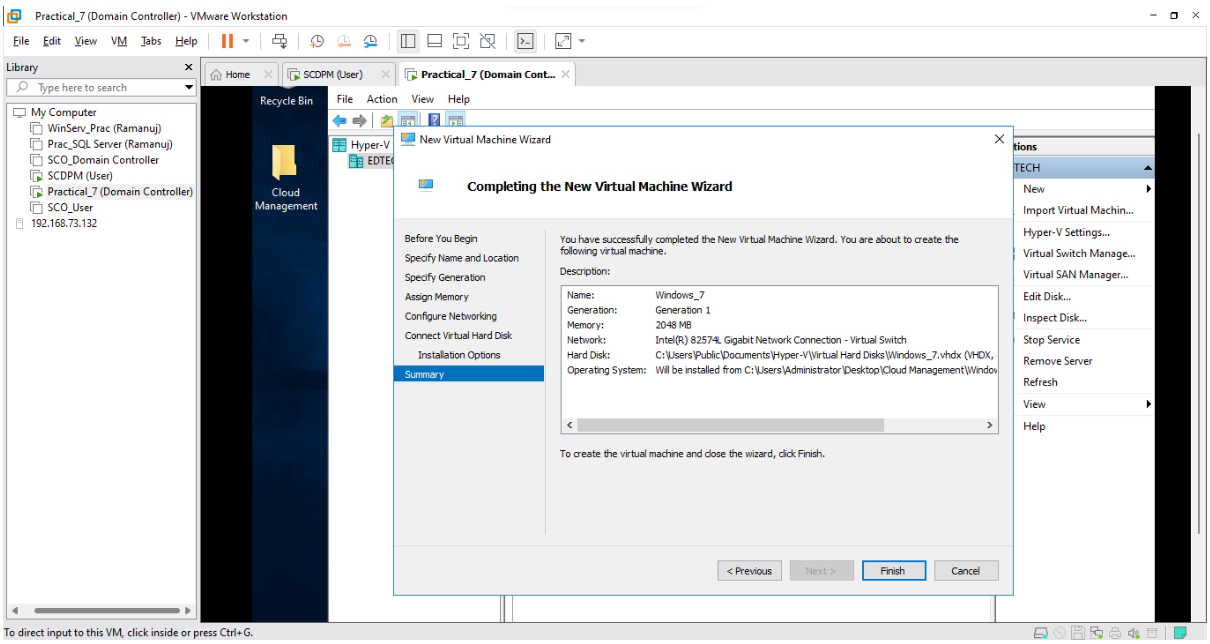
- Keep default values and Click **Next**



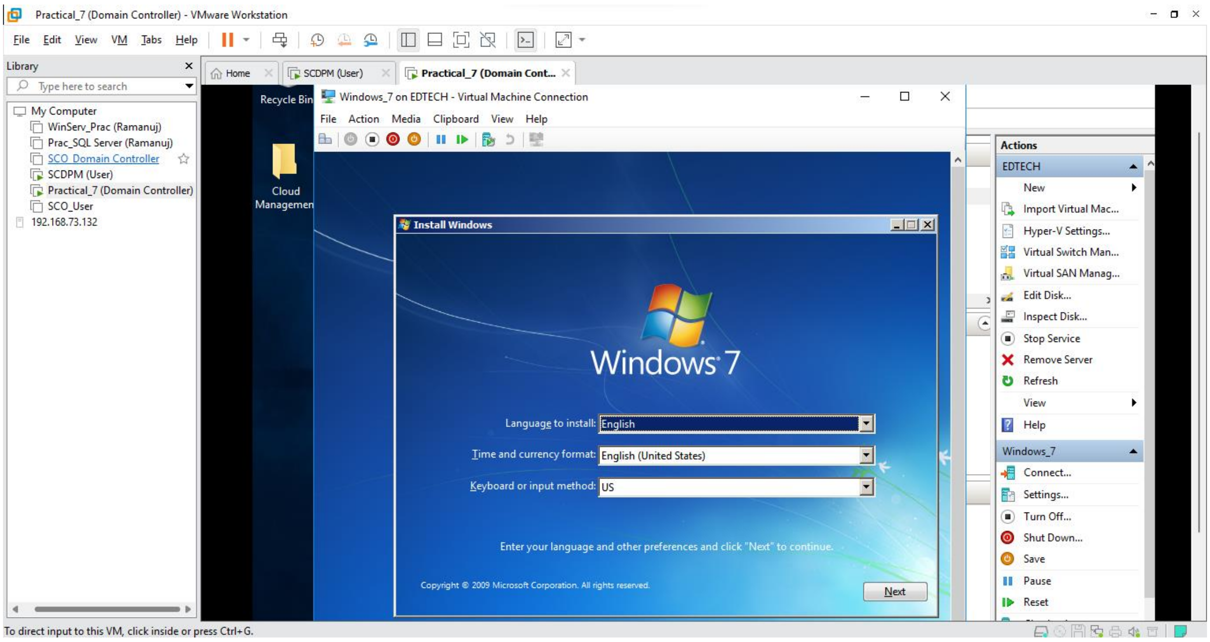
- Select **Install an operating system from a bootable CD/DVD ROM** and Click on **Image file** (.iso) and browse for your Windows iso file and Click **Next**



- View the summary of your virtual machine and Click **Next**

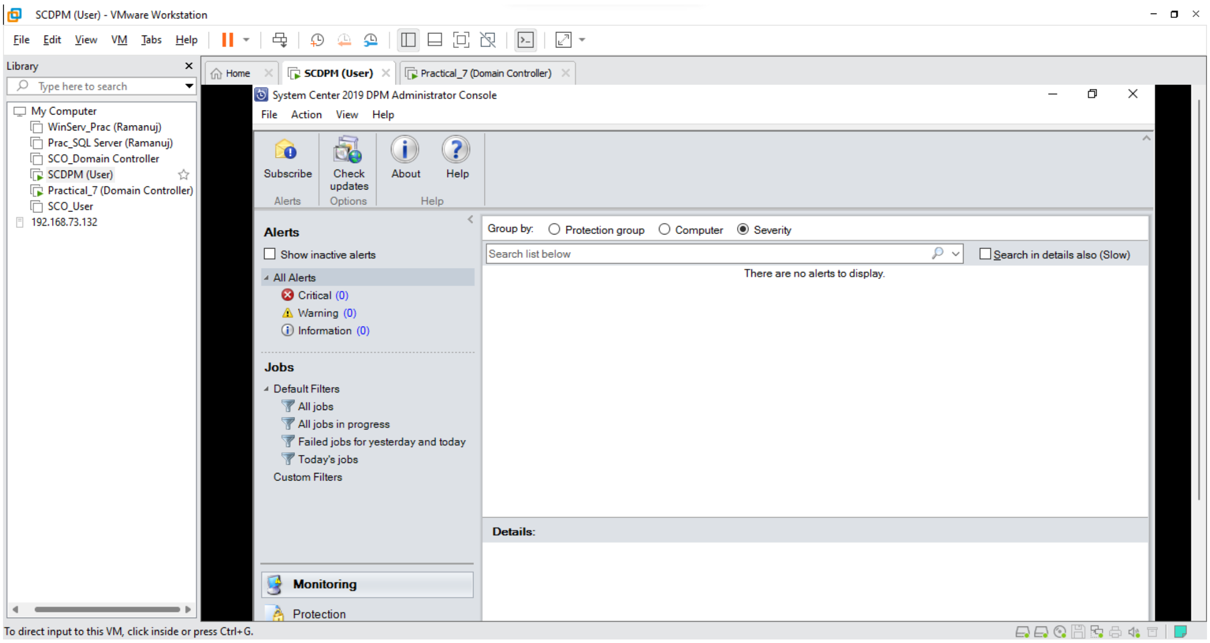


- To ensure it works **Power On** the Machine

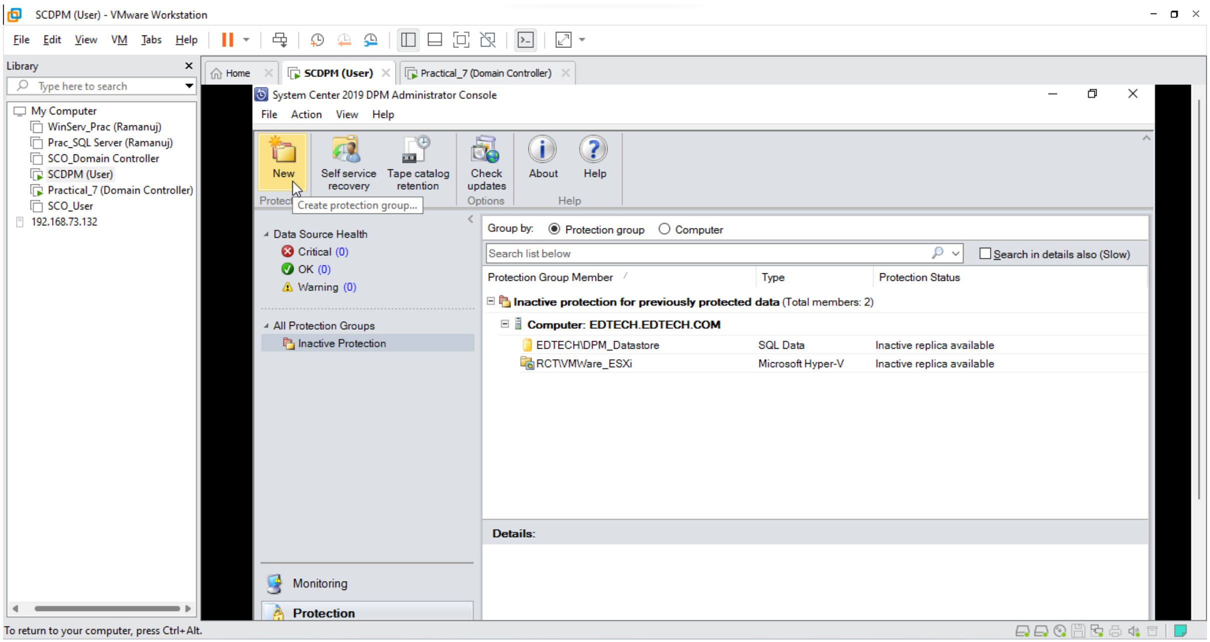


Step 3: Creating a Protection Group for your Hyper-V Virtual Machine

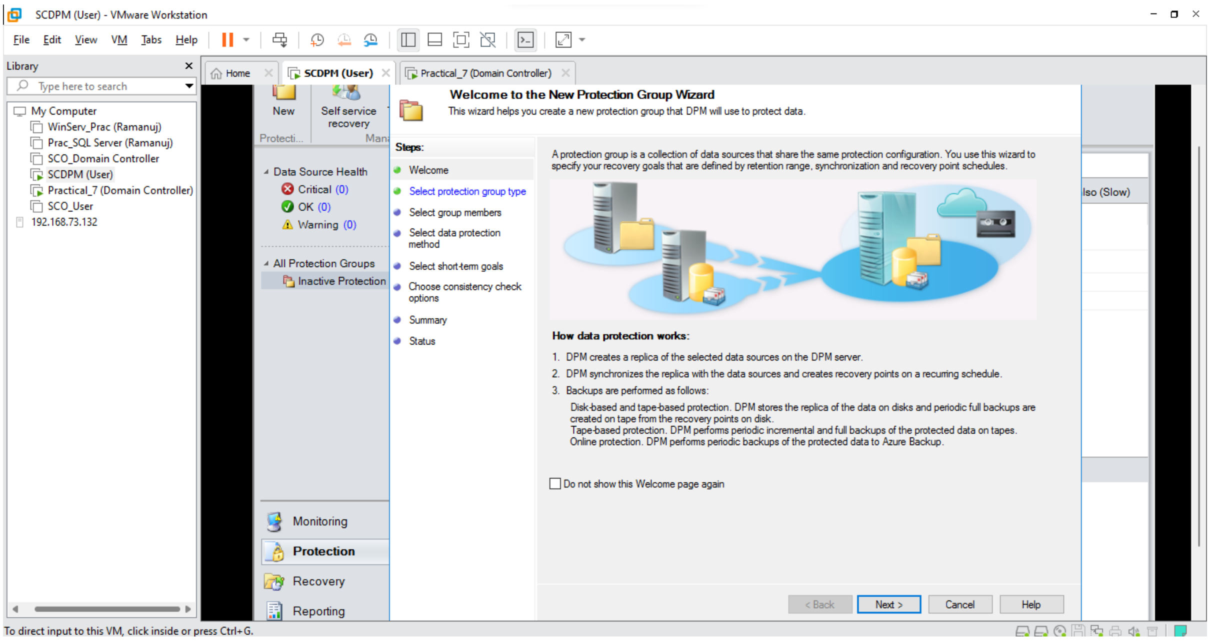
- **Power Off** your Machine and Switch over to the **SCDPM (User)** VM and Open the **DPM manager**



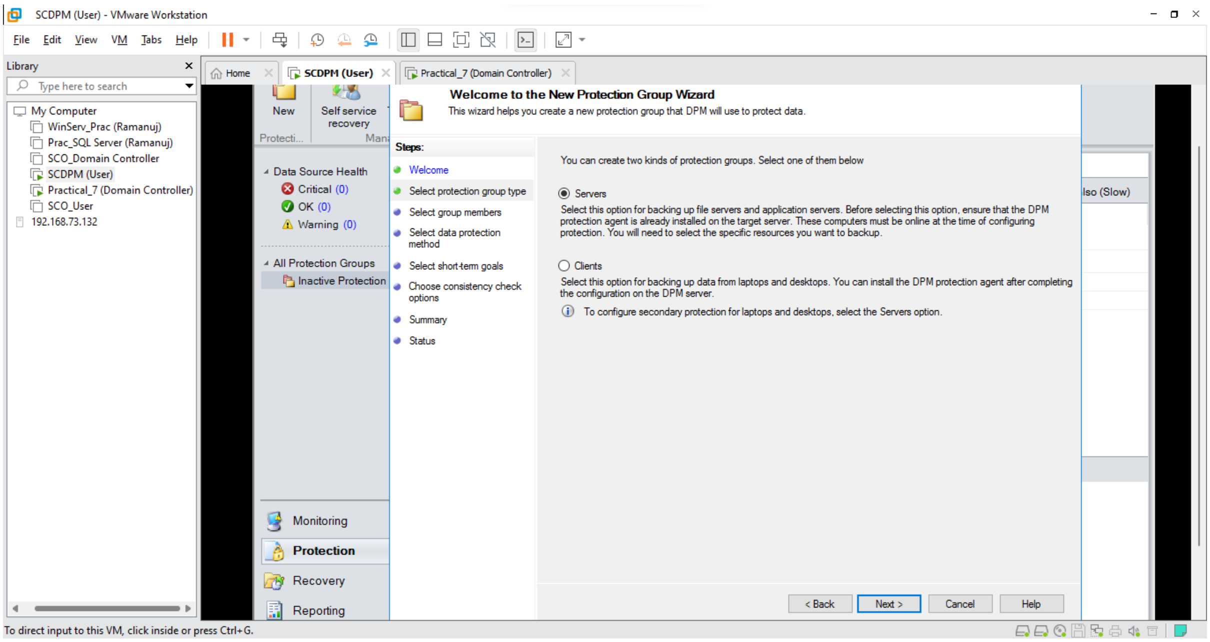
- Click on the **Protection** Tab and Select **New**



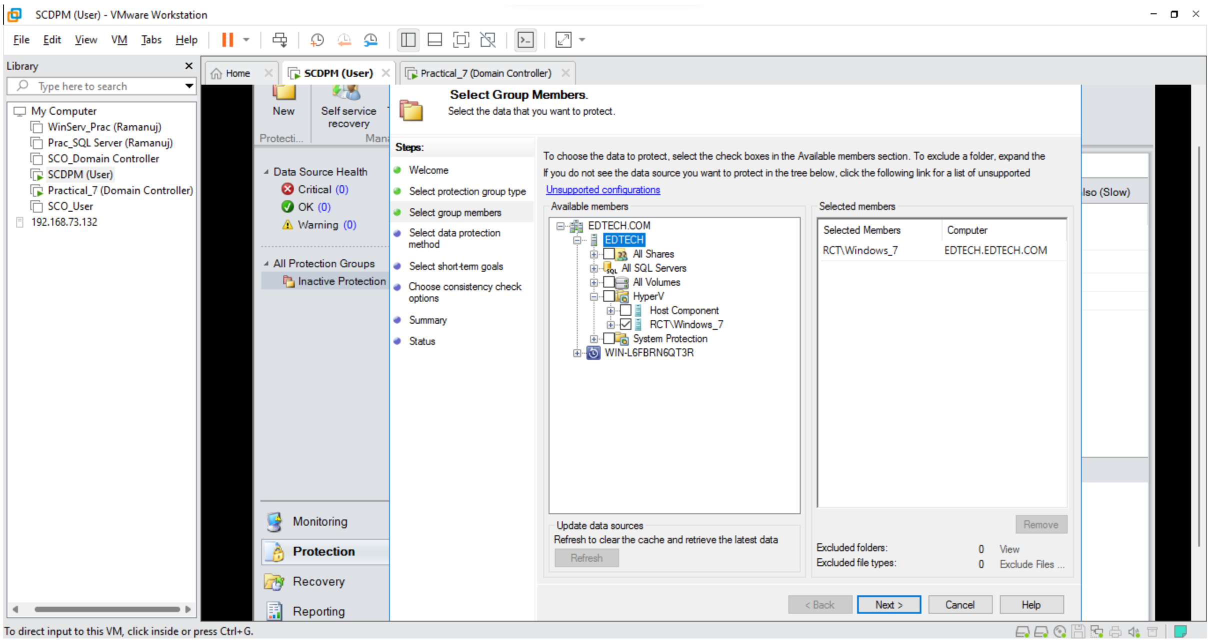
- Click **Next**



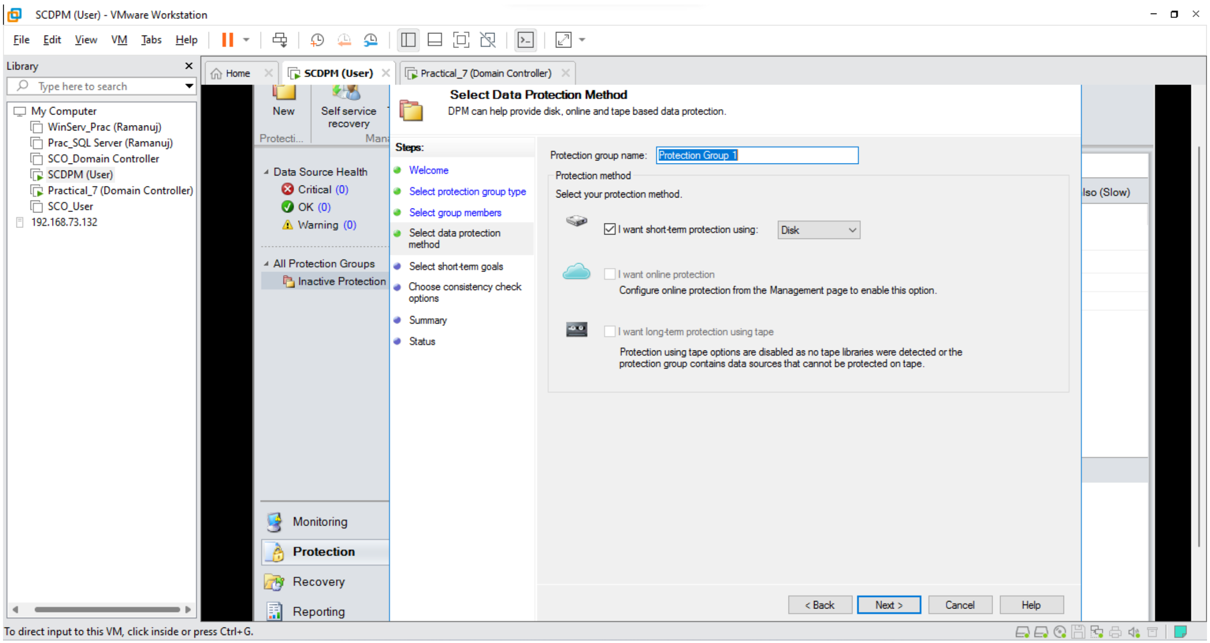
- Select **Servers** and Click **Next**



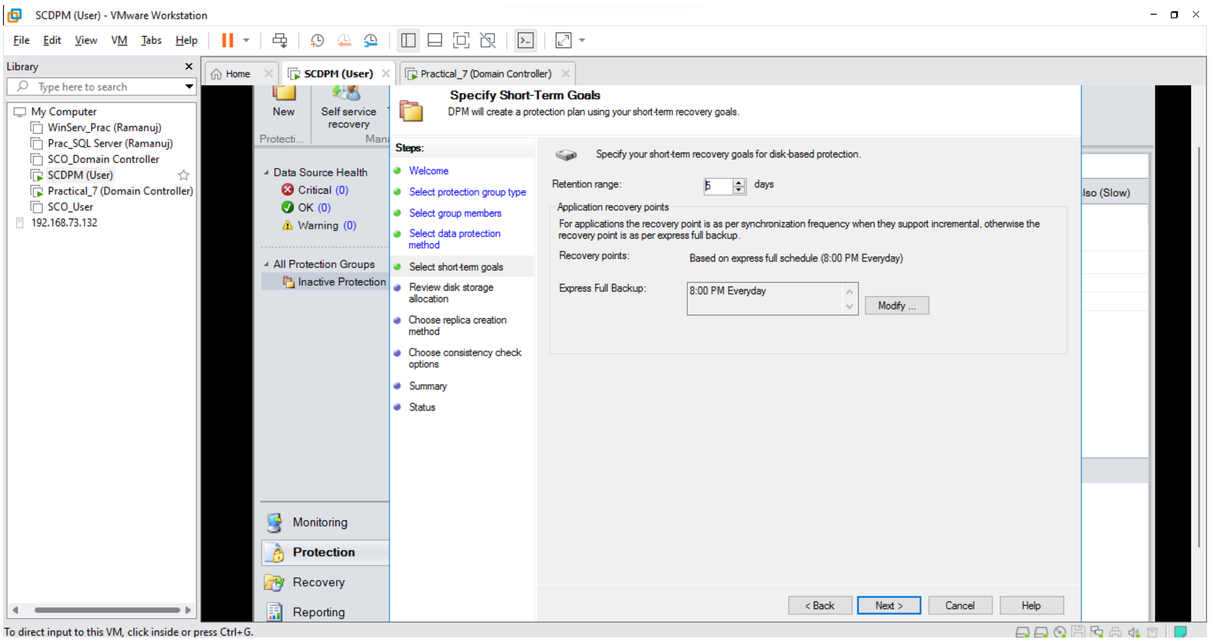
- Select **EDTECH** and Click on **Refresh,** after refresh Open **Hyper-V** and Click on your **virtual machine** (Here it is RCT\Windows_7) and Click **Next**



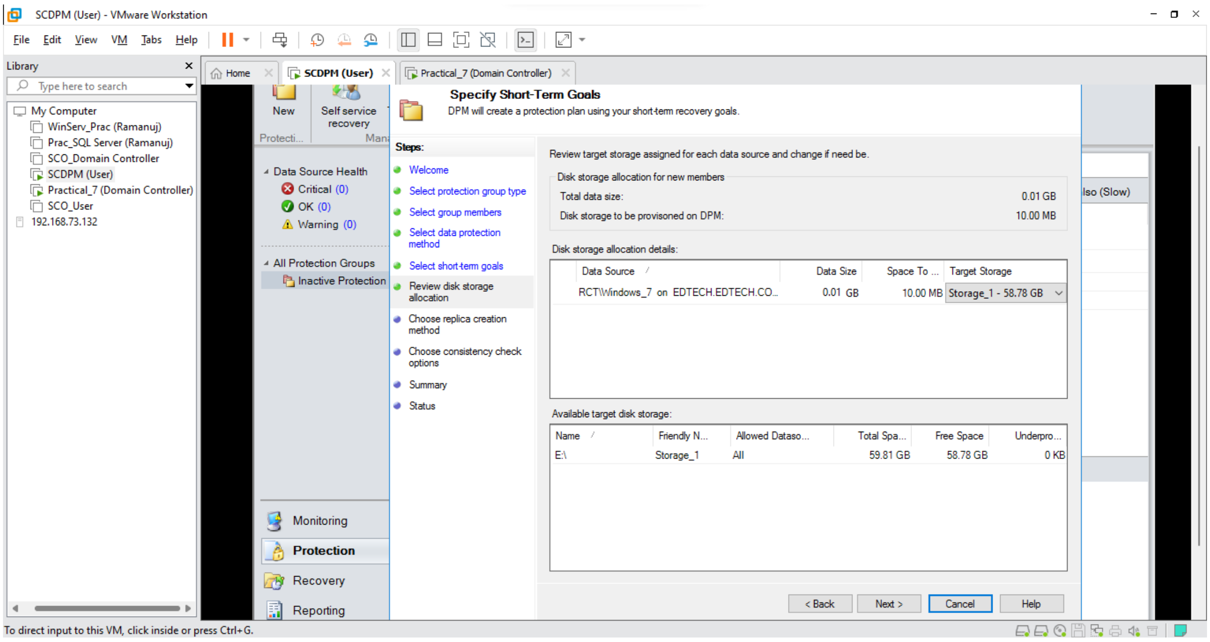
- Name your **Protection Group** and Select **I want short term protection** and Select **Disk** and Click **Next**



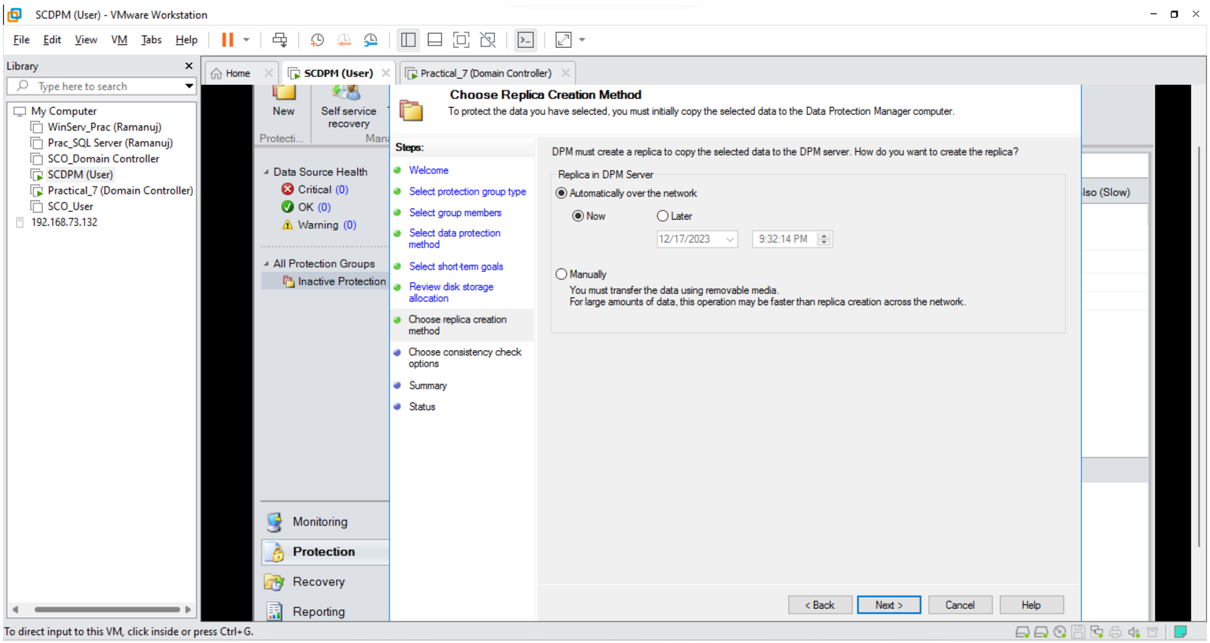
- Keep default values and Click **Next**



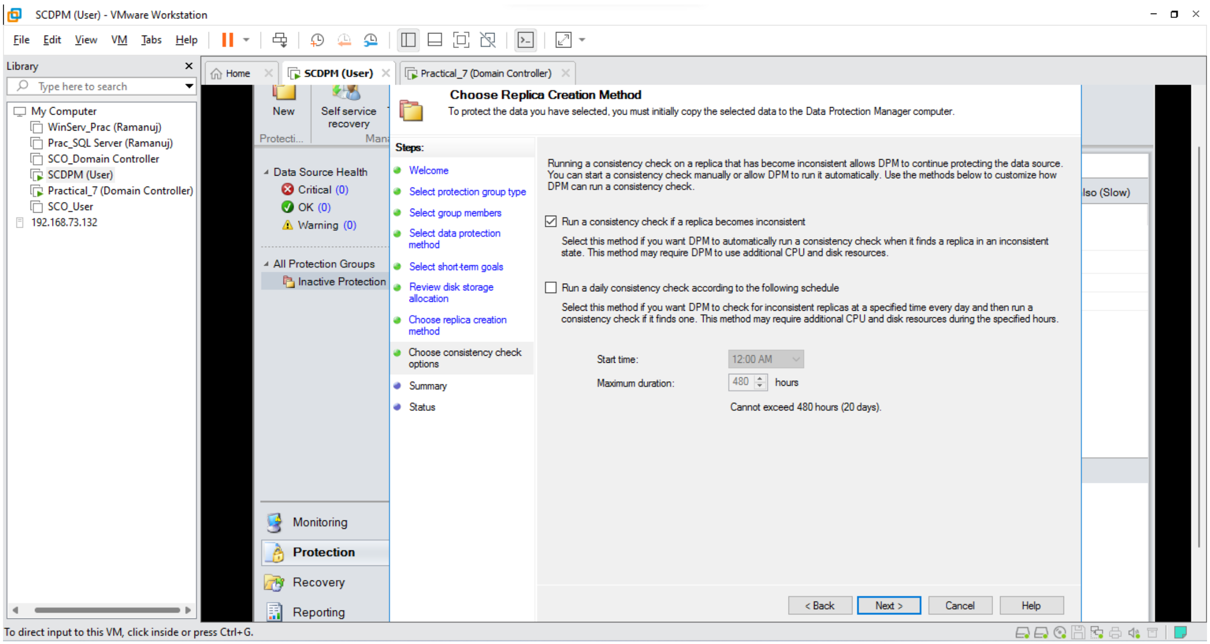
- Click **Next**



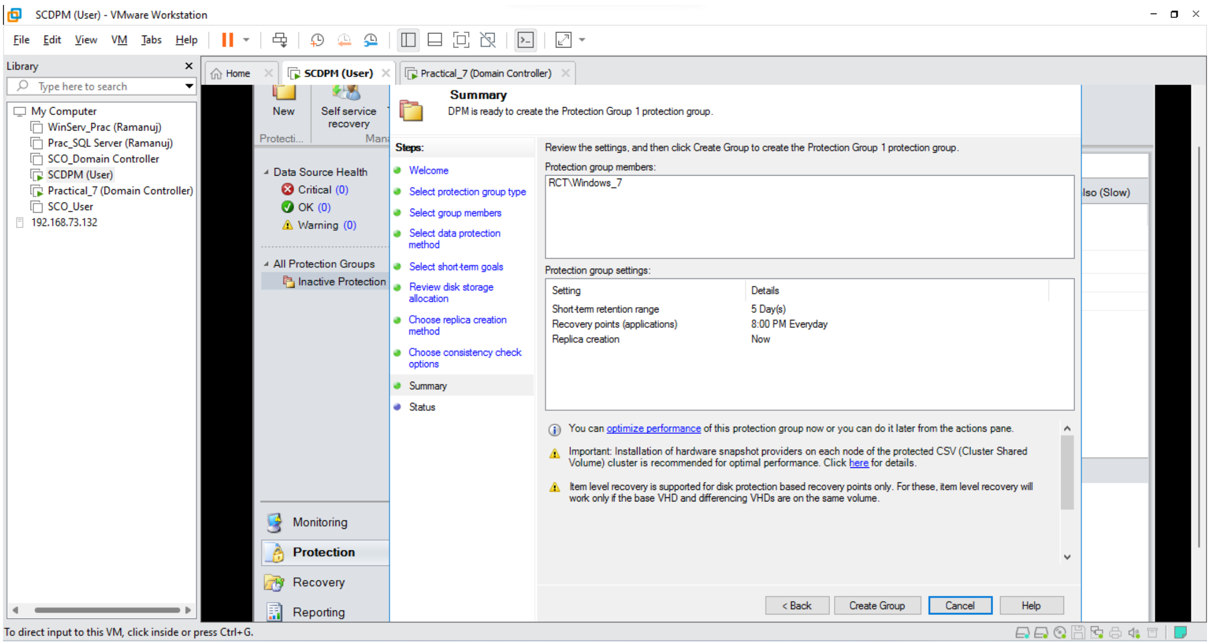
- Click **Next**



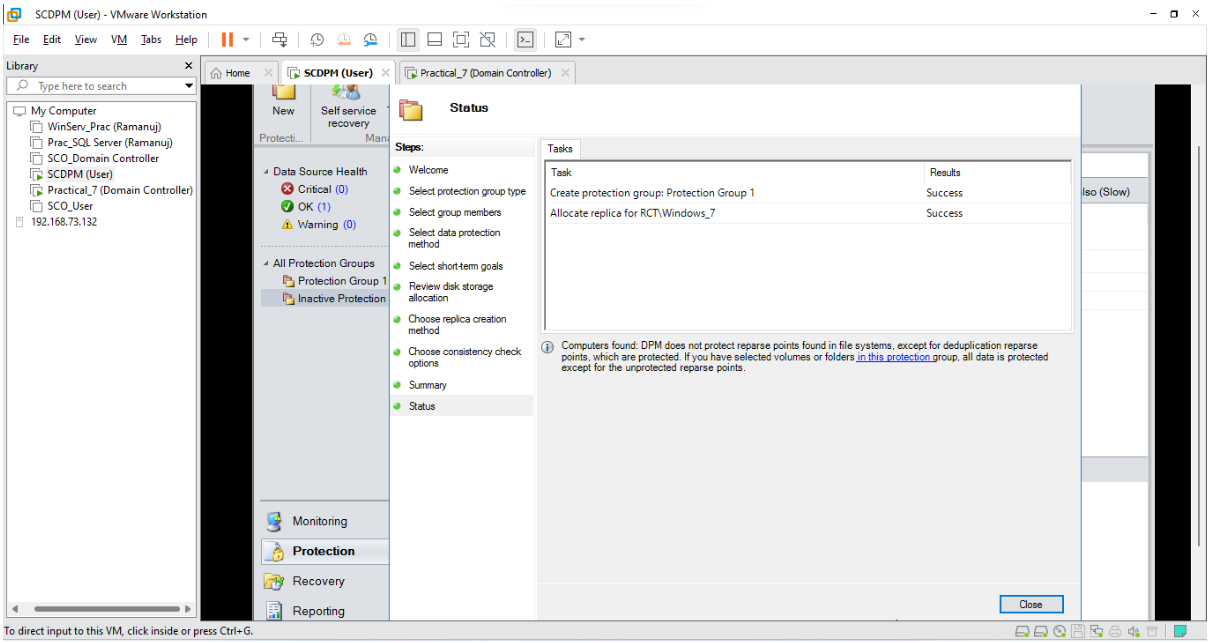
- Select **Run a consistency check if a replica becomes inconsistent** and Click **Next**



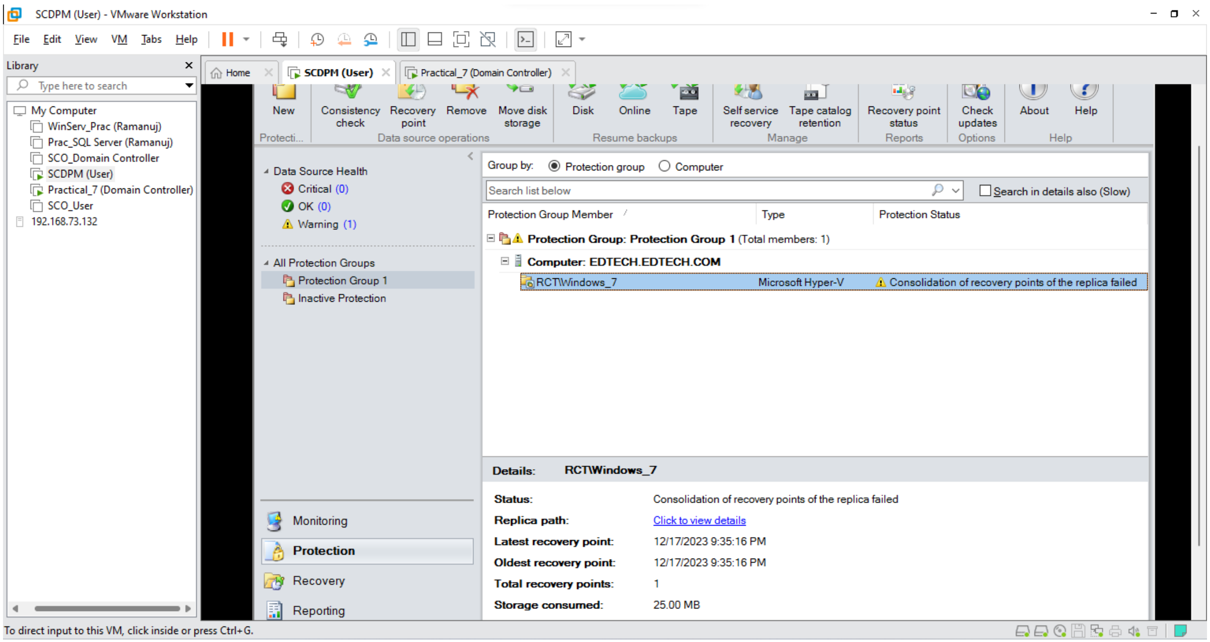
- View the summary of your group and Click **Create Group**



- After your Protection Group has been created, Click **Close**



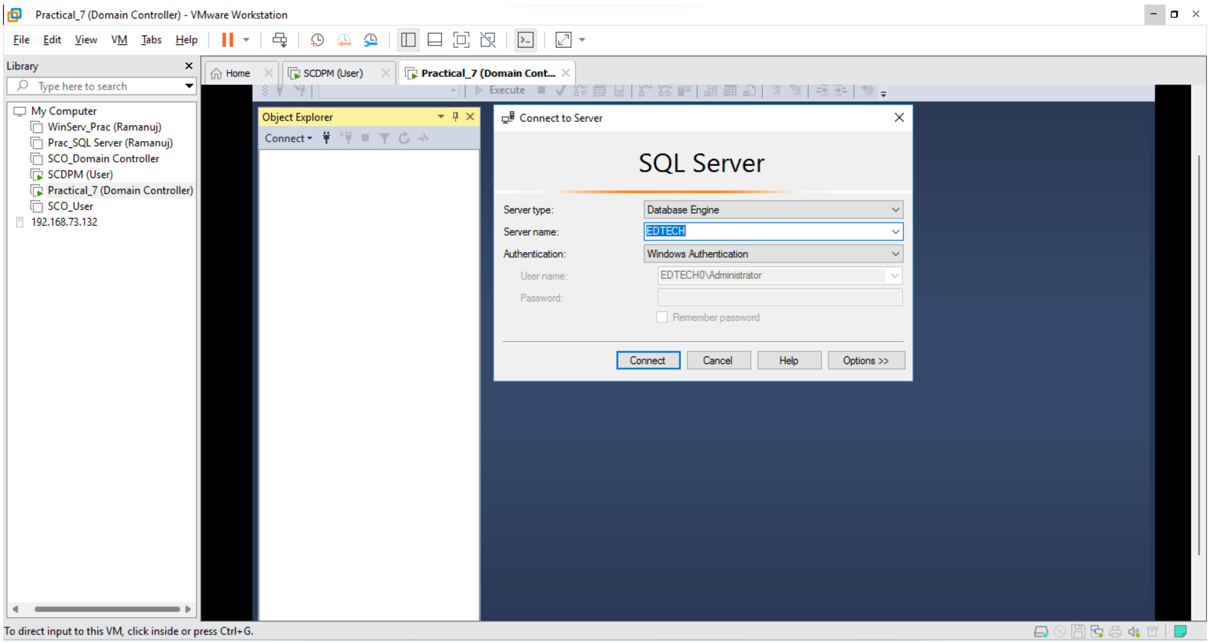
- You can now see that your Protection Group for Hyper-V has been created



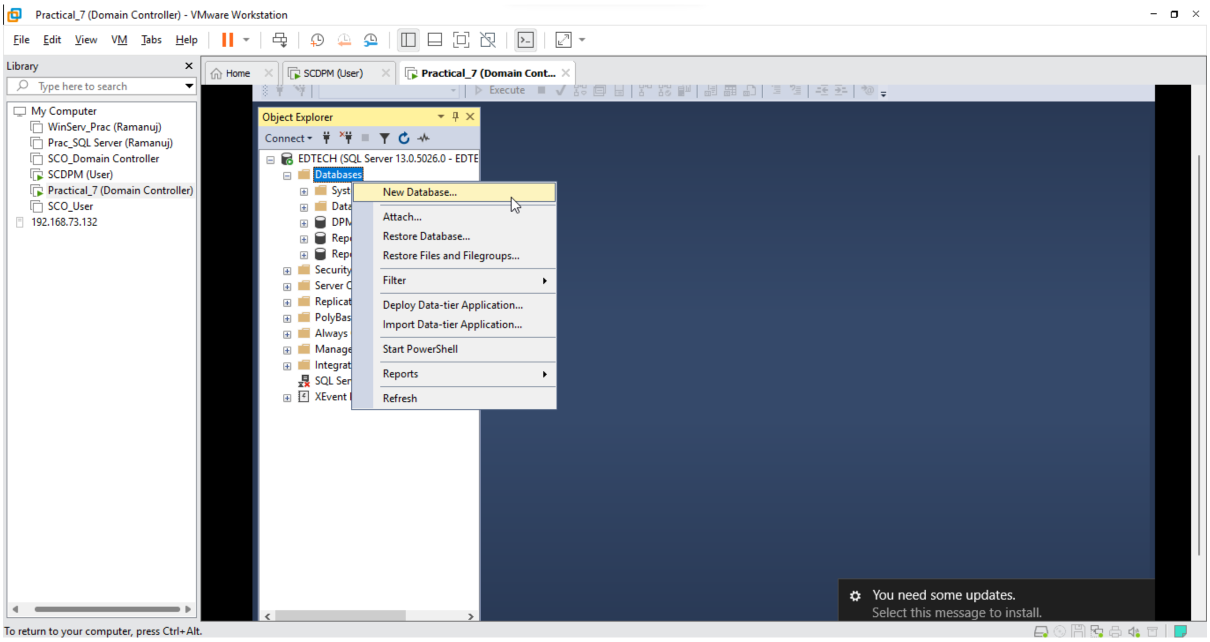
7-B: Back up SQL Server with DPM

Step 1: Create an SQL Database in Domain Controller

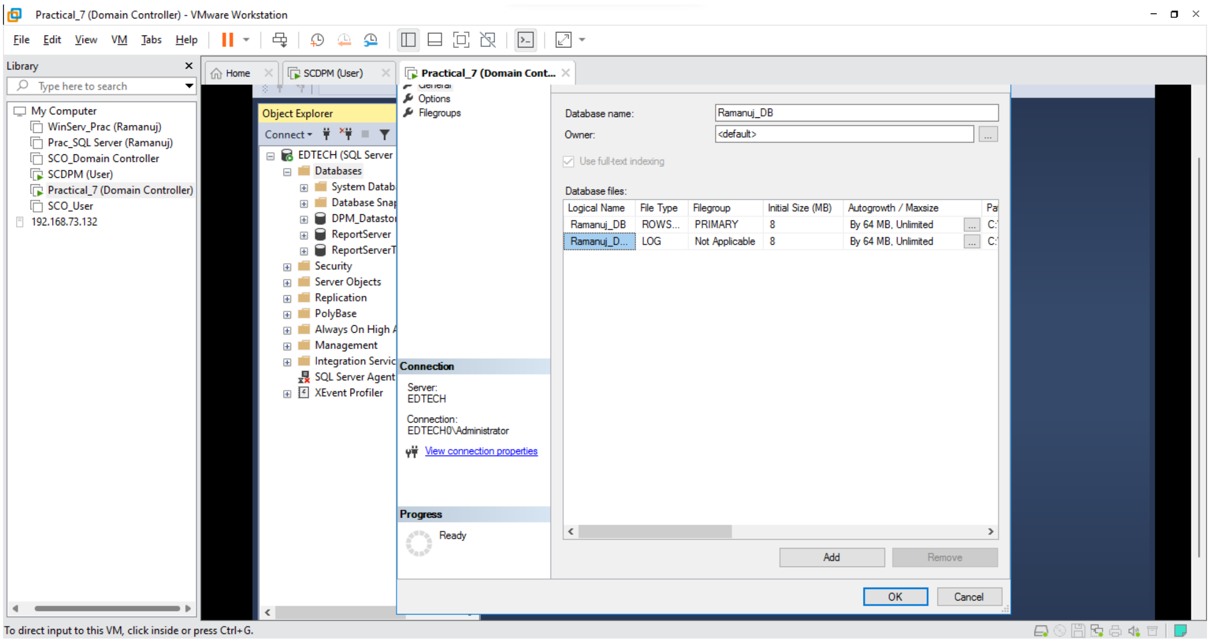
- Open **SQL Server Management Studio 18** and Click **Connect**



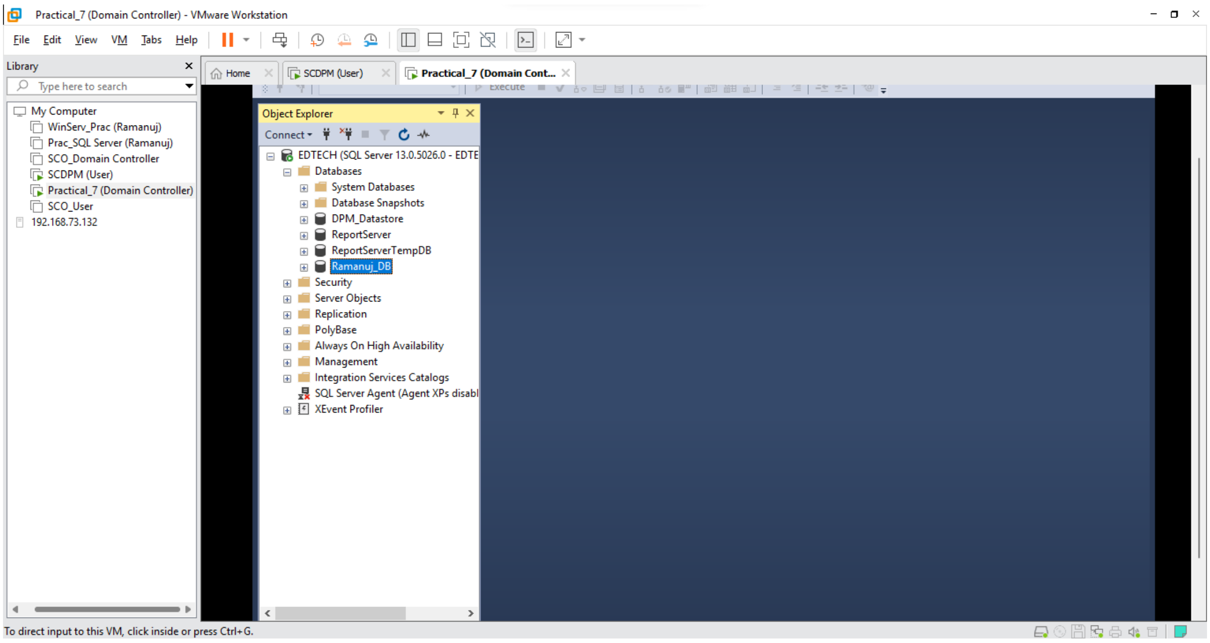
- Now Right Click on **Databases** and Select **New Database**



- Give the database a **name** (Here it is Ramanuj_DB) and Click **Ok**

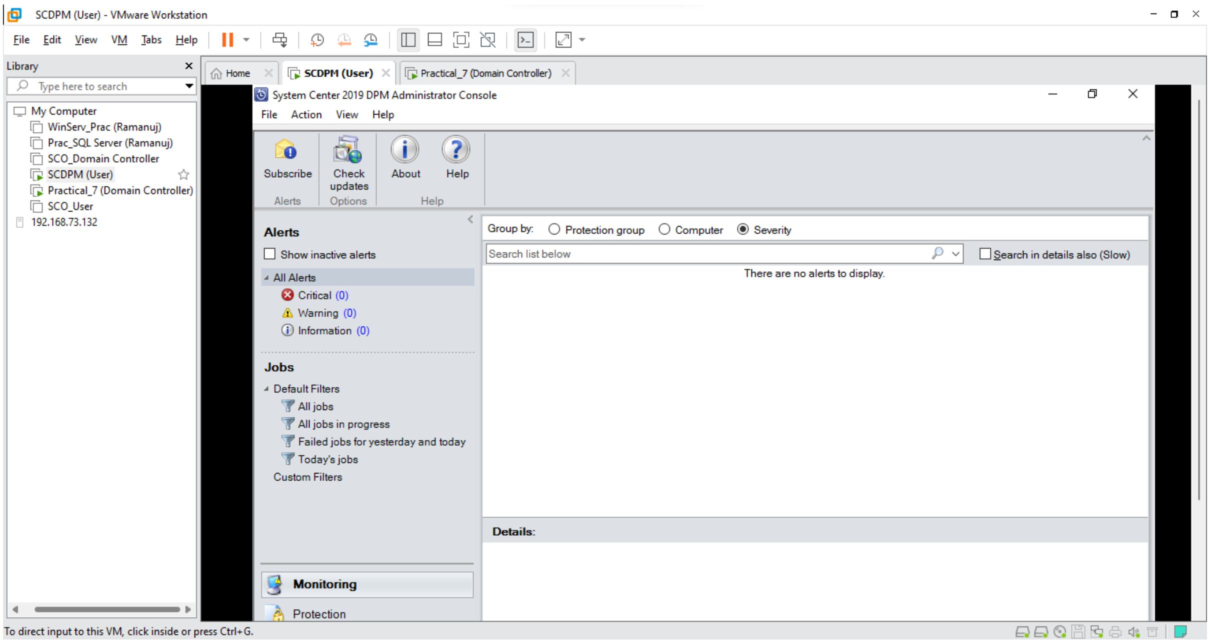


- Your database has been successfully created

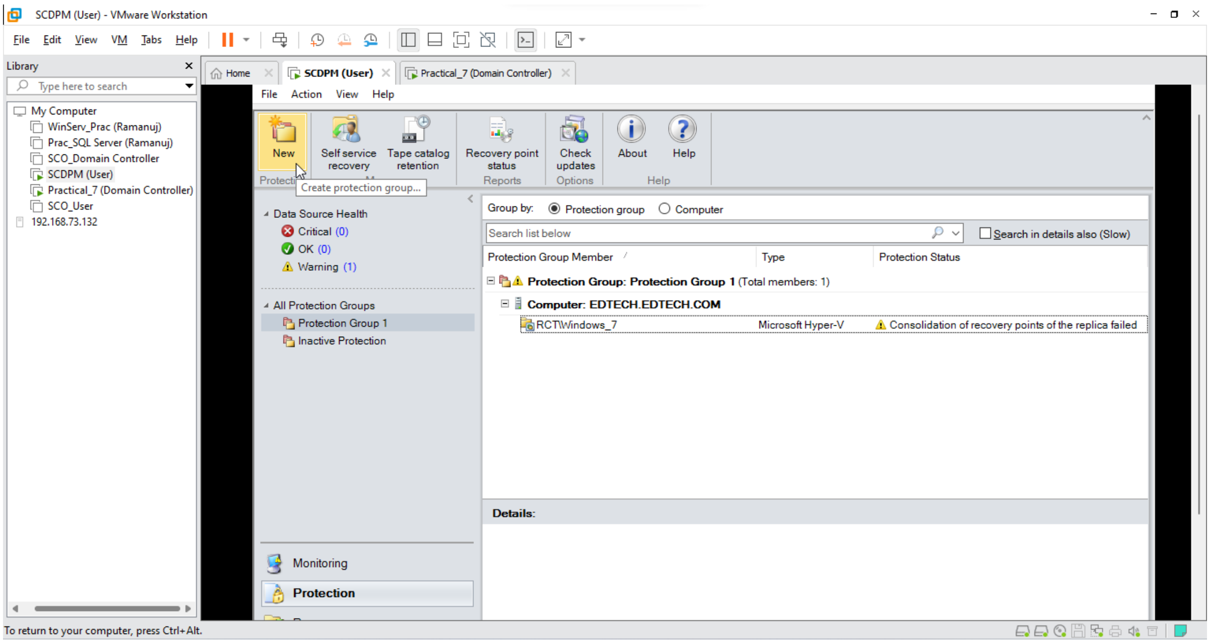


Step 2: Creating a Protection Group for the Newly Created SQL Database

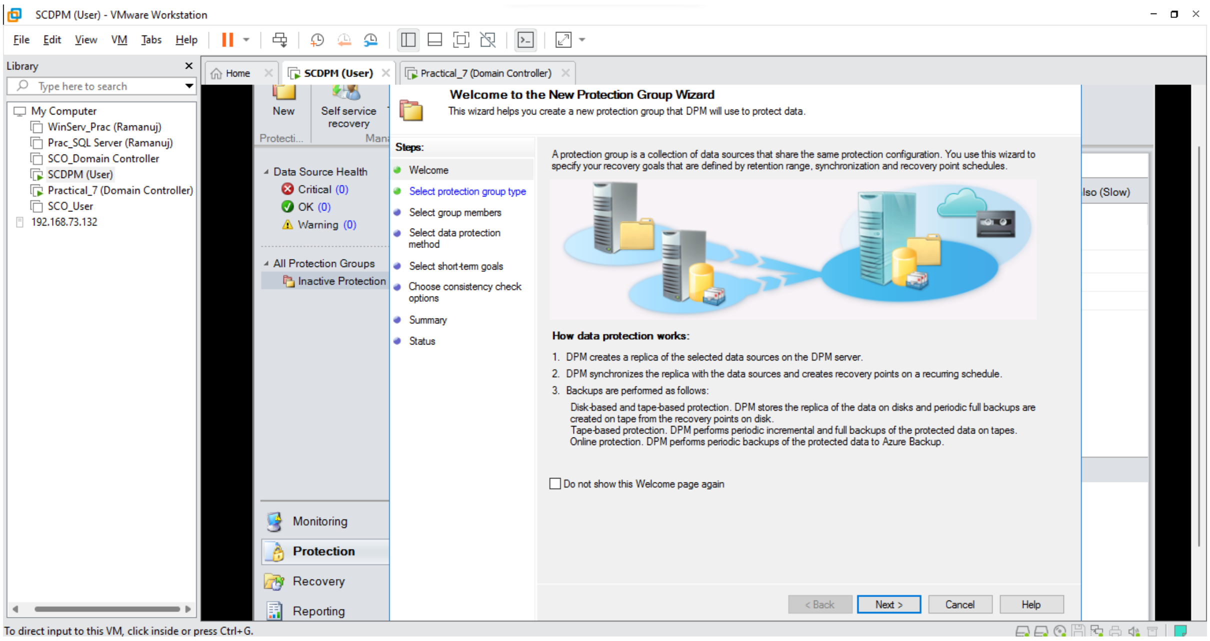
- Switch over to the **SCDPM (User)** VM and Open the **DPM manager**



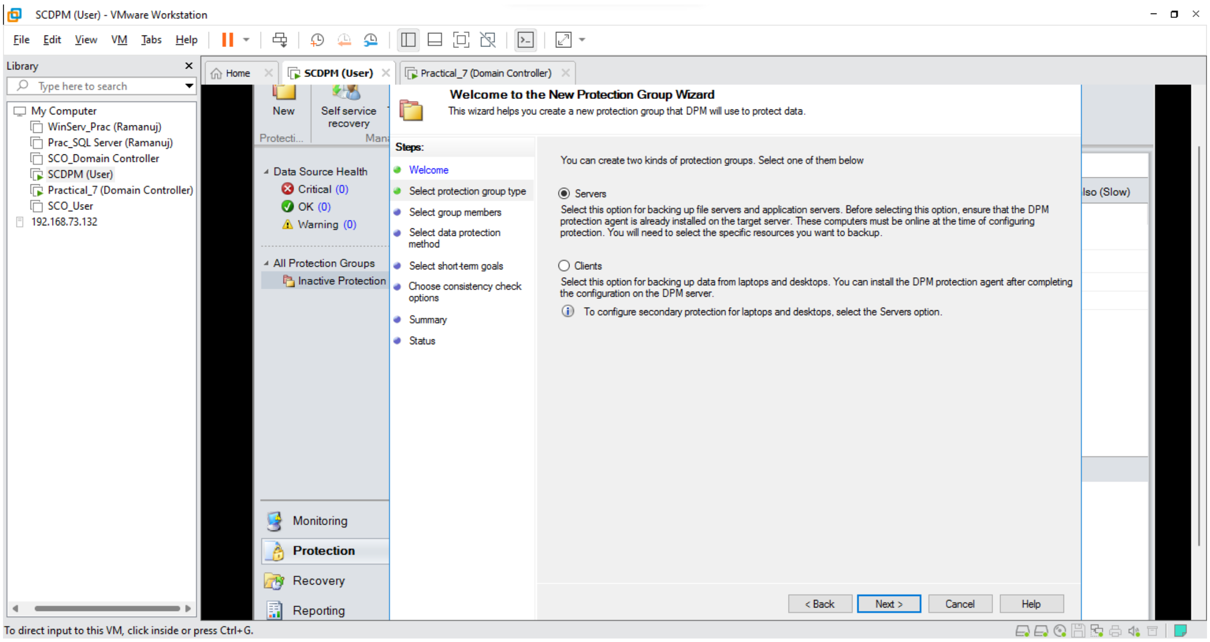
- Click on the **Protection** Tab and Select **New**



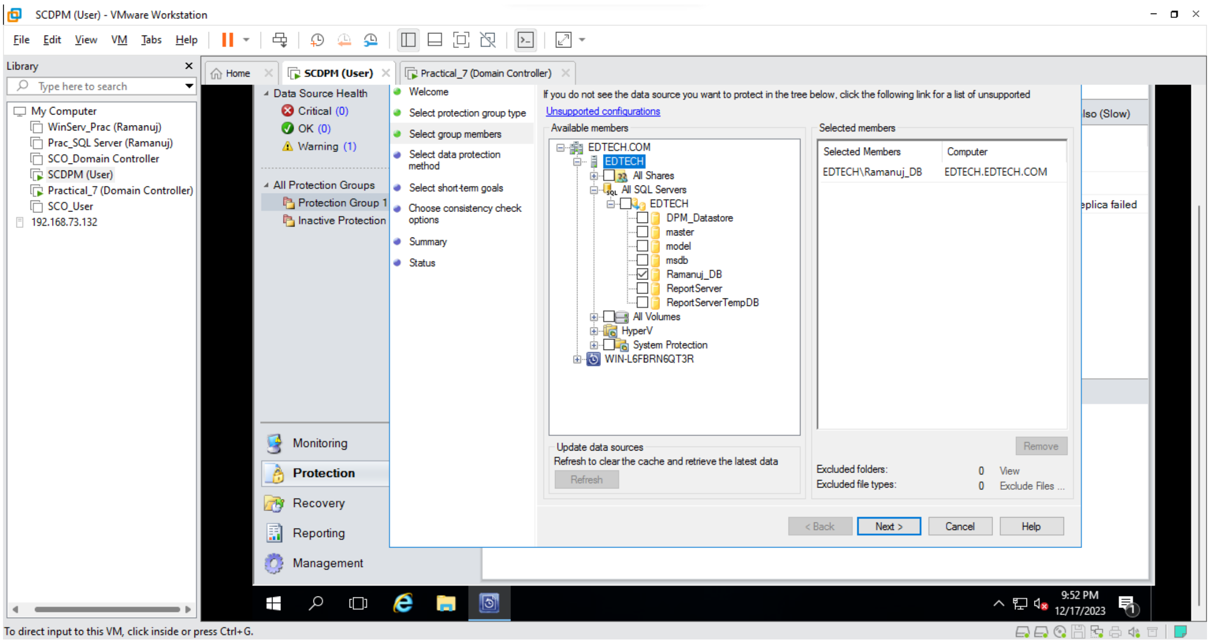
- Click **Next**



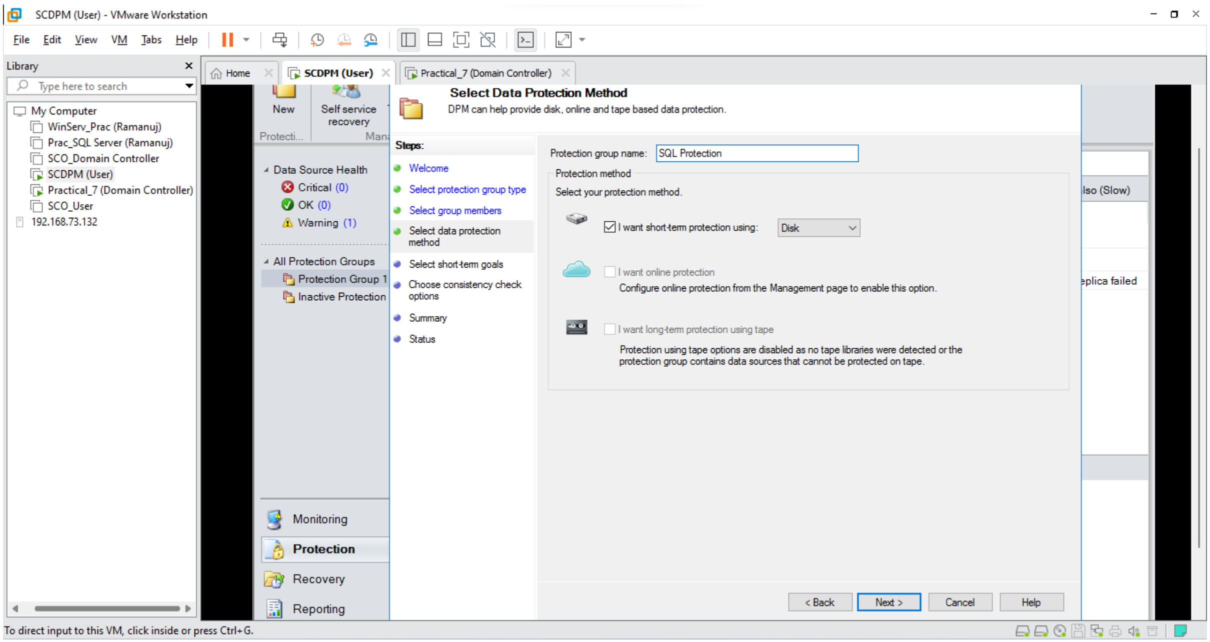
- Select **Servers** and Click **Next**



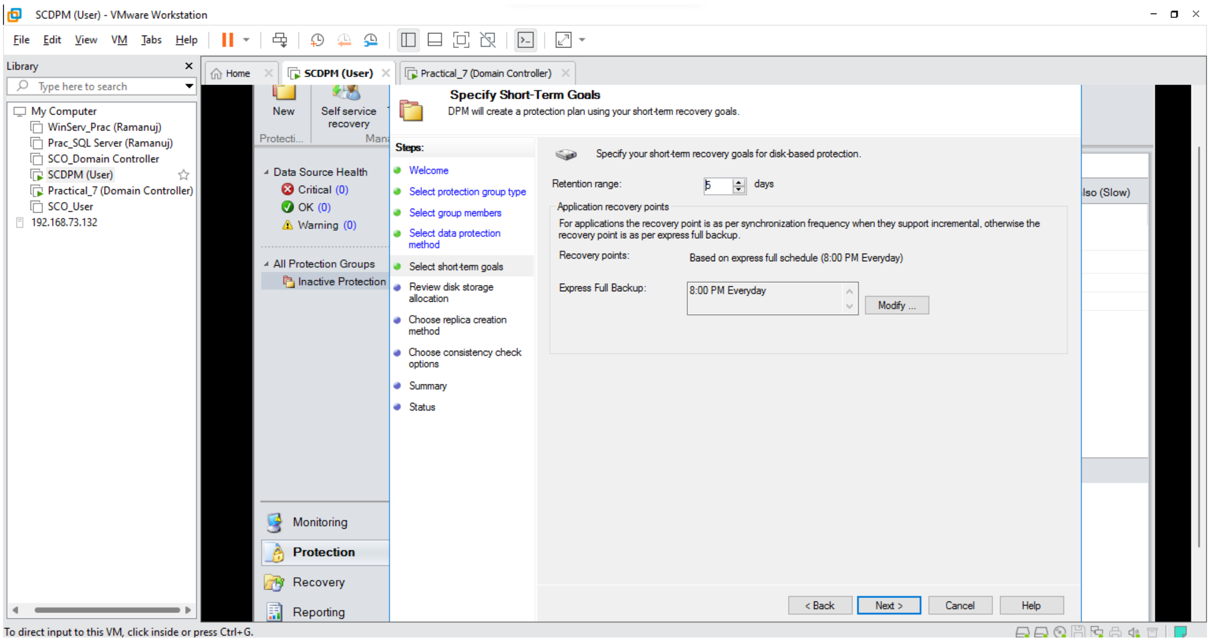
- Select **EDTECH** and Click on **Refresh,** after refresh Open **All SQL Server** and Click on **EDTECH** and Click on your newly created **SQL Database** (Here it is Ramanuj_DB) and Click **Next**



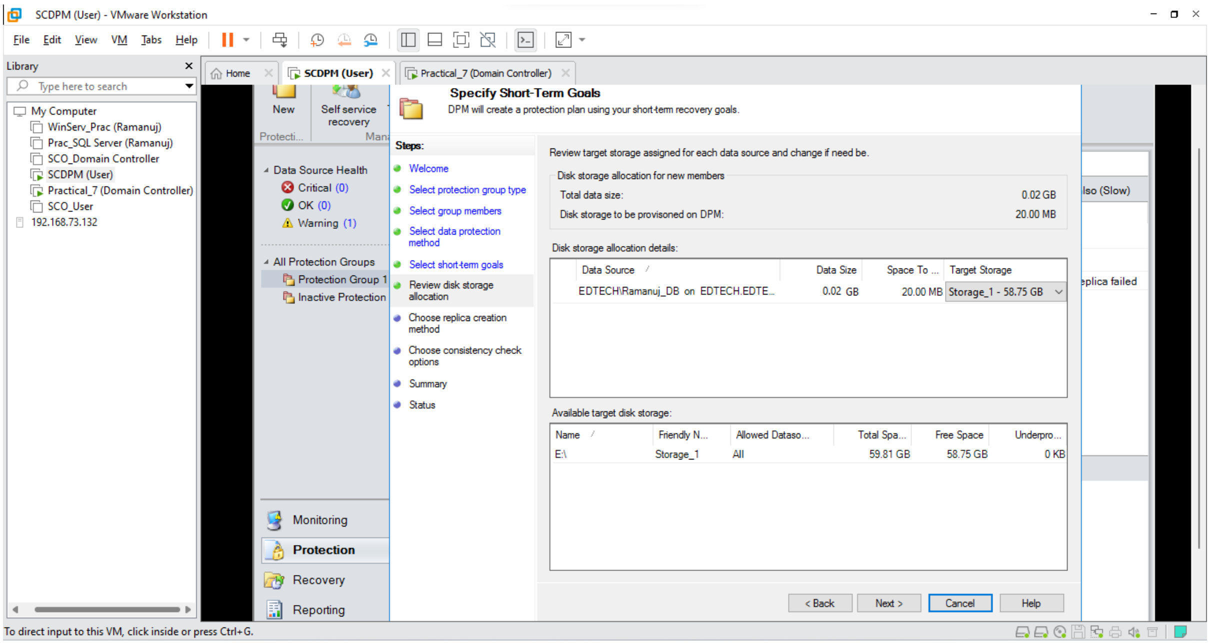
- Name your **Protection Group** (Here it is SQL Protection) and Select **I want short term protection** and Select **Disk** and Click **Next**



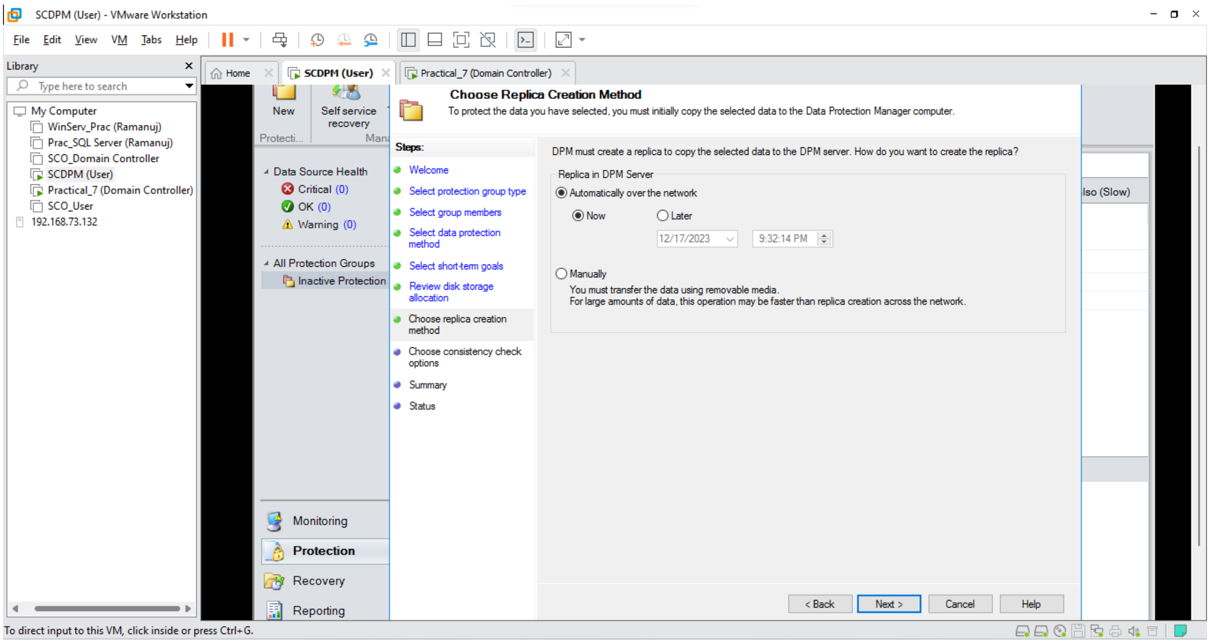
- Keep default values and Click **Next**



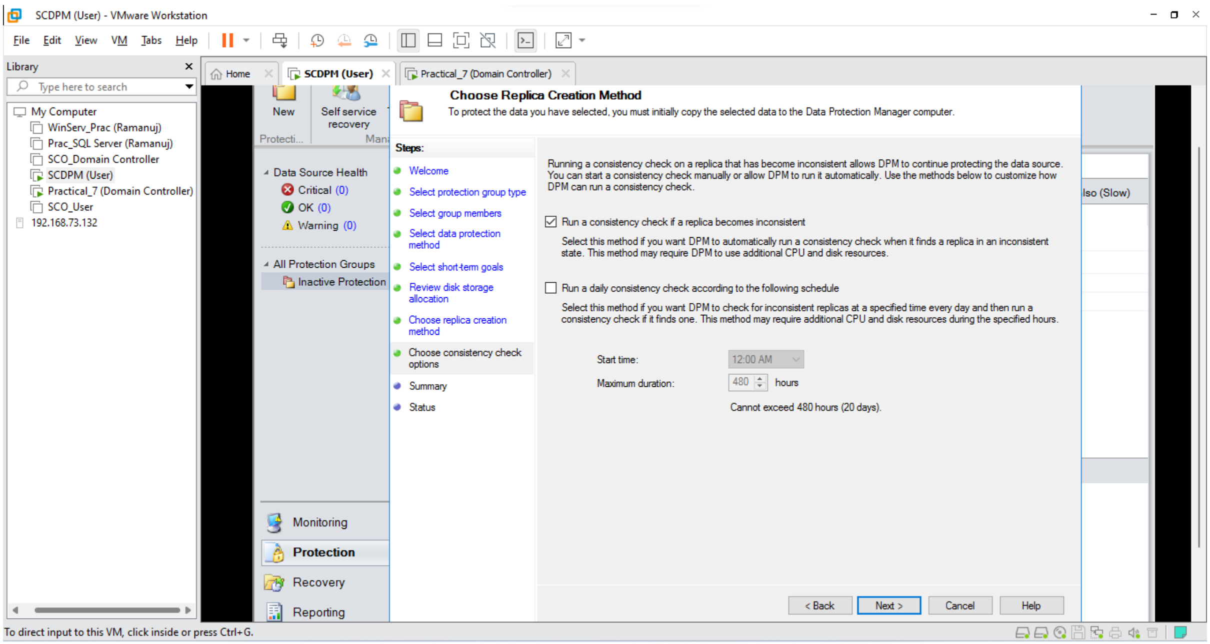
- Click **Next**



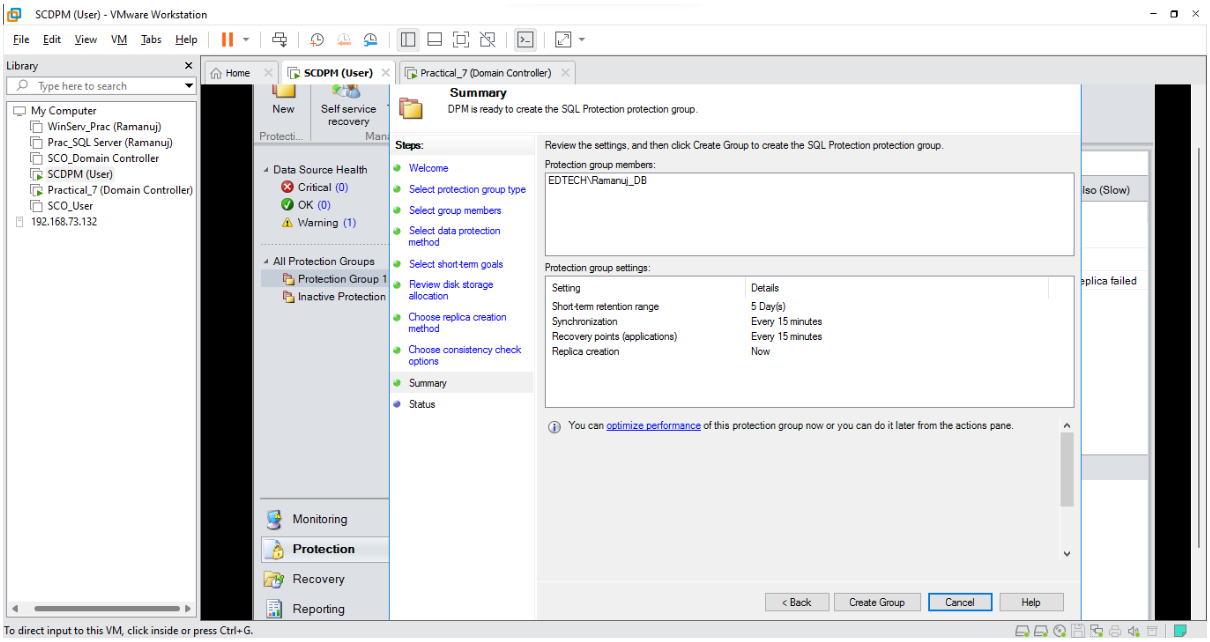
- Click **Next**



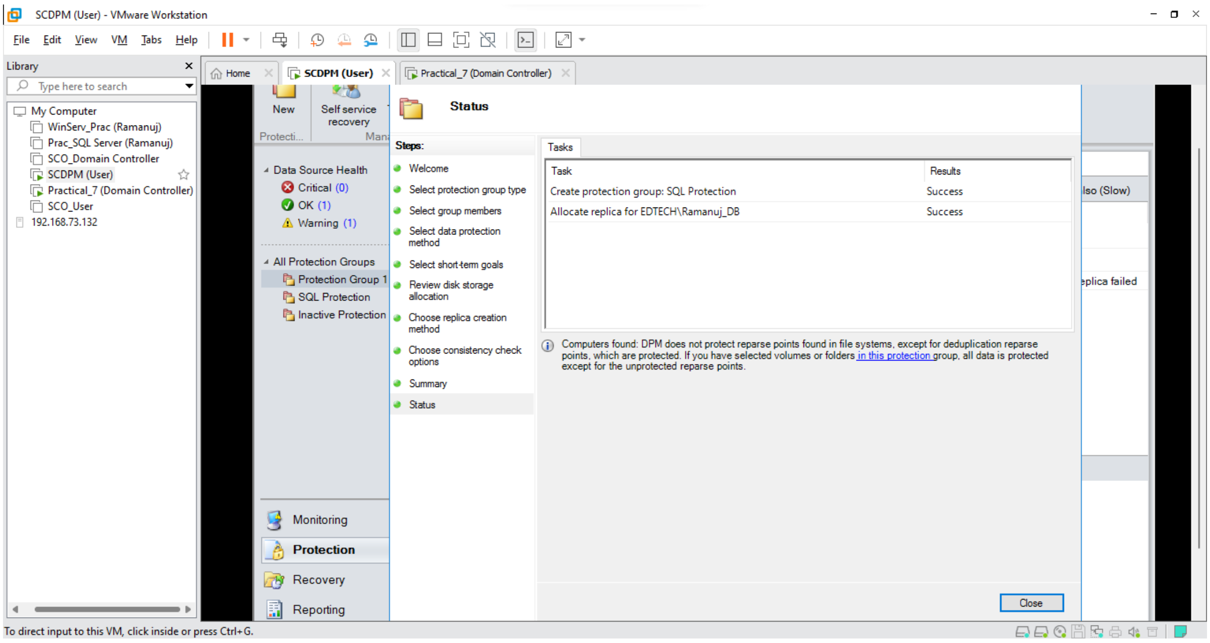
- Select **Run a consistency check if a replica becomes inconsistent** and Click **Next**



- View the summary of your group and Click **Create Group**



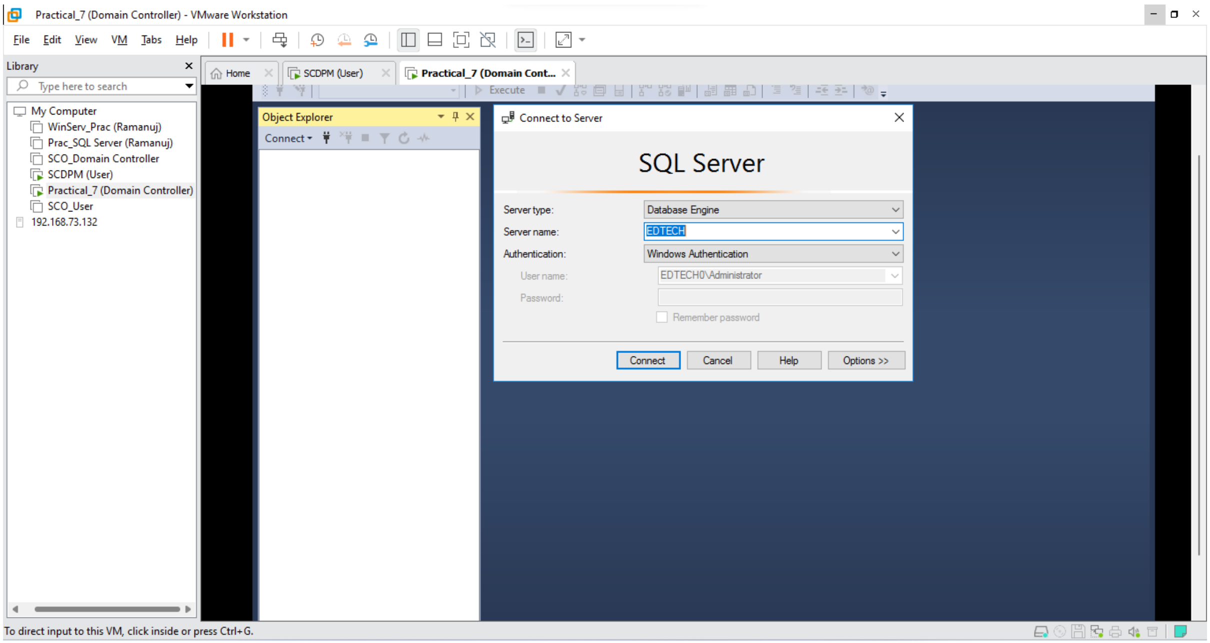
- After it is successfully created Click **Close**



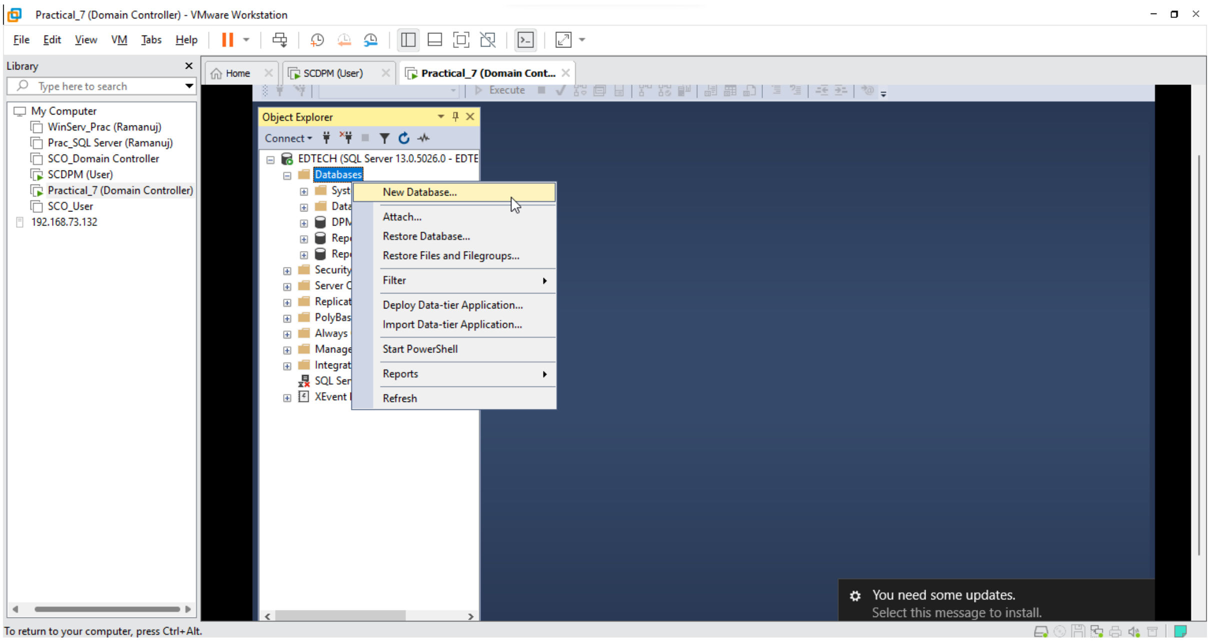
7-C: Back up file data (Database)

Step 1: Create an SQL Database in Domain Controller

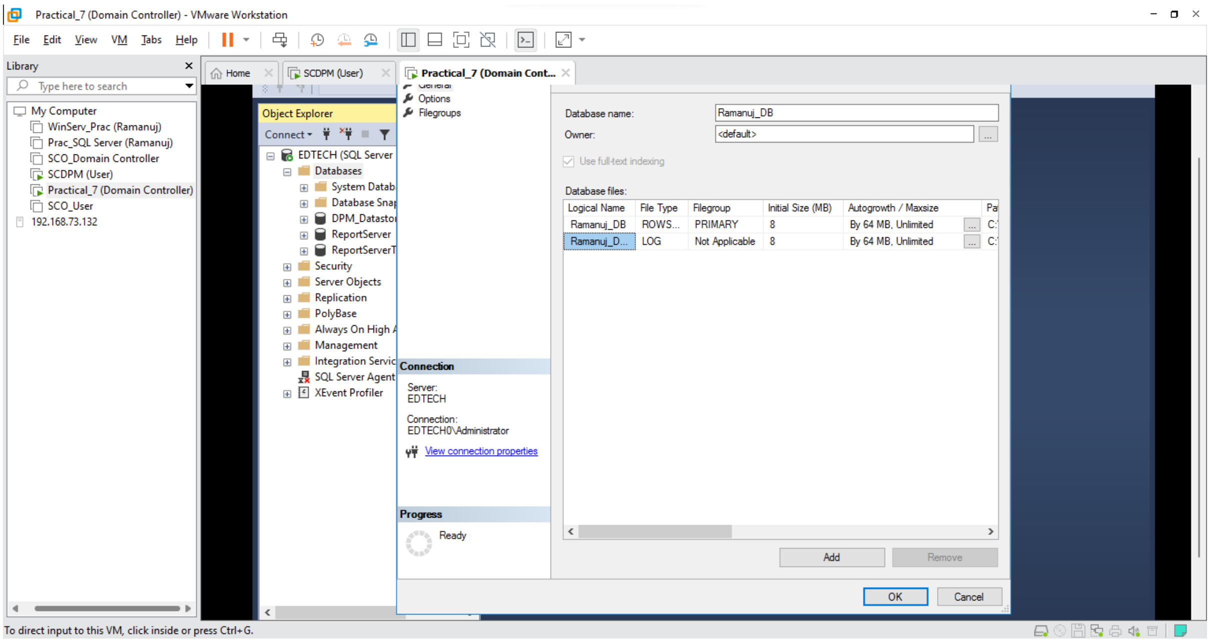
- Open **SQL Server Management Studio 18** and Click **Connect**



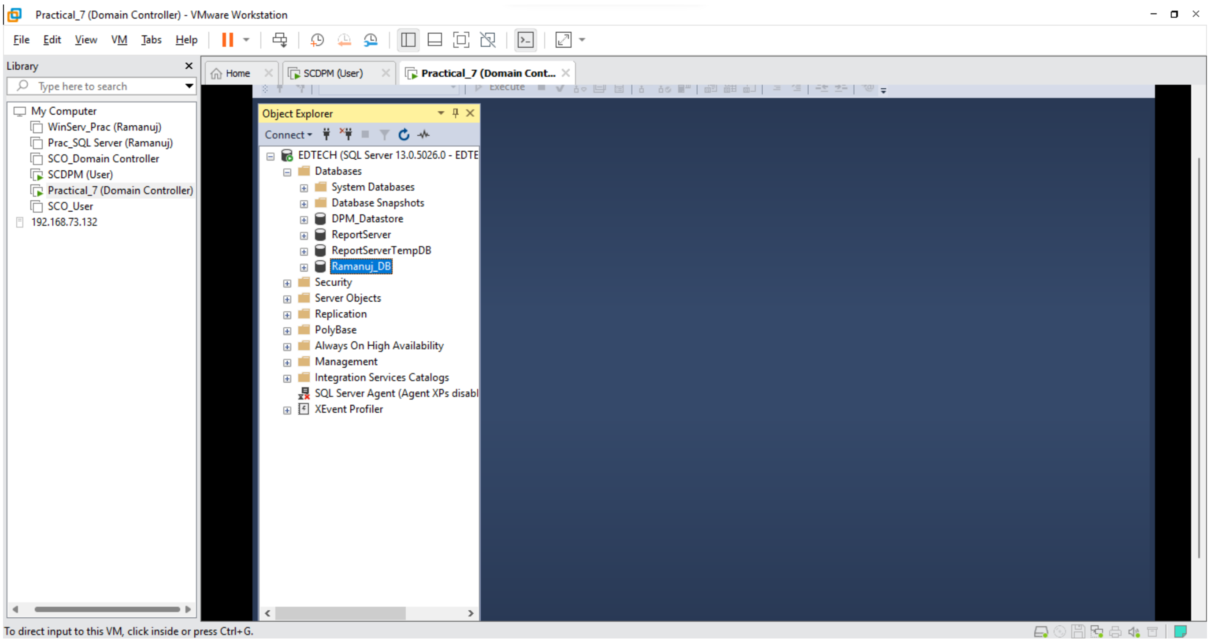
- Now Right Click on **Databases** and Select **New Database**



- Give the database a **name** (Here it is Ramanuj_DB) and Click **Ok**

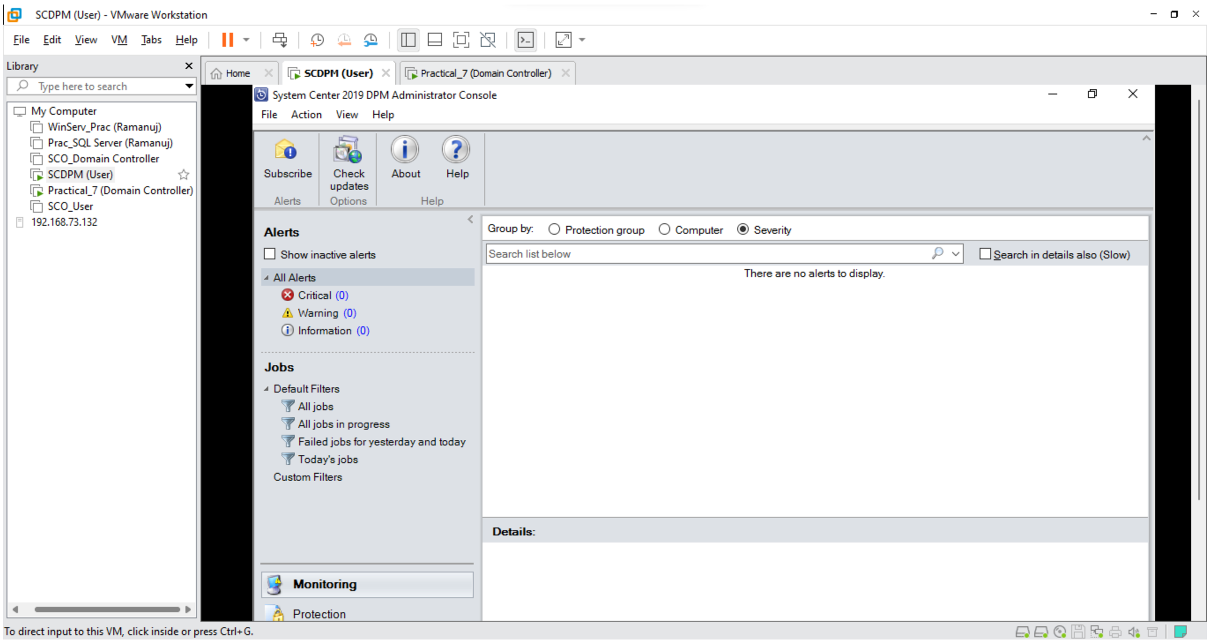


- Your database has been successfully created

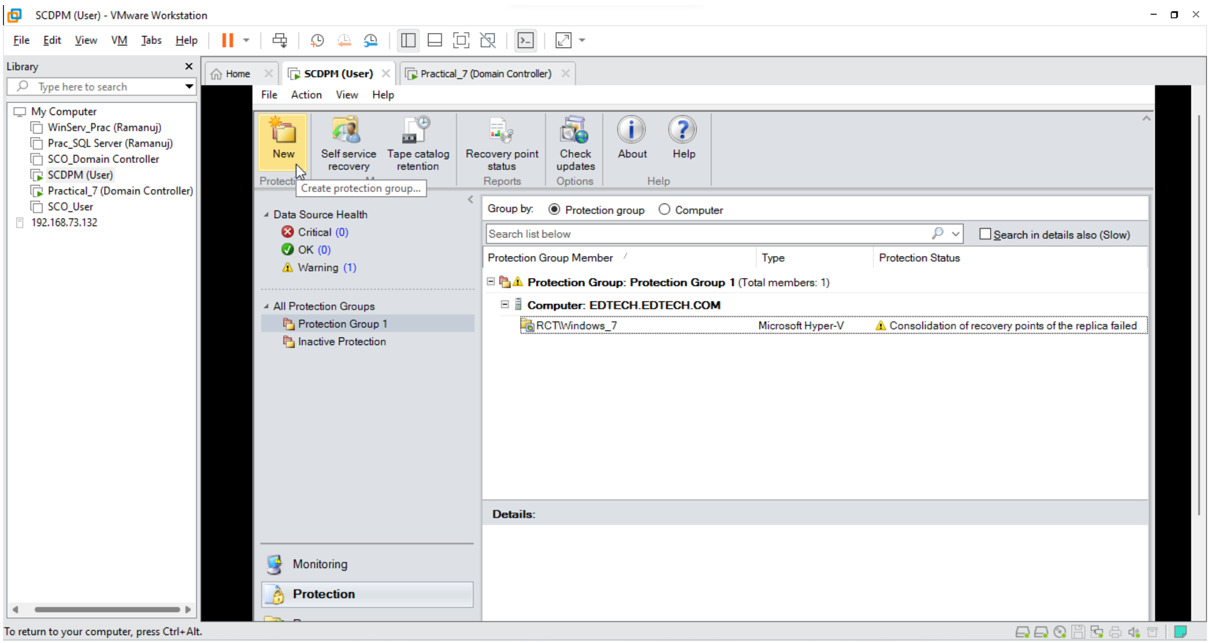


Step 2: Creating a Protection Group for the Newly Created SQL Database

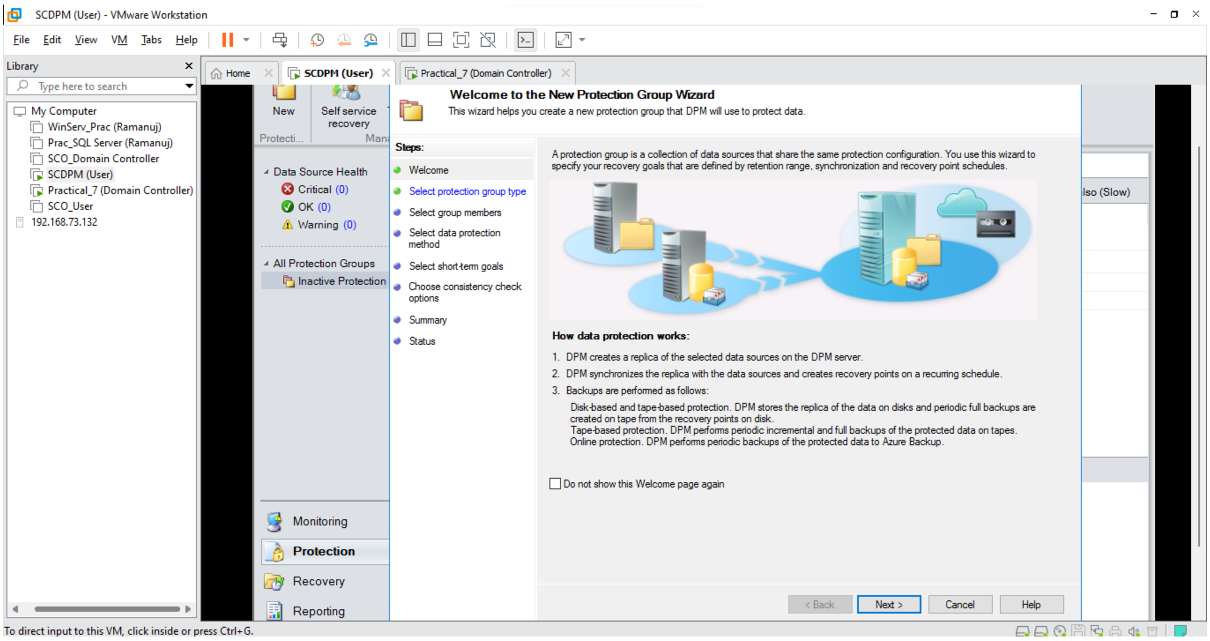
- Switch over to the **SCDPM (User)** VM and Open the **DPM manager**



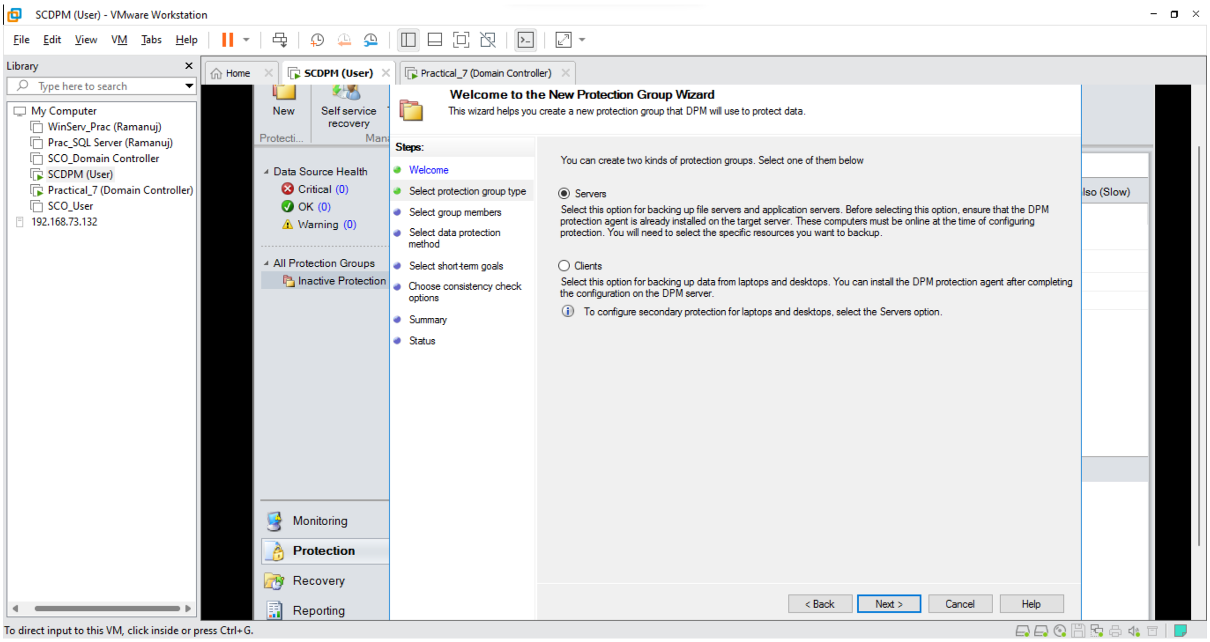
- Click on the **Protection** Tab and Select **New**



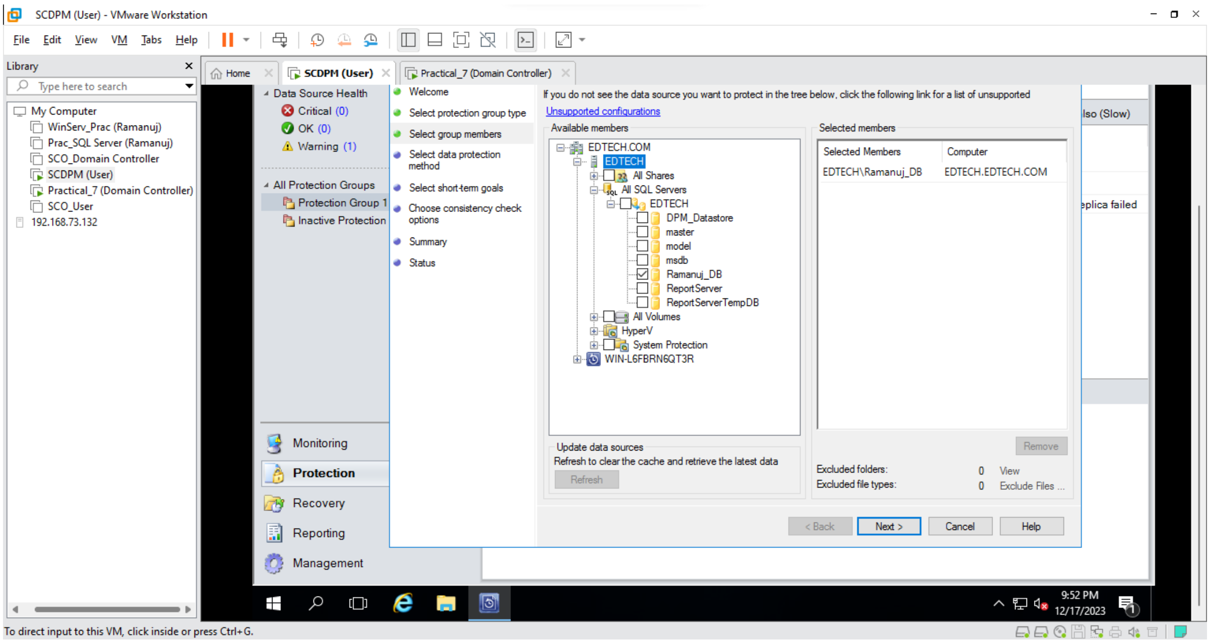
- Click **Next**



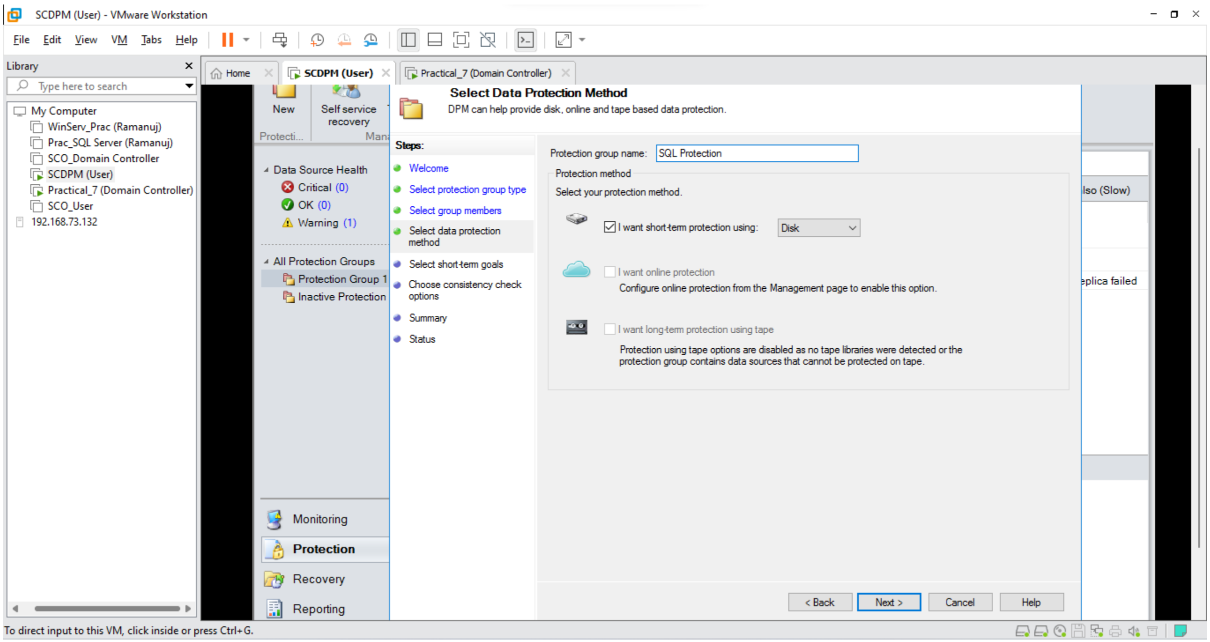
- Select **Servers** and Click **Next**



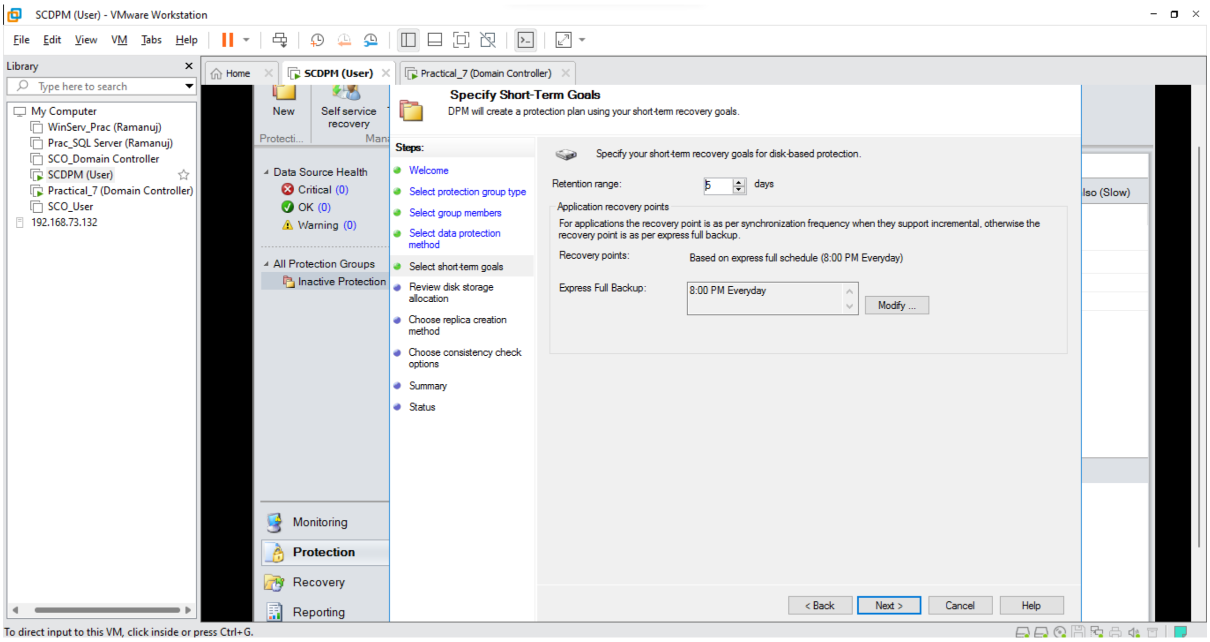
- Select **EDTECH** and Click on **Refresh,** after refresh Open **All SQL Server** and Click on **EDTECH** and Click on your newly created **SQL Database** (Here it is Ramanuj_DB) and Click **Next**



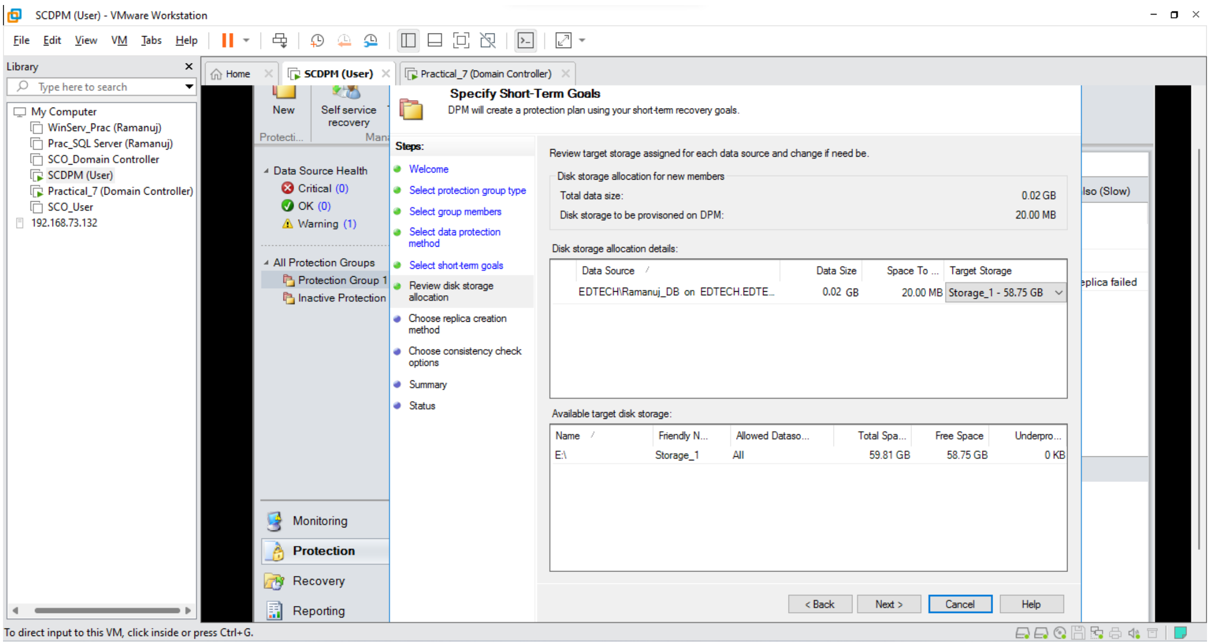
- Name your **Protection Group** (Here it is SQL Protection) and Select **I want short term protection** and Select **Disk** and Click **Next**



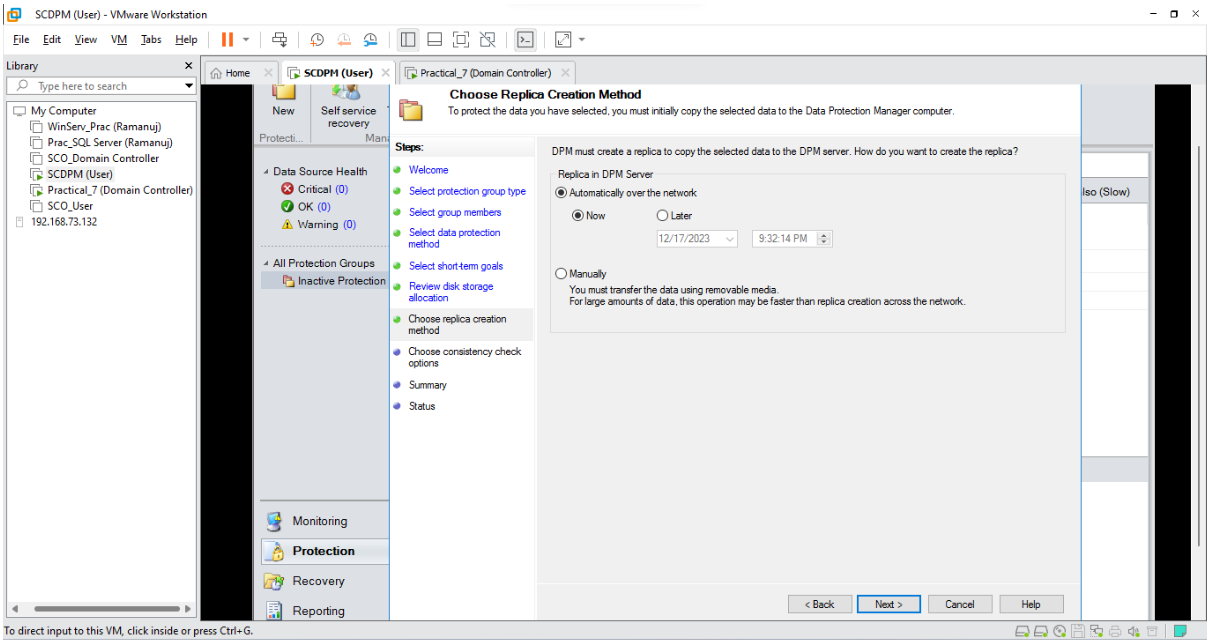
- Keep default values and Click **Next**



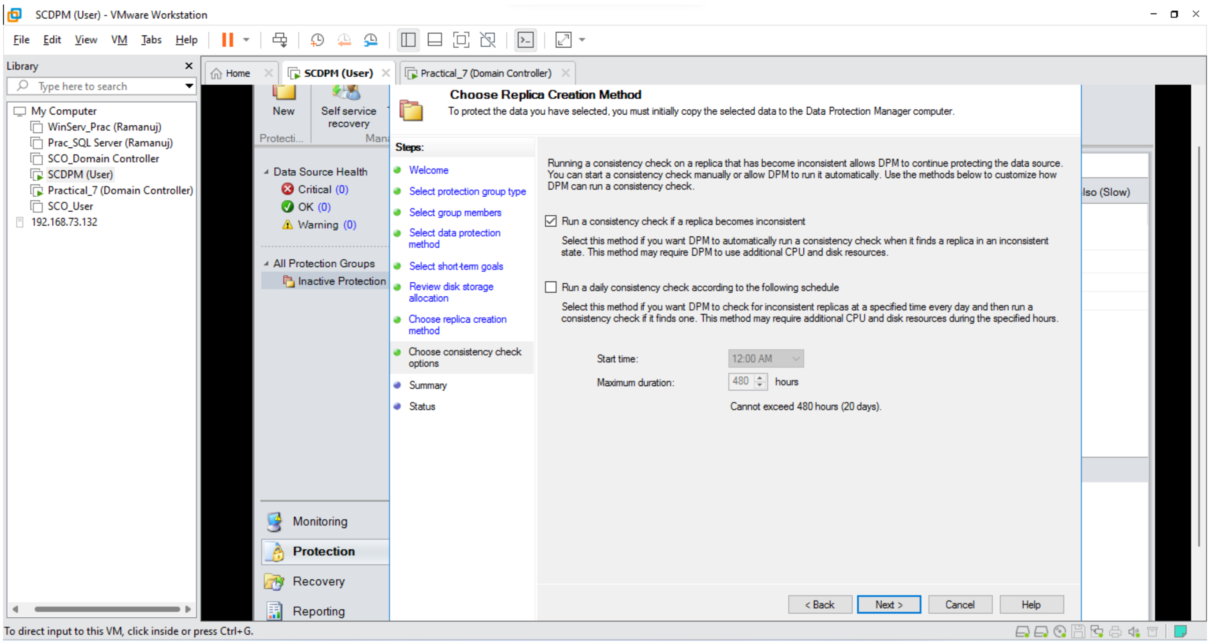
- Click **Next**



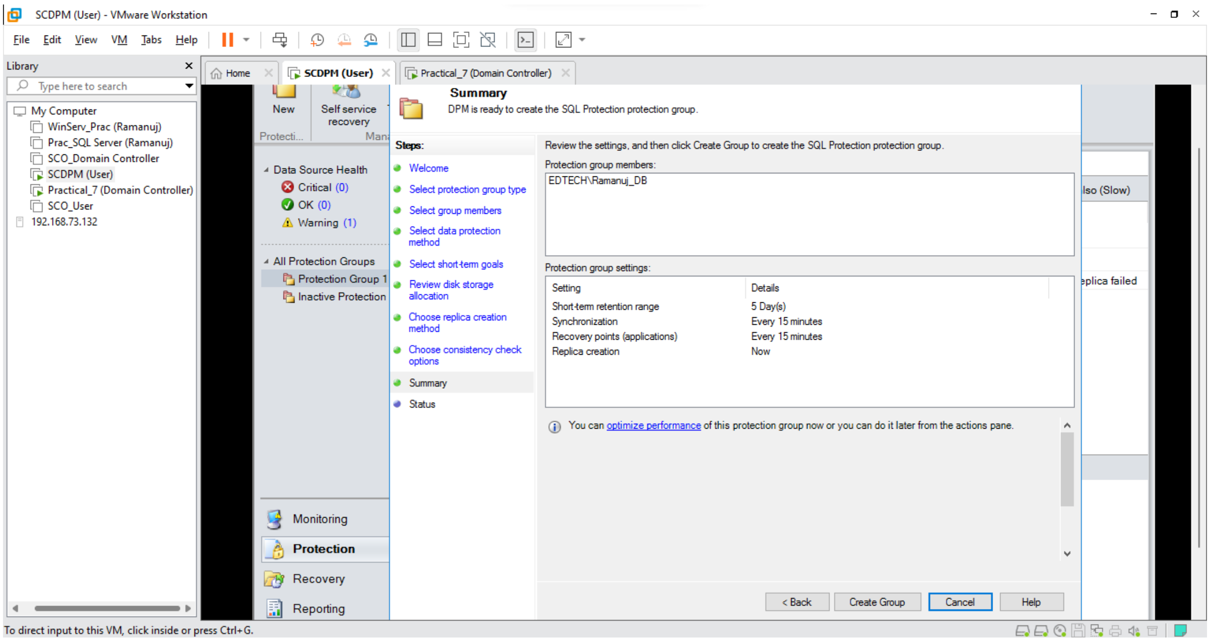
- Click **Next**



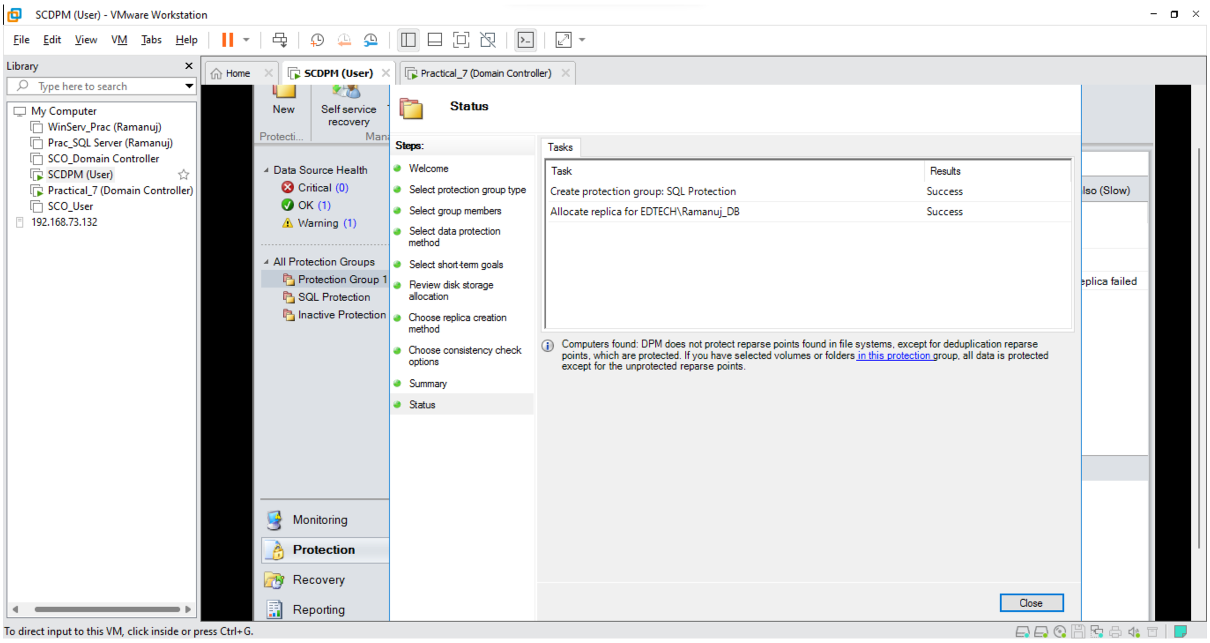
- Select **Run a consistency check if a replica becomes inconsistent** and Click **Next**



- View the summary of your group and Click **Create Group**



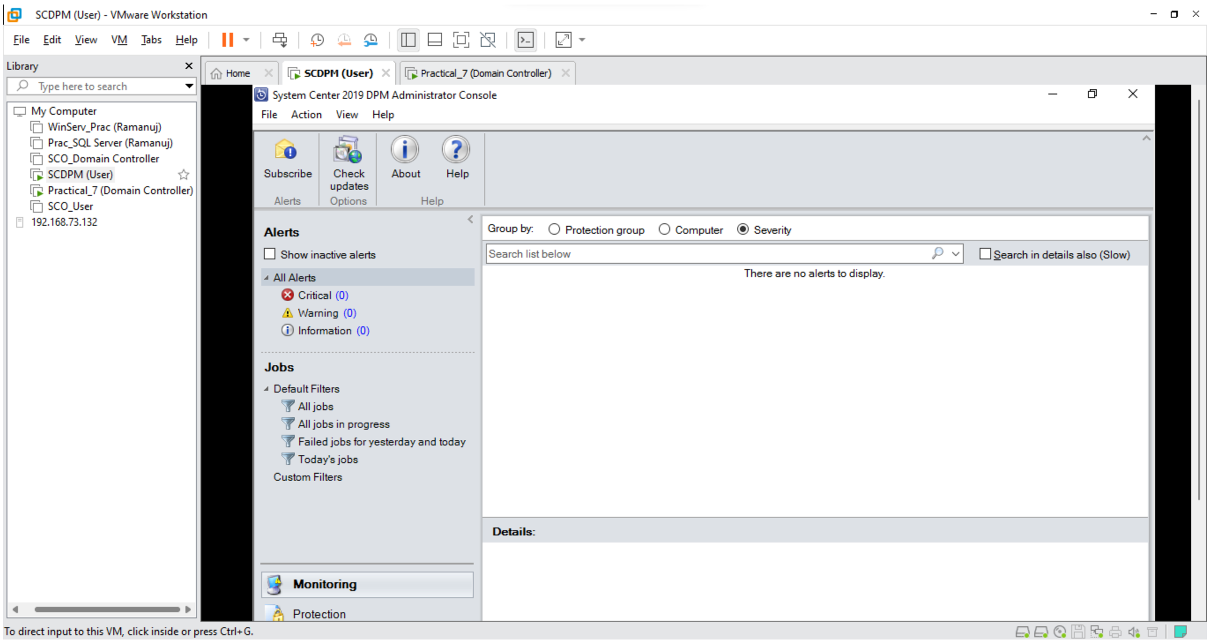
- After it is successfully created Click **Close**



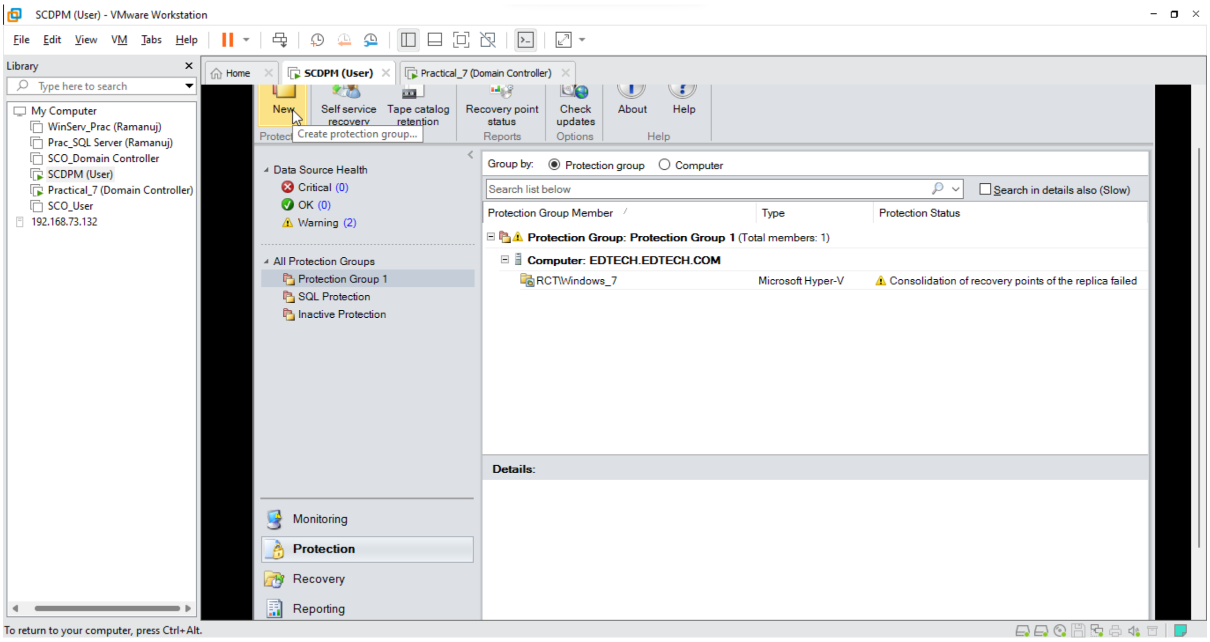
7-D: Backup System State

Step 1: Creating Protection Group for System State

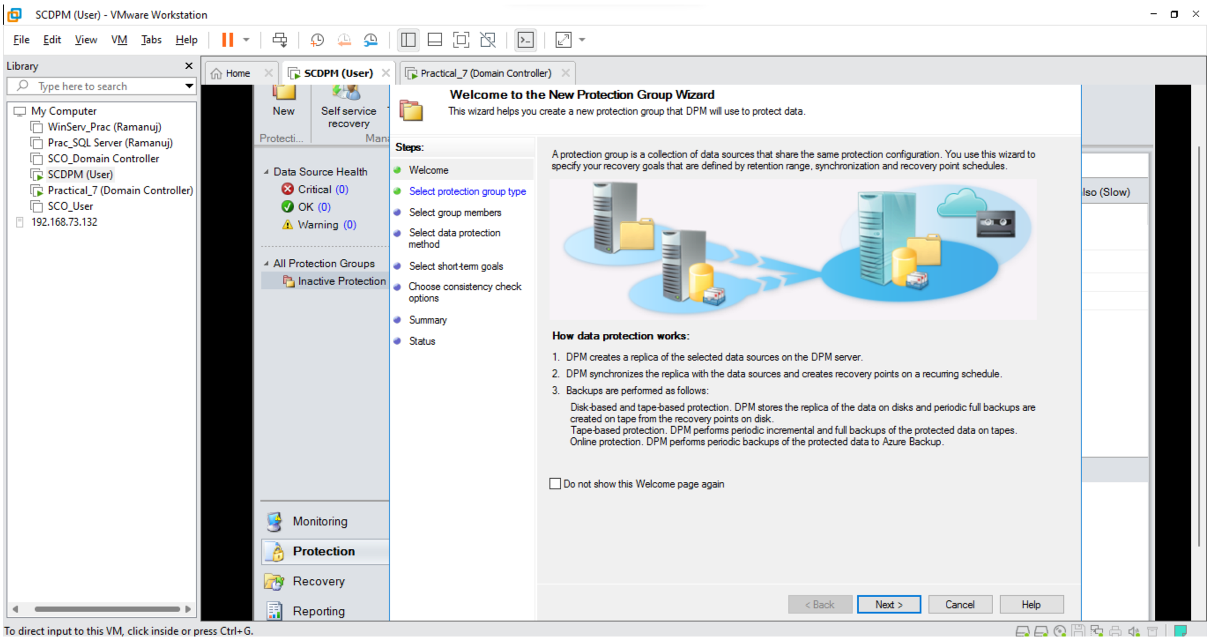
- In the **SCDPM (User)** VM and Open the **DPM manager**



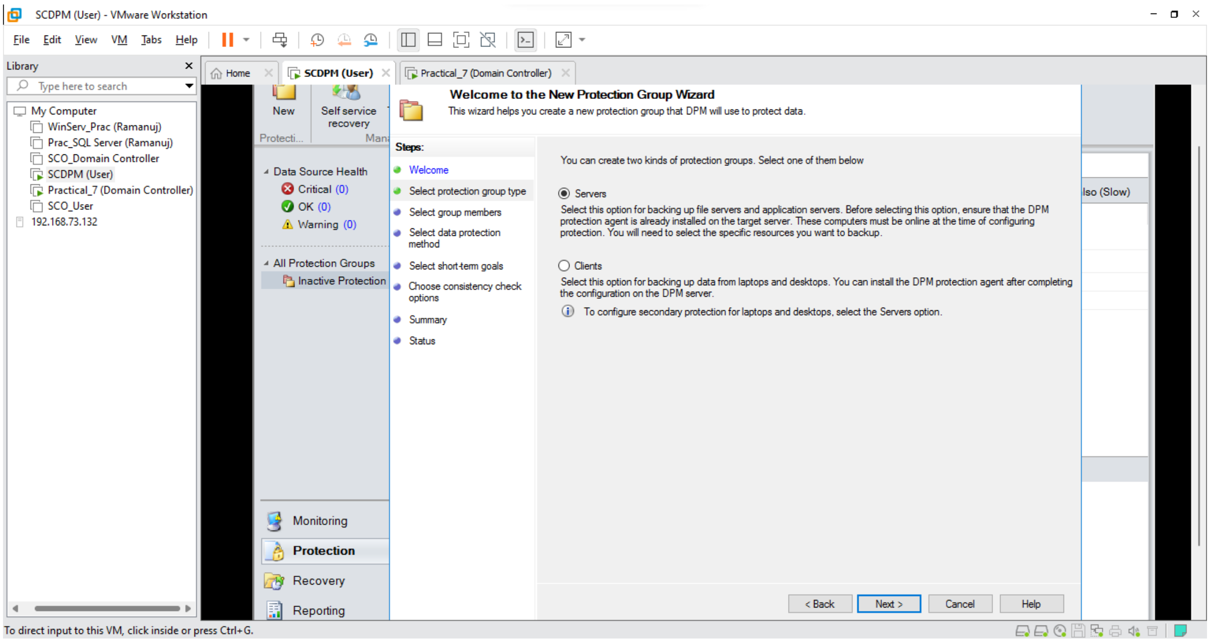
- Click on the **Protection** Tab and Select **New**



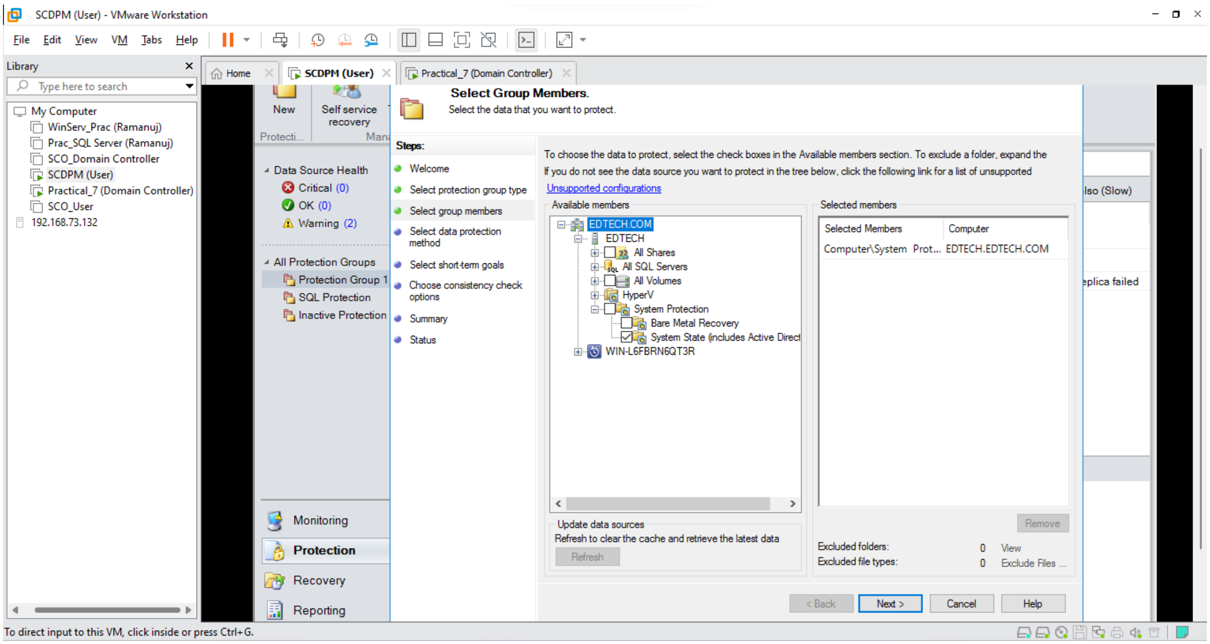
- Click **Next**



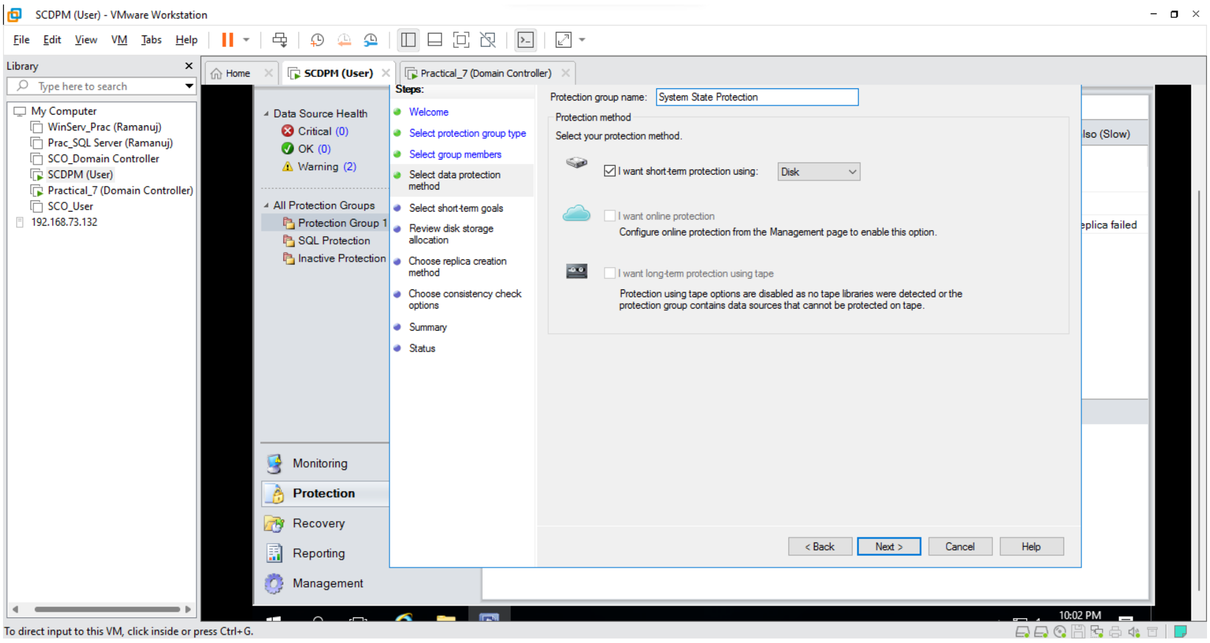
- Select **Servers** and Click **Next**



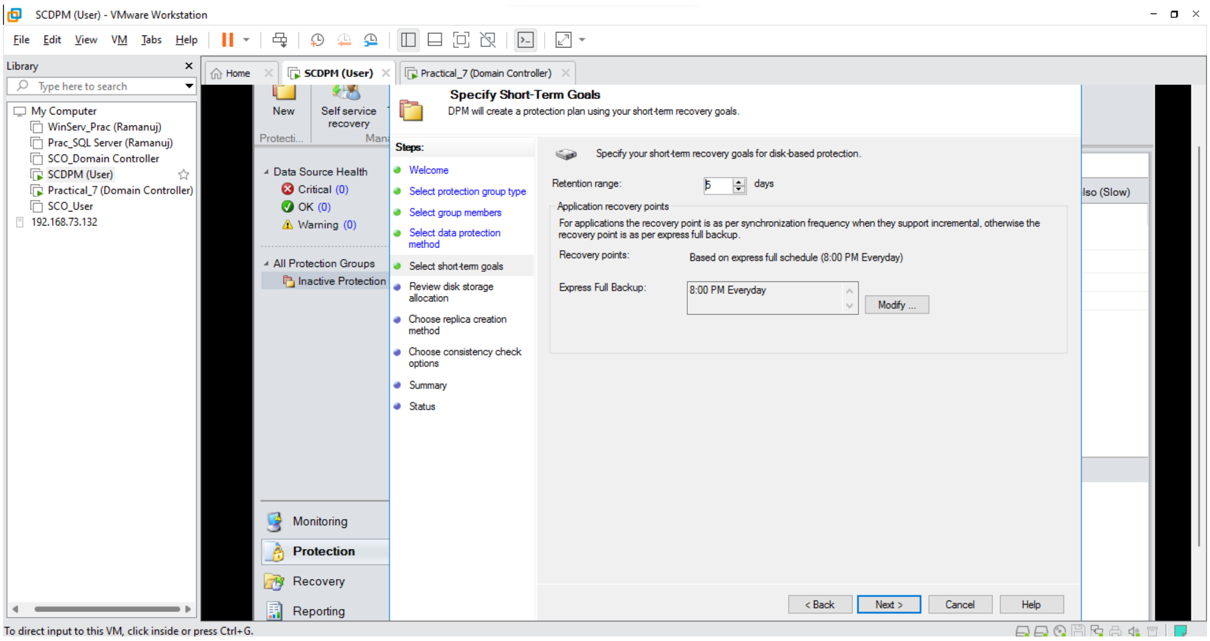
- Select **EDTECH** and Click on **Refresh,** after refresh Open **System Protection** and Click on **System State (including Active Directories)** and Click **Next**



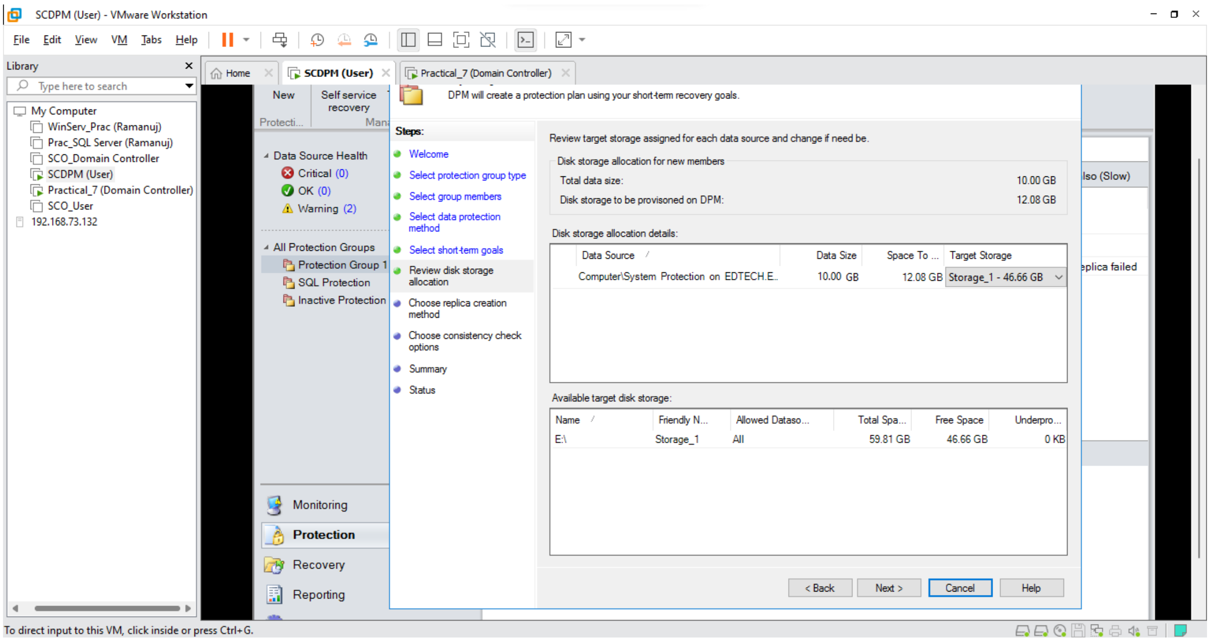
- Name your **Protection Group** (Here it is System State Protection) and Select **I want short term protection** and Select **Disk** and Click **Next**



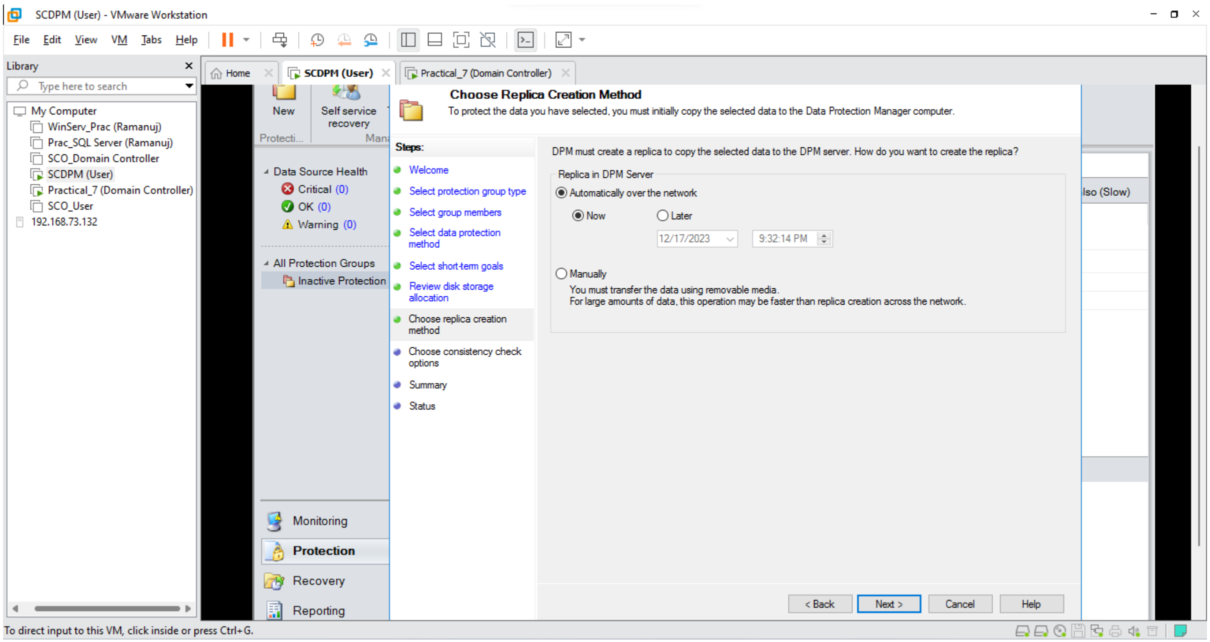
- Keep default values and Click **Next**



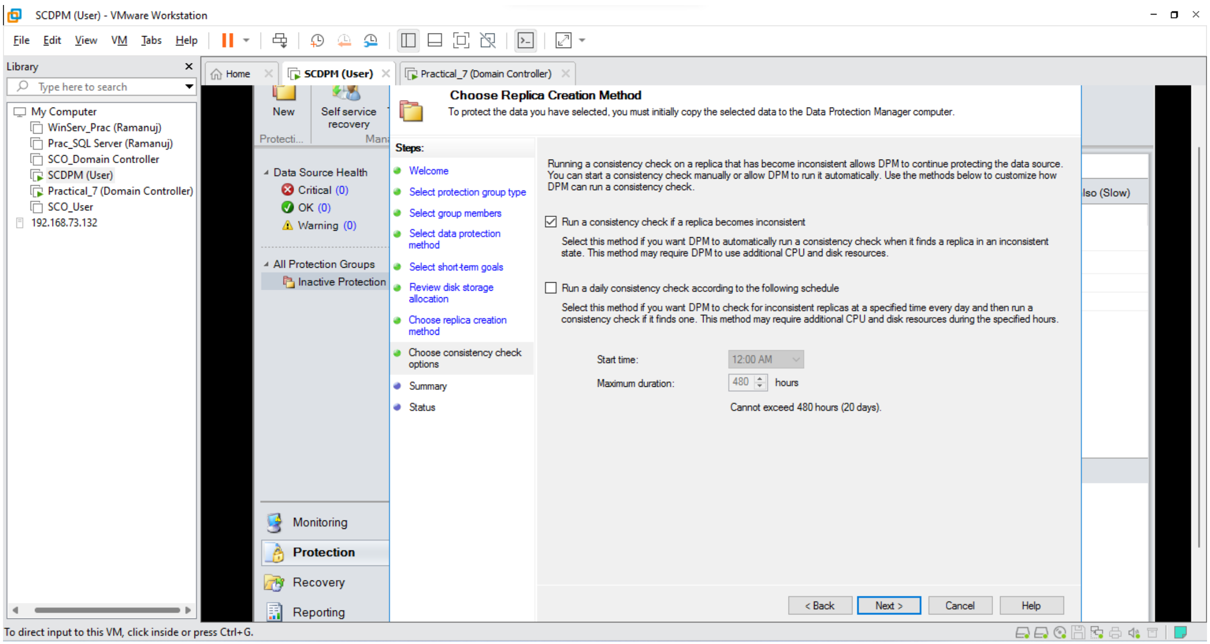
- Click **Next**



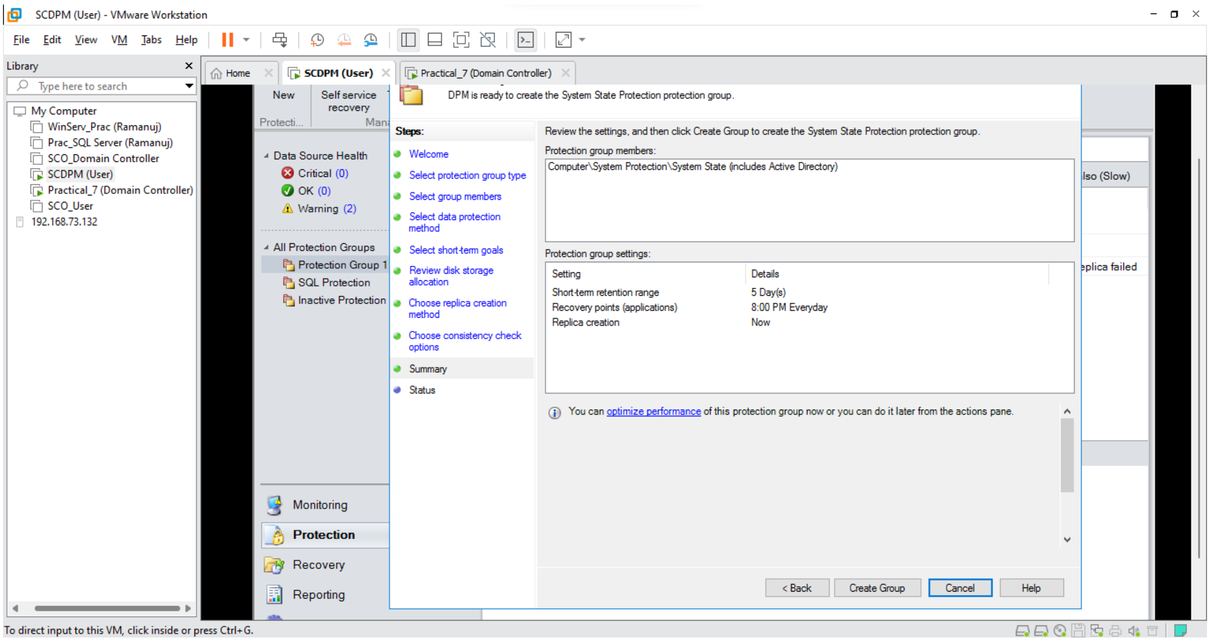
- Click **Next**



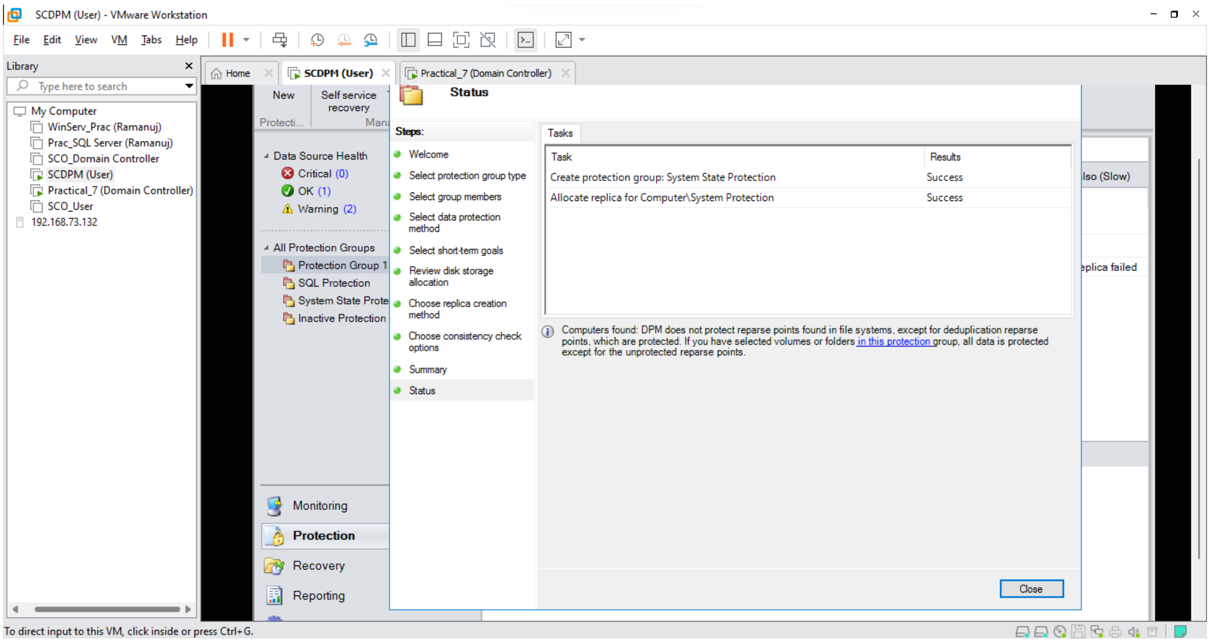
- Select **Run a consistency check if a replica becomes inconsistent** and Click **Next**



- View the summary of your group and Click **Create Group**

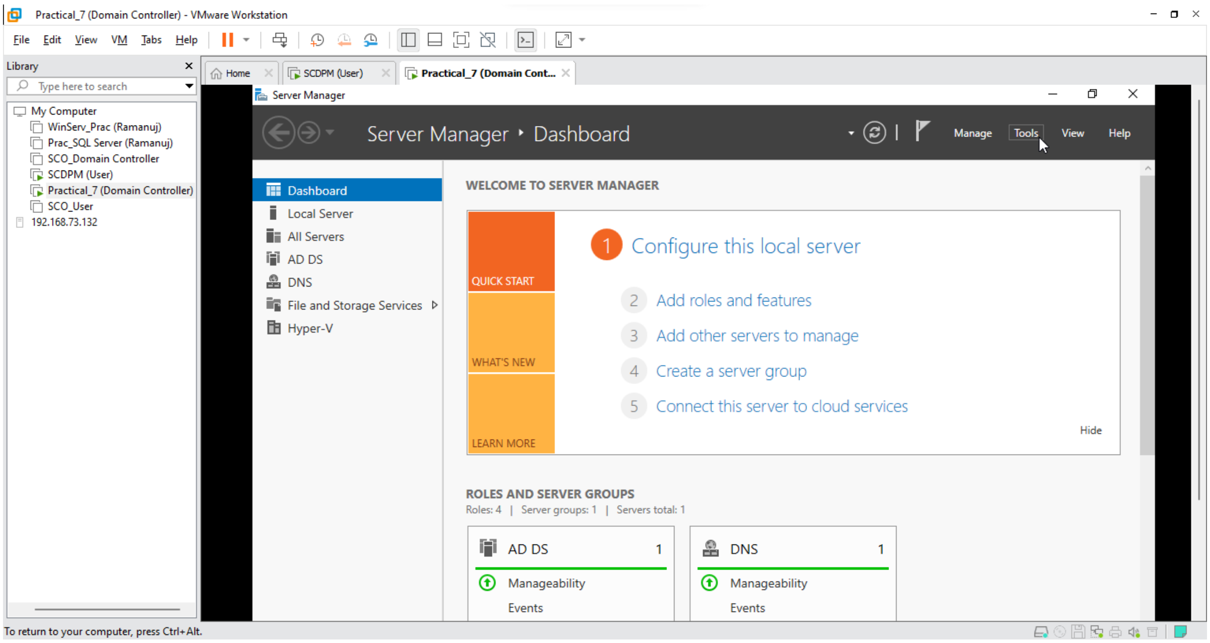


- After it is successfully created Click **Close**

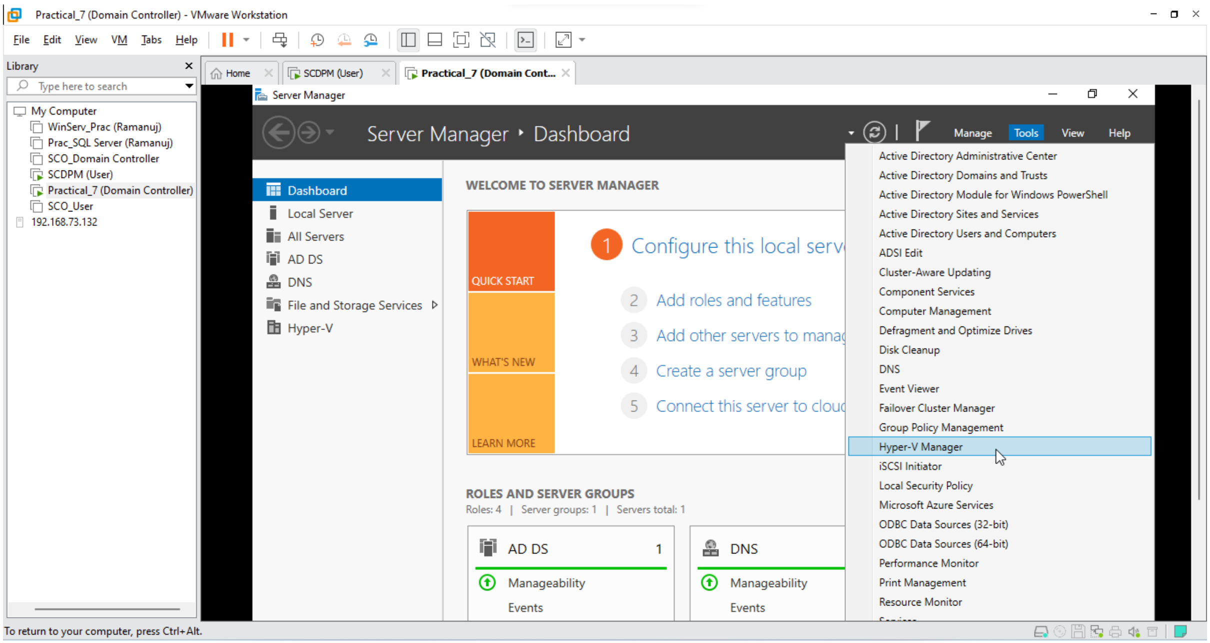


7-E: Backup VMware Servers
Step 1: Creating a VMware Server in Domain Controller

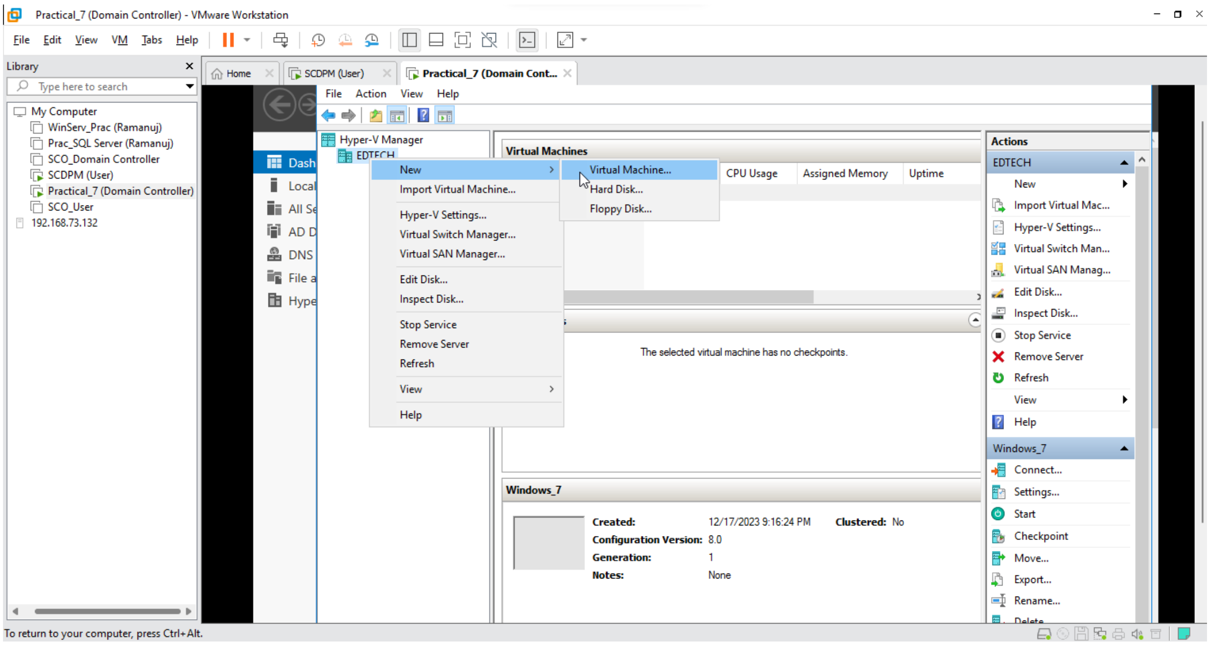
- Within Server Manager Click on **Tools**



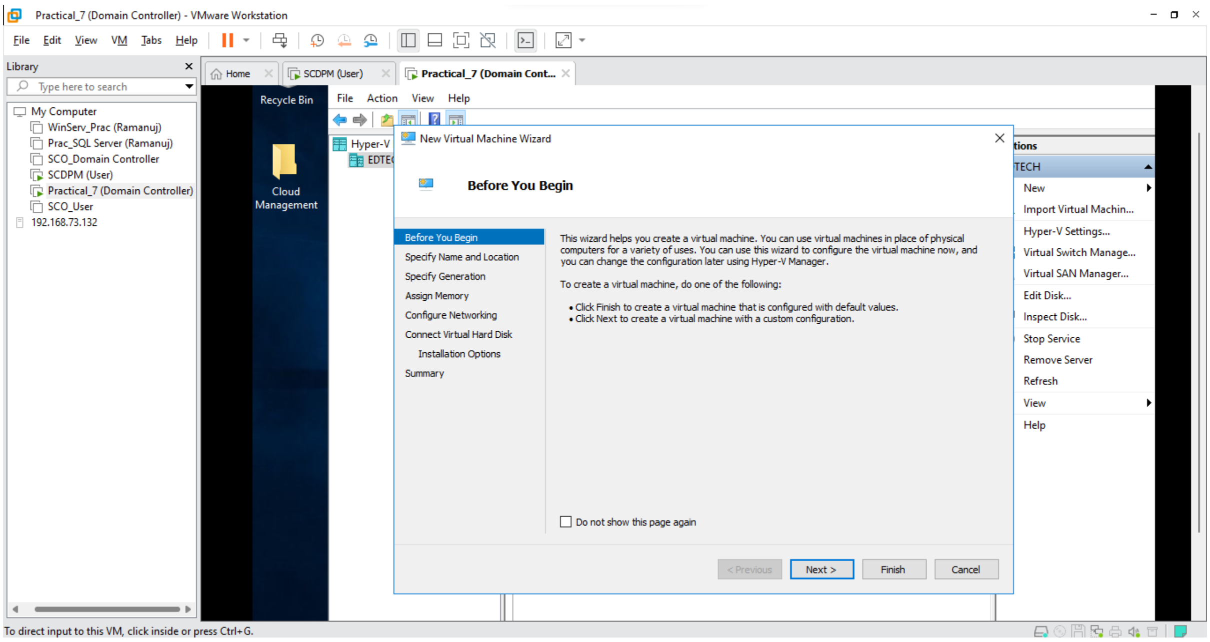
- Select **Hyper-V Manager**



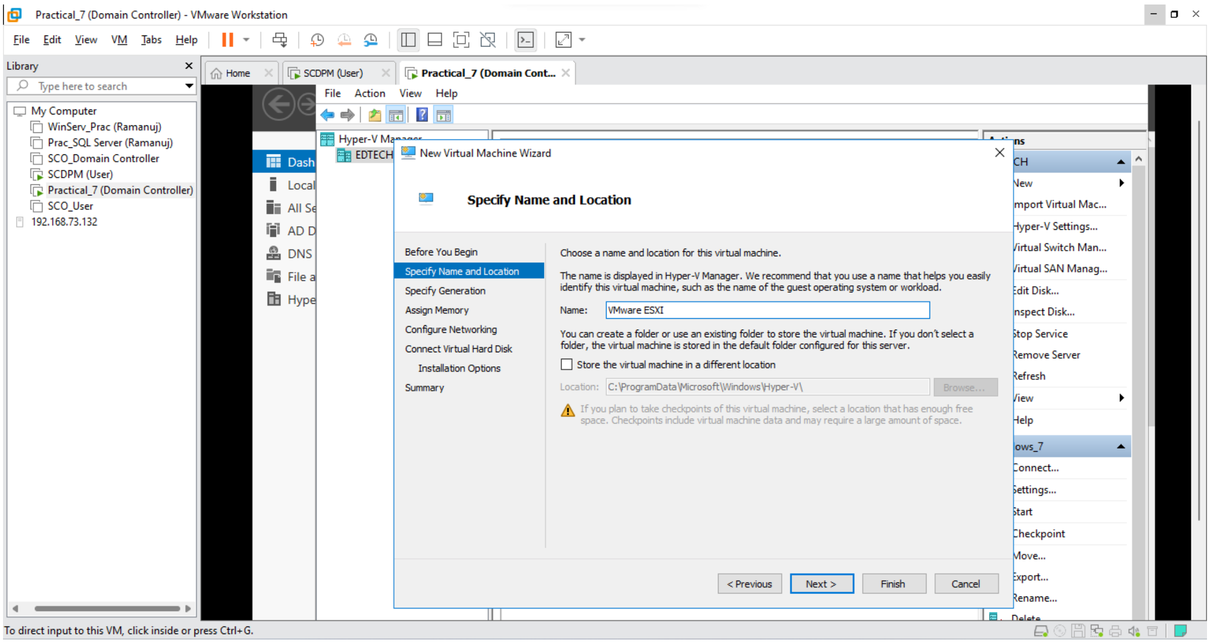
- Within Hyper-V Manager Right-Click on your **Domain Controller** and Select **New** and Click on **Virtual Machine**



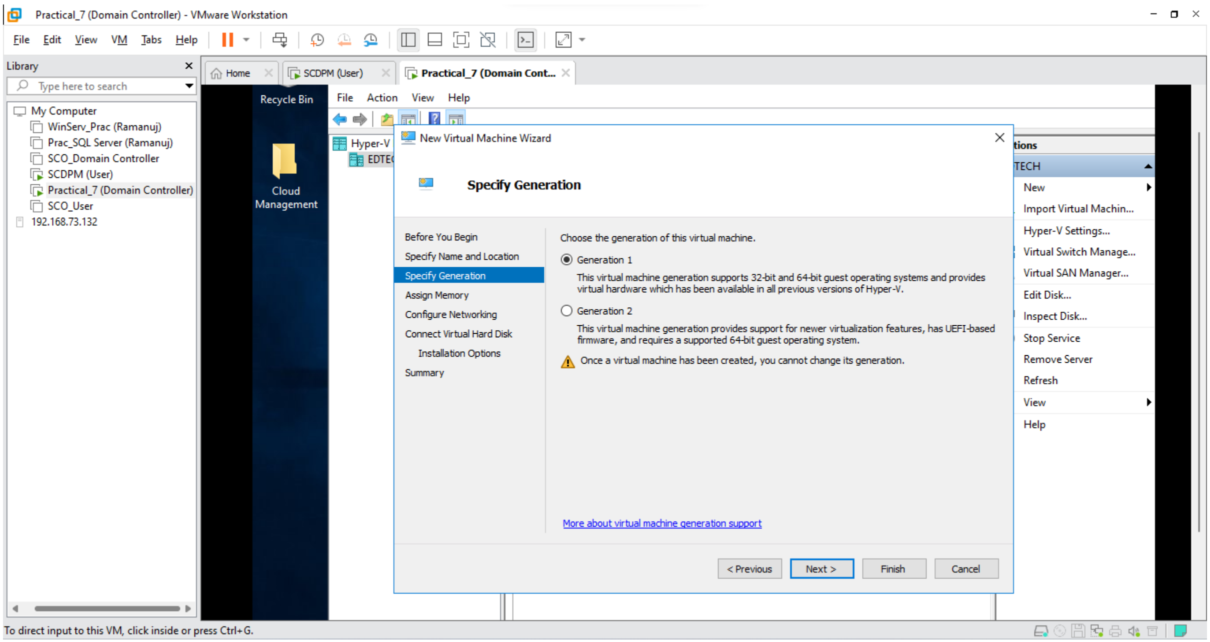
- Click **Next**



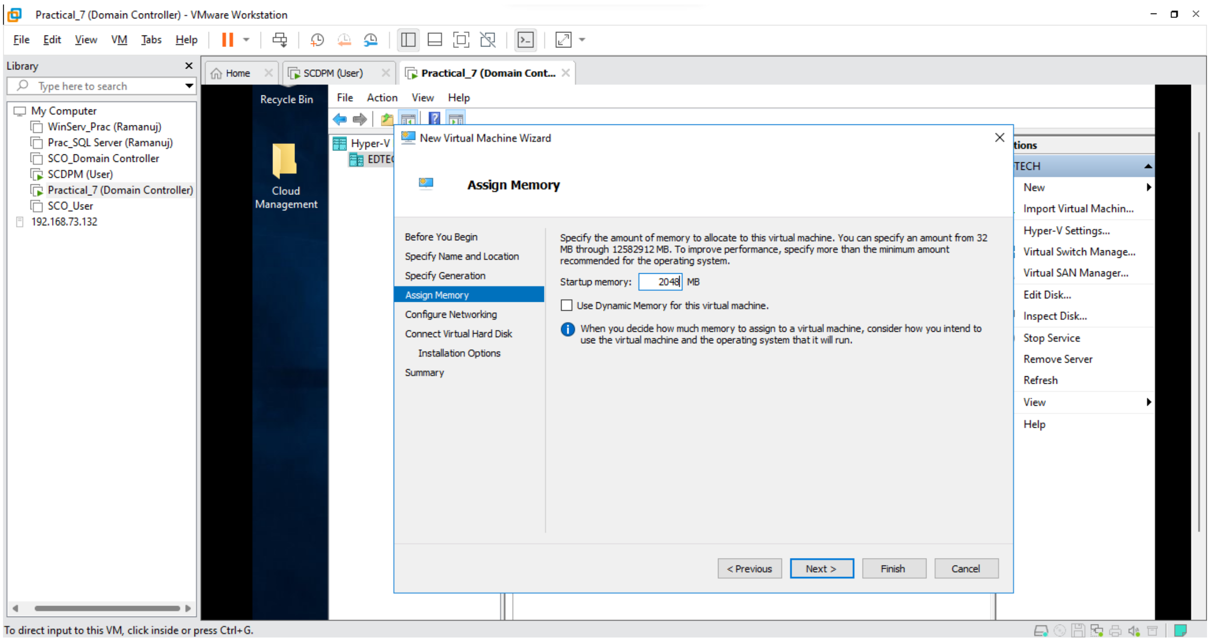
- Name your virtual machine (Here it is VMware ESXI) and Click **Next**



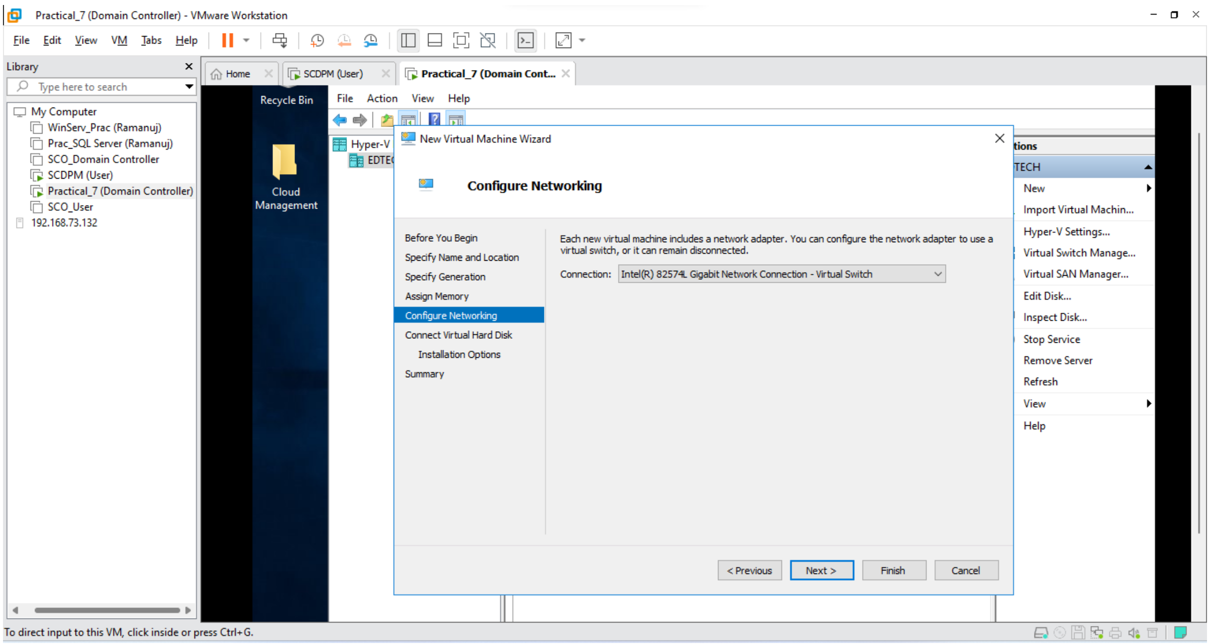
- Select **Generation 1** and Click **Next**



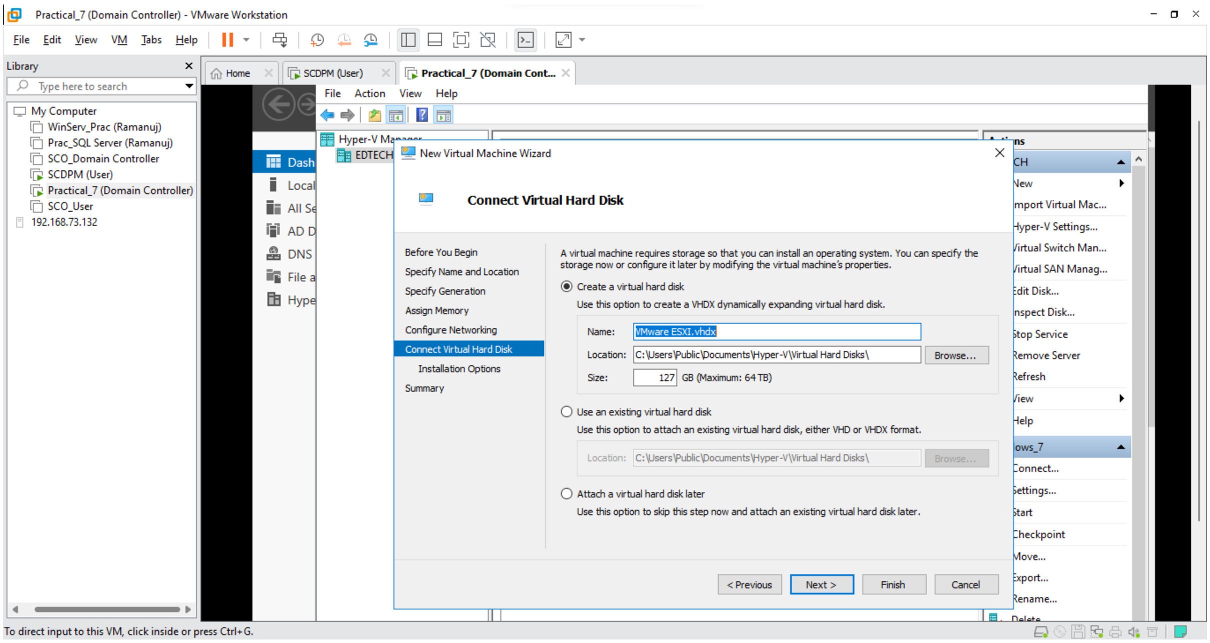
- Select Start-up memory as **2048 MB** and Click **Next**



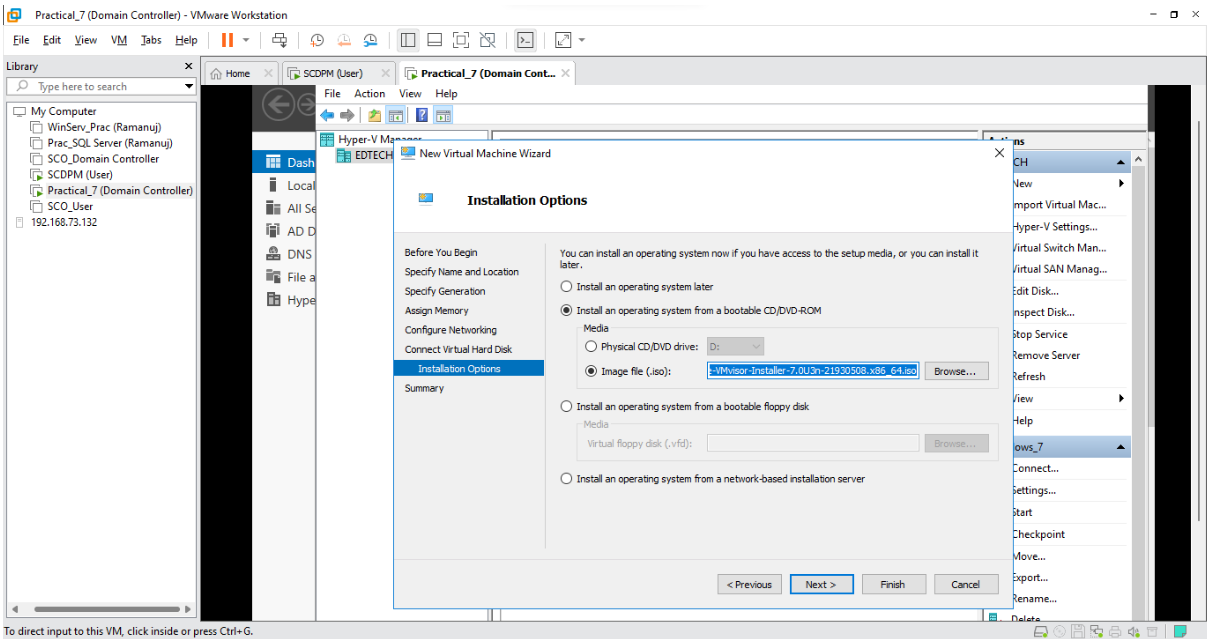
- In Connection Select your **network adapter** (Here it is Intel (R) Gigabit) and Click **Next**



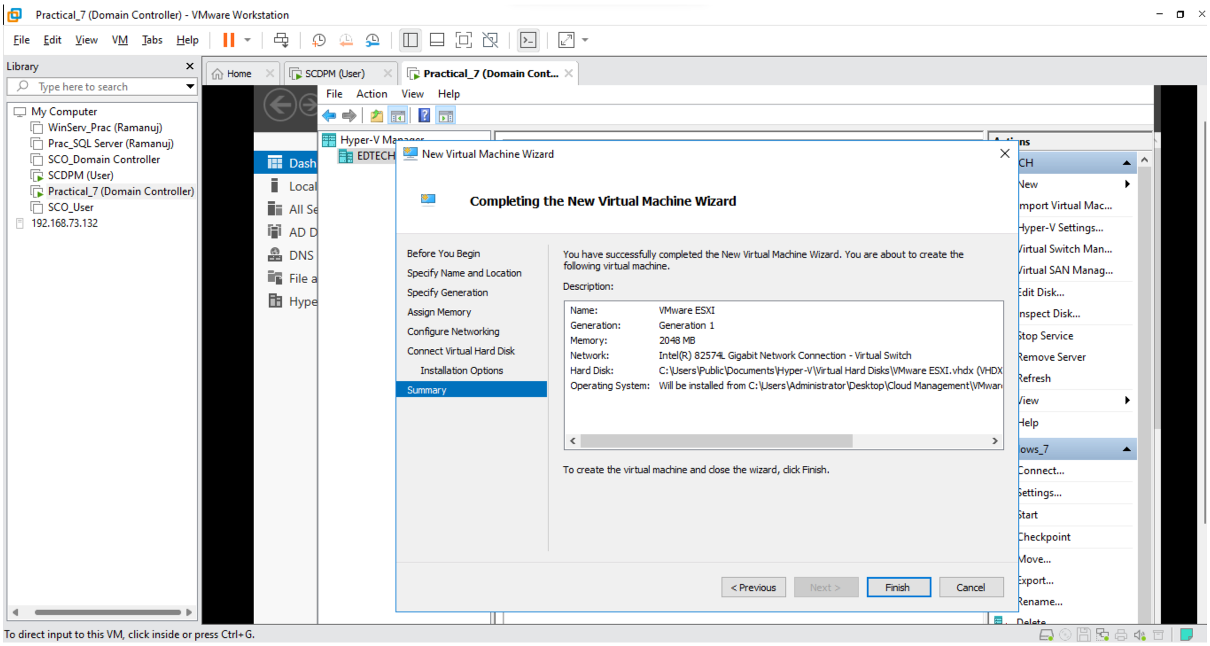
- Keep default values and Click **Next**



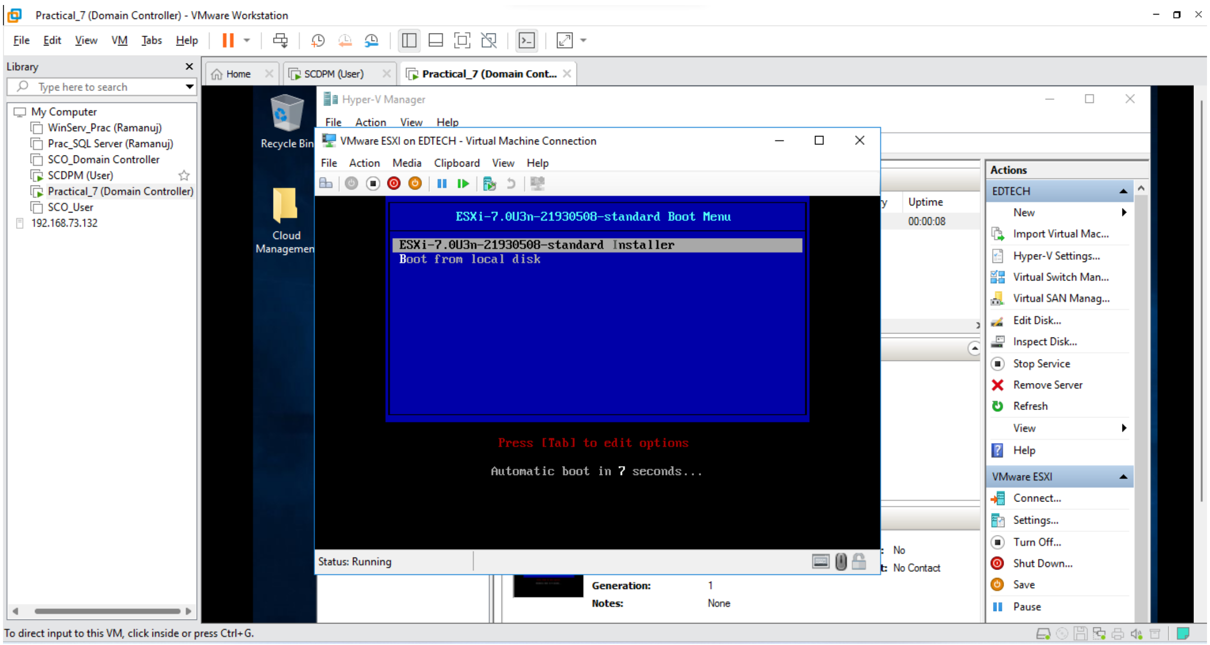
- Select **Install an operating system from a bootable CD/DVD ROM** and Click on **Image file** (.iso) and browse for your VMvisor7 iso file and Click **Next**



- View the summary of your virtual machine and Click **Next**

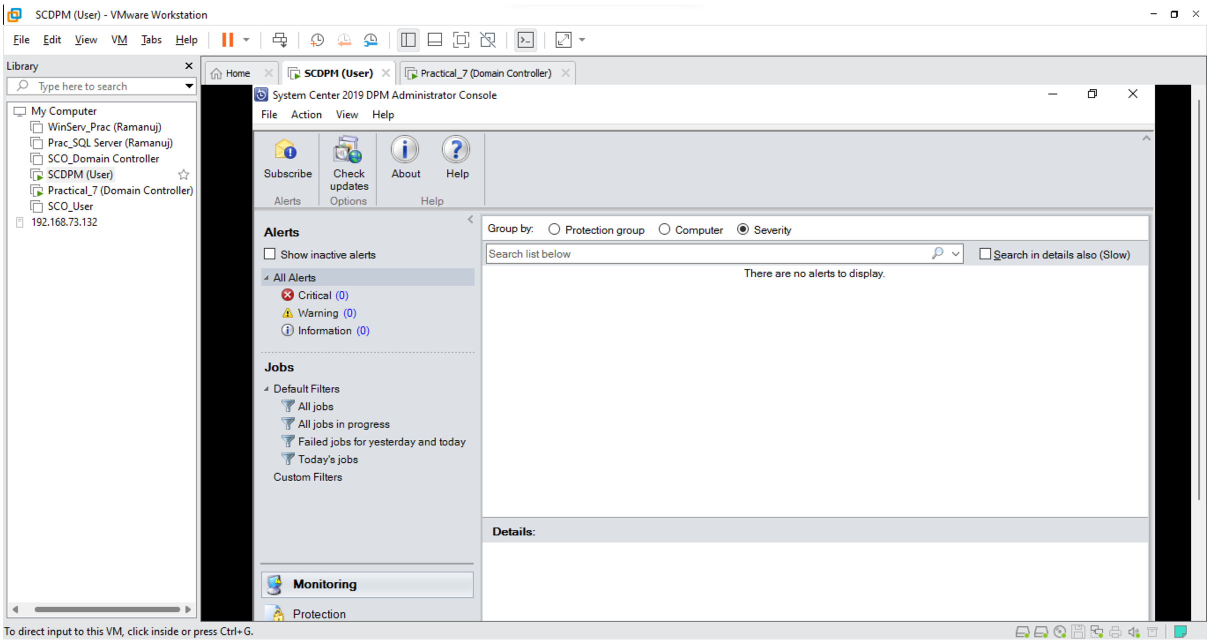


- To ensure it works **Power On** the Machine

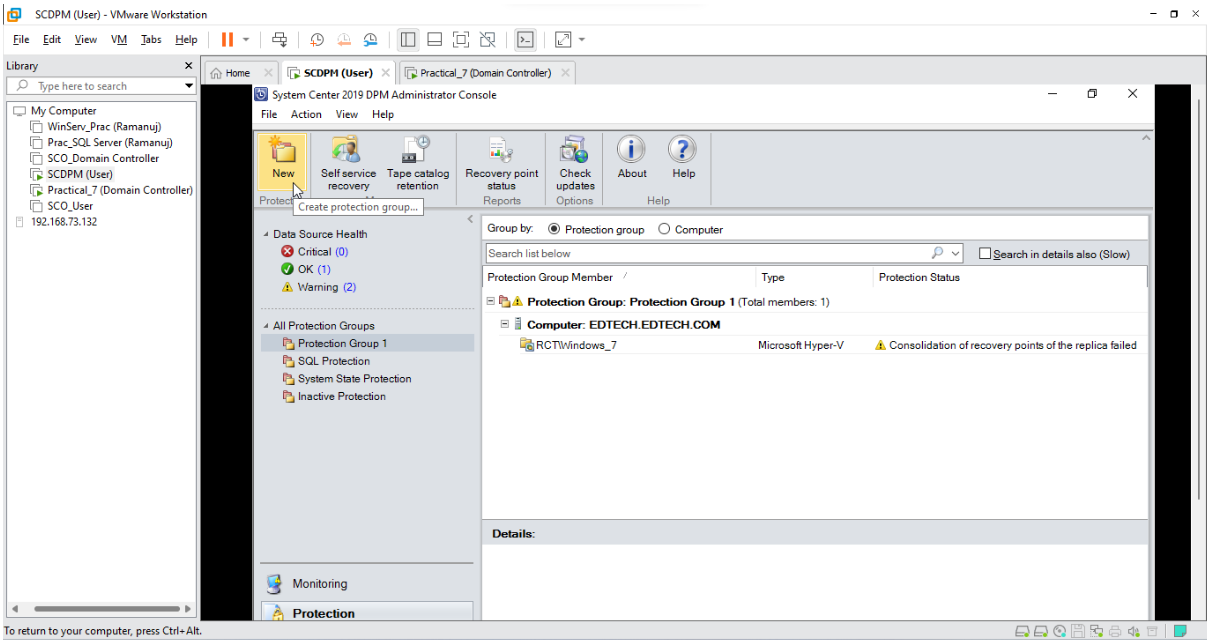


Step 2: Creating a Protection Group for the VMware Server

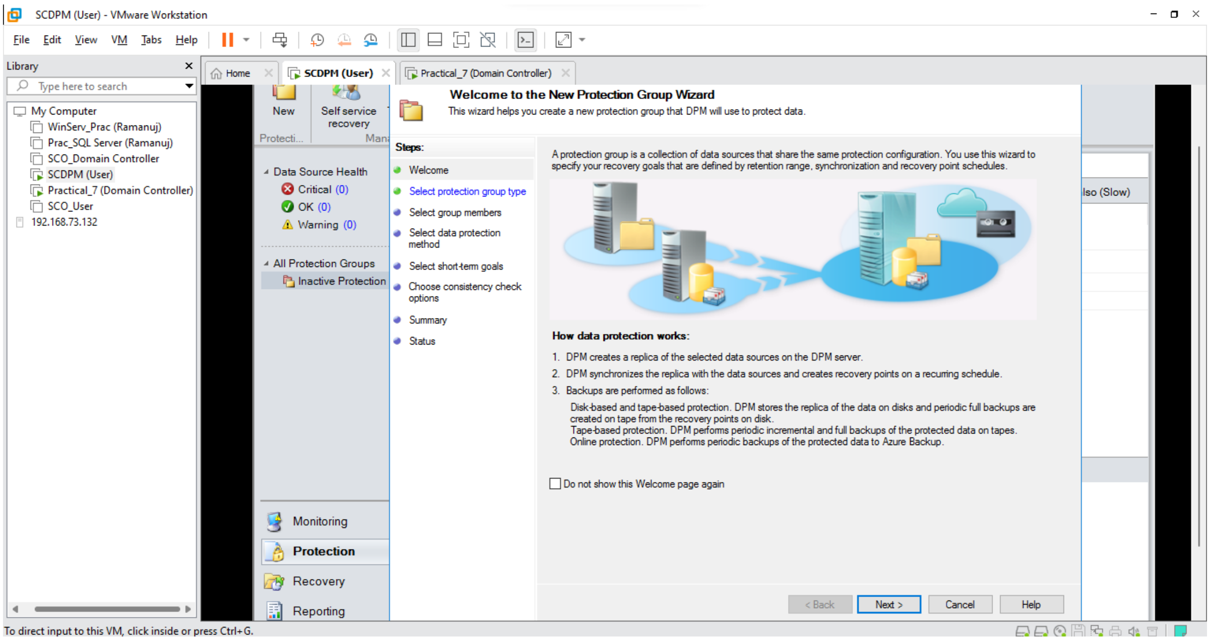
- Power off your VMware Server and Switch over to the **SCDPM (User)** VM and Open the **DPM manager**



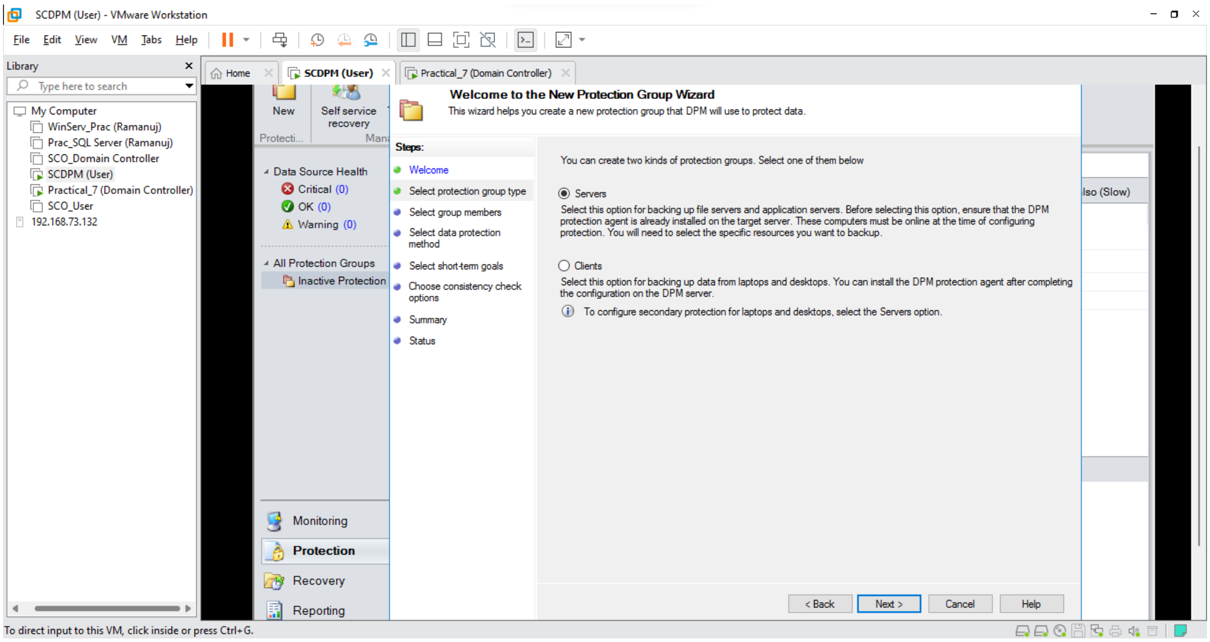
- Click on the **Protection** Tab and Select **New**



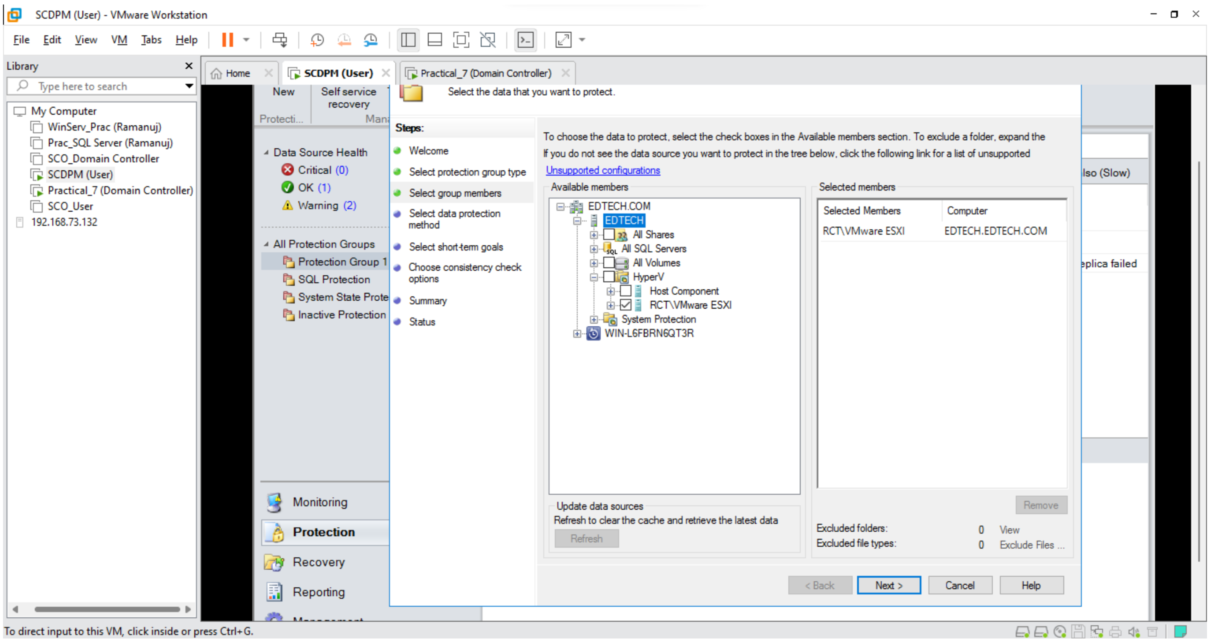
- Click **Next**



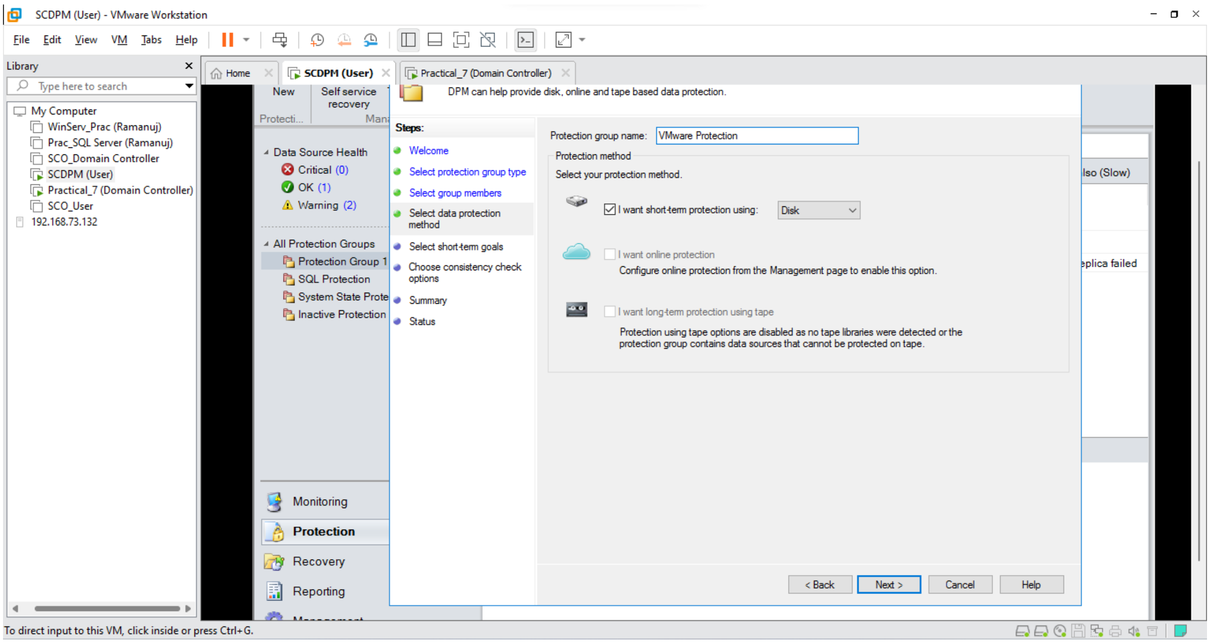
- Select **Servers** and Click **Next**



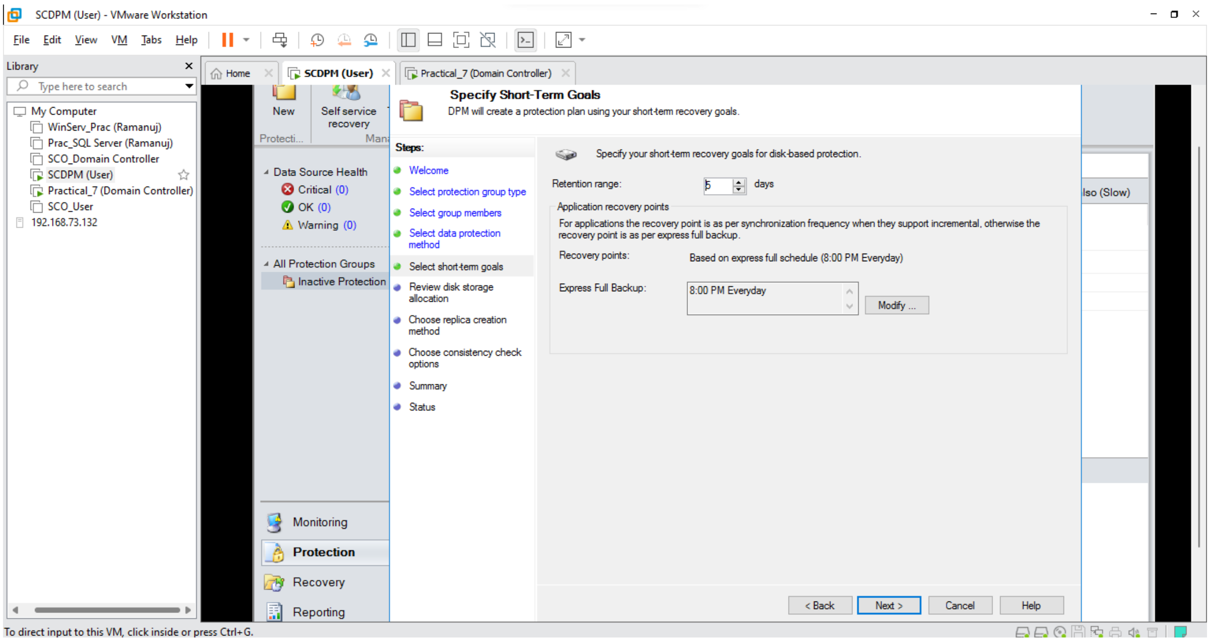
- Select **EDTECH** and Click on **Refresh,** after refresh Open **Hyper-V** and Click on your **virtual machine** (Here it is RCT\VMware ESXi) and Click **Next**



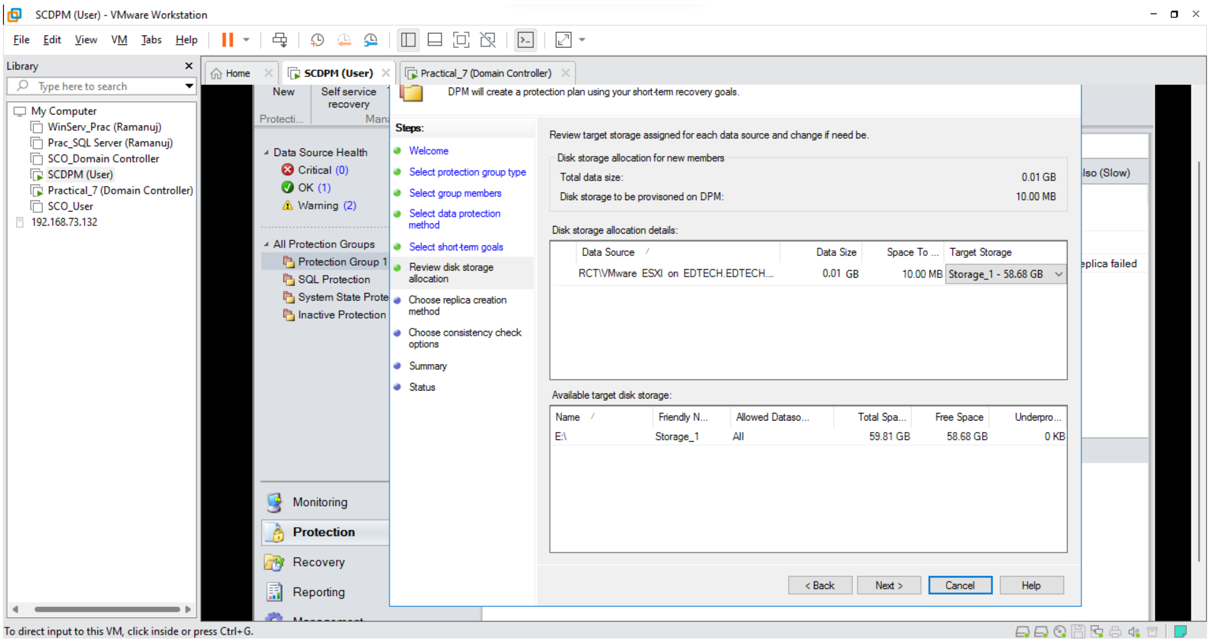
- Name your **Protection Group** (Here it is VMware Protection) and Select **I want short term protection** and Select **Disk** and Click **Next**



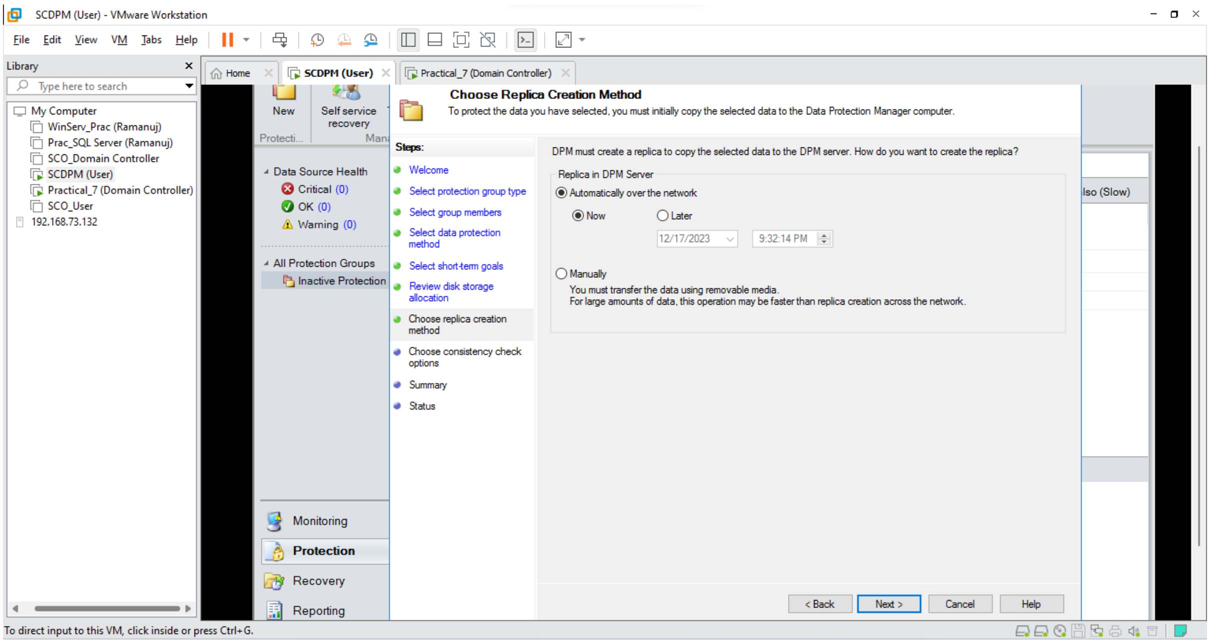
- Keep default values and Click **Next**



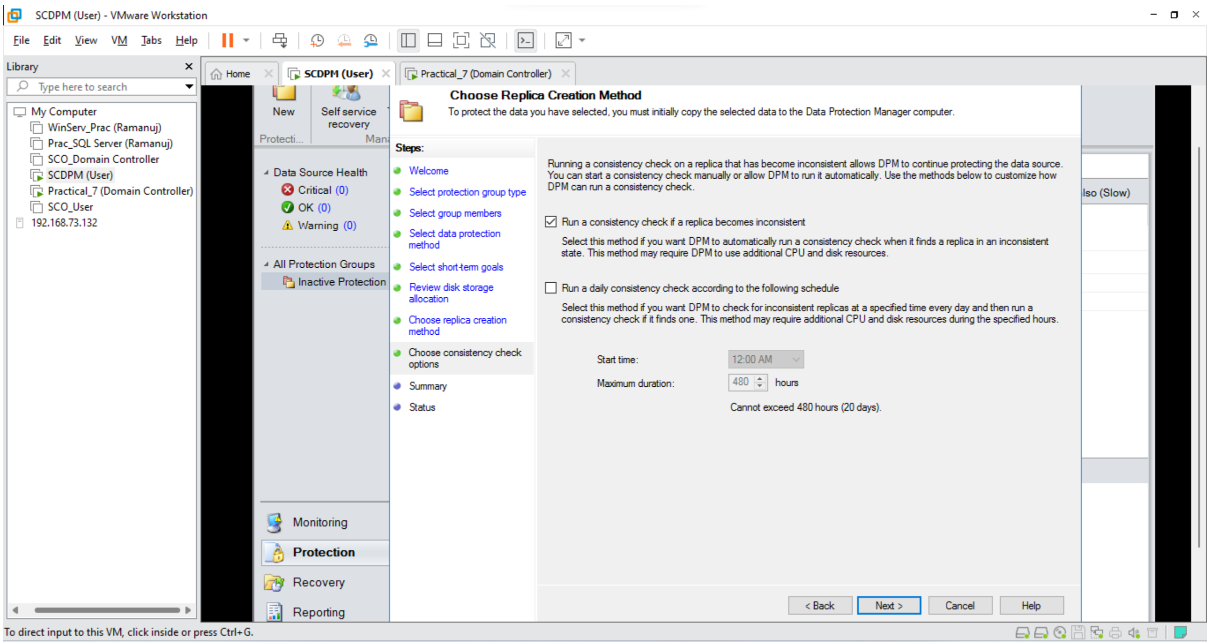
- Click **Next**



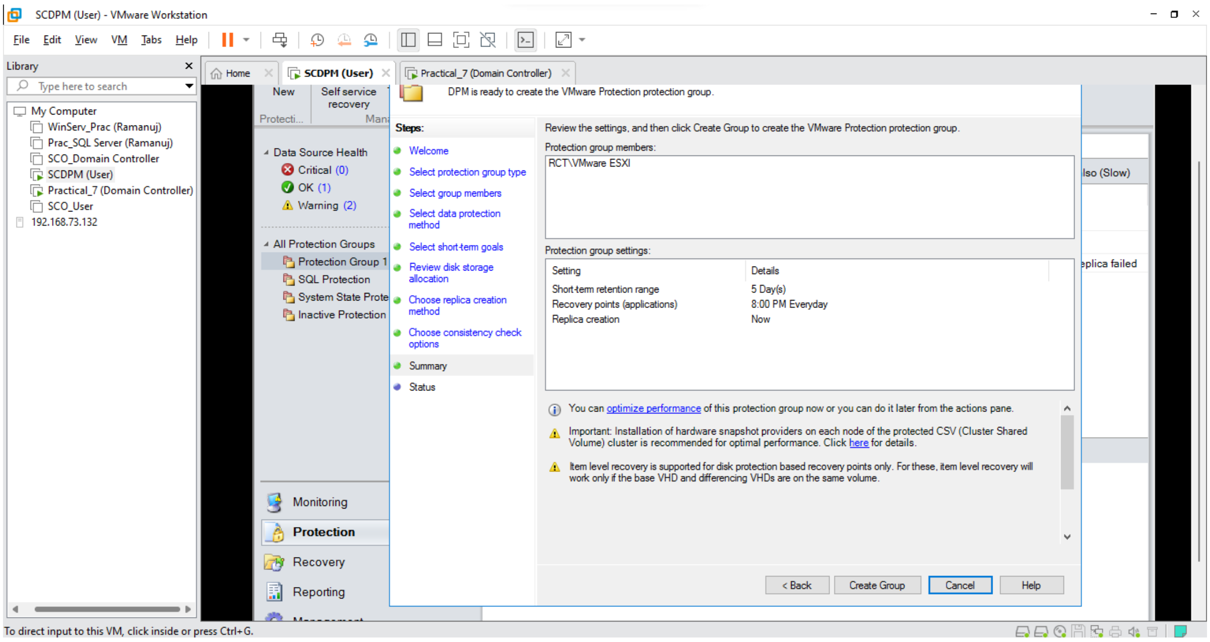
- Click **Next**



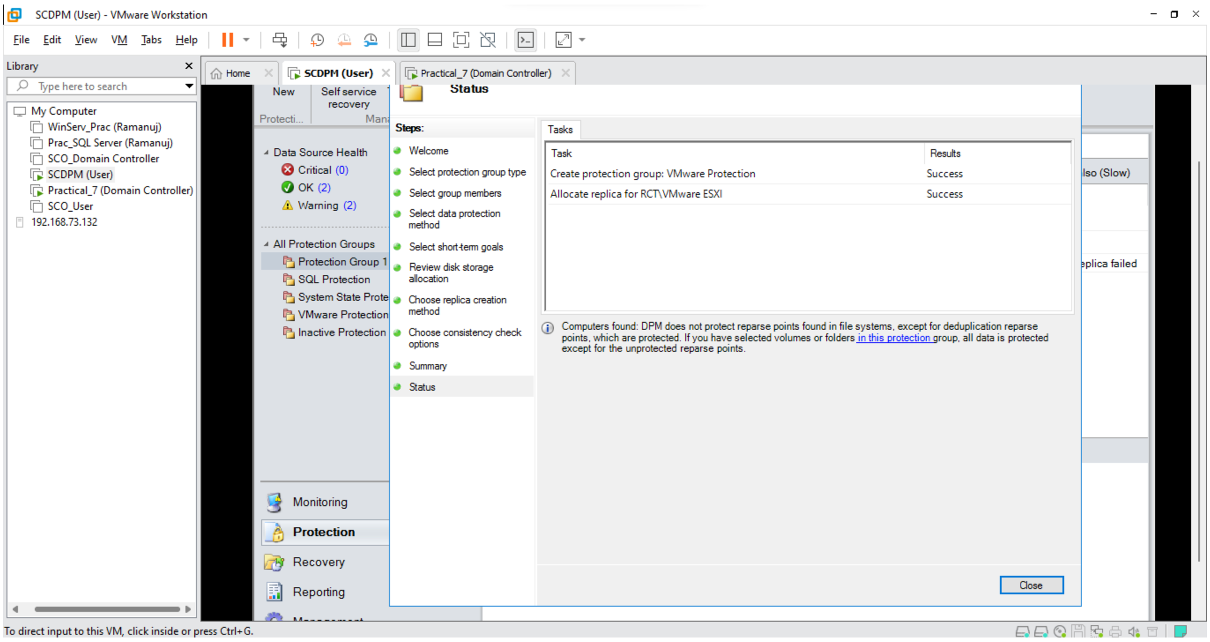
- Select **Run a consistency check if a replica becomes inconsistent** and Click **Next**



- View the summary of your group and Click **Create Group**



- After it is successfully created Click **Close**



- Now all your Protection Groups have been created successfully